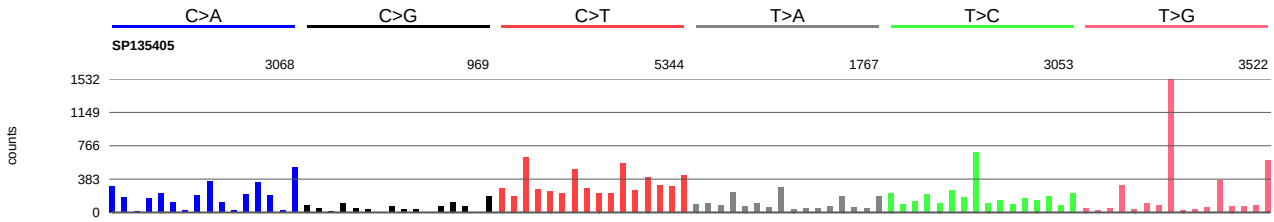
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SP135405
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5344
1767
3053
3522
counts
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383
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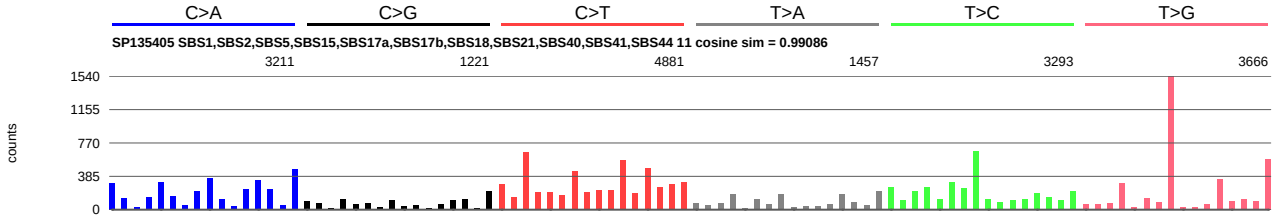
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1221
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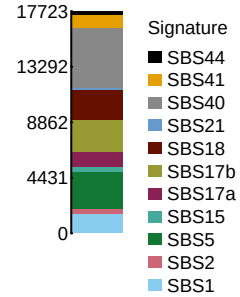
1457
3293
3666
counts
1540
1155
770
385
0


17723
13292
8862
4431
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Signature
SBS44
SBS41
SBS40
SBS21
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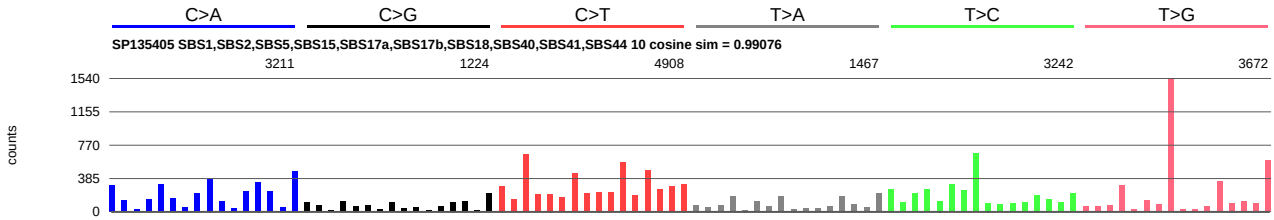
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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1224
4908
1467
3242
3672
counts
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1155
770
385
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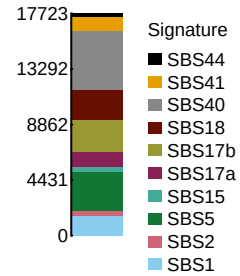
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13292
8862
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SBS2
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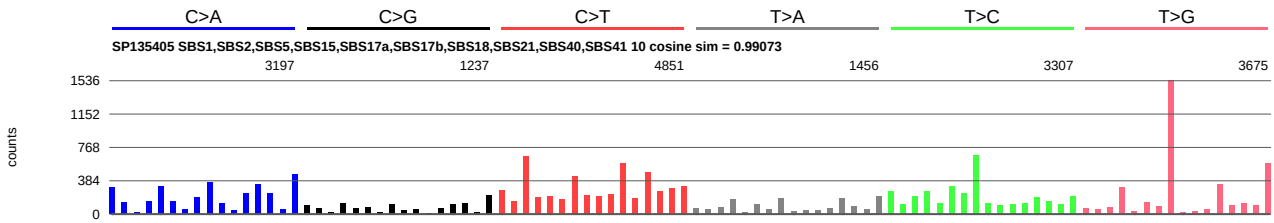
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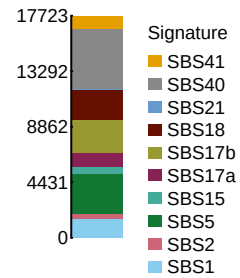
1237
4851
1456
3307
3675
counts
1536
1152
768
384
0


17723
13292
8862
4431
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Signature
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SBS21
SBS18
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SBS17a
SBS15
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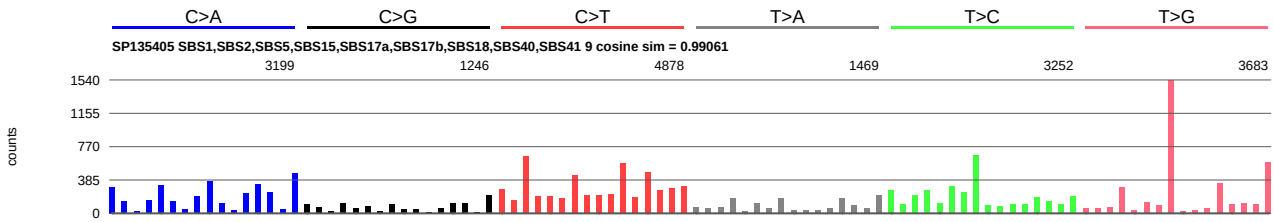
SBS2
SBS1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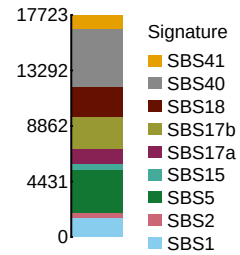
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99061
3199
1246
4878
1469
3252
3683
counts
1540
1155
770
385
0


17723
13292
8862
4431
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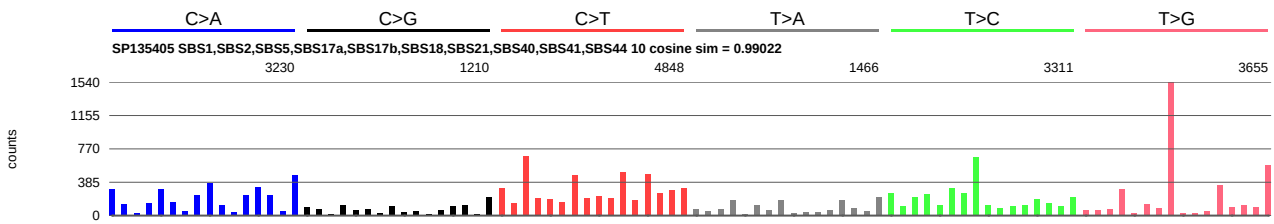
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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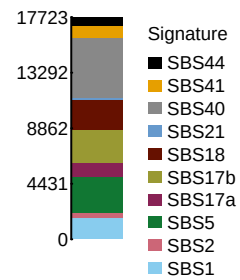
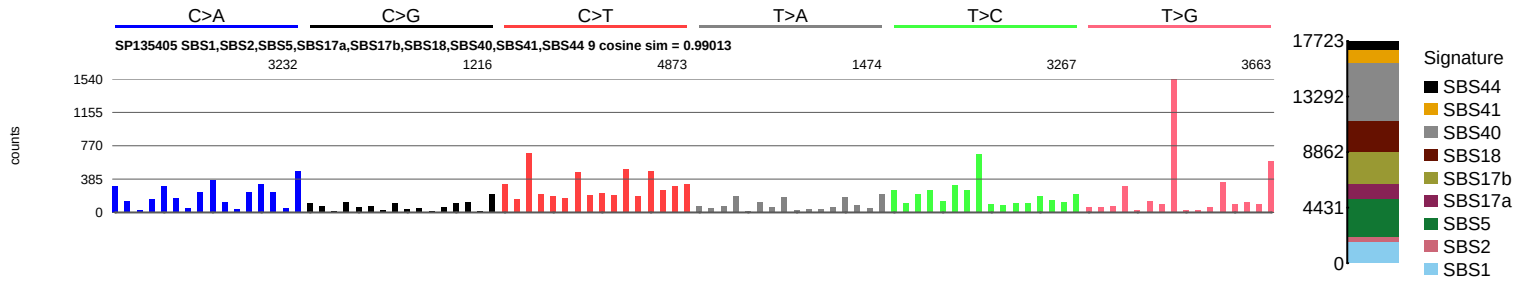
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99022
3230
1210
4848
1466
3311
3655
counts
1540
1155
770
385
0


17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99013
3232
1216
4873
1474
3267
3663
counts
1540
1155
770
385
0
17723
13292
8862
4431
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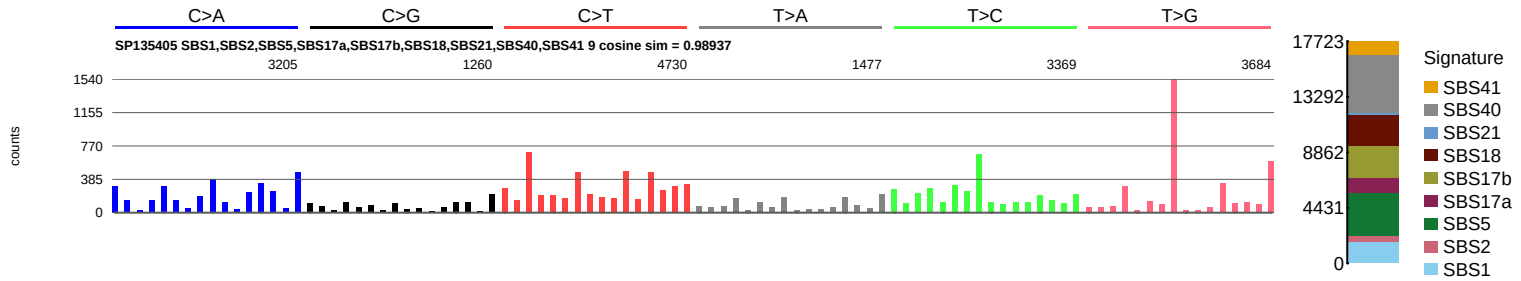
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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T>C
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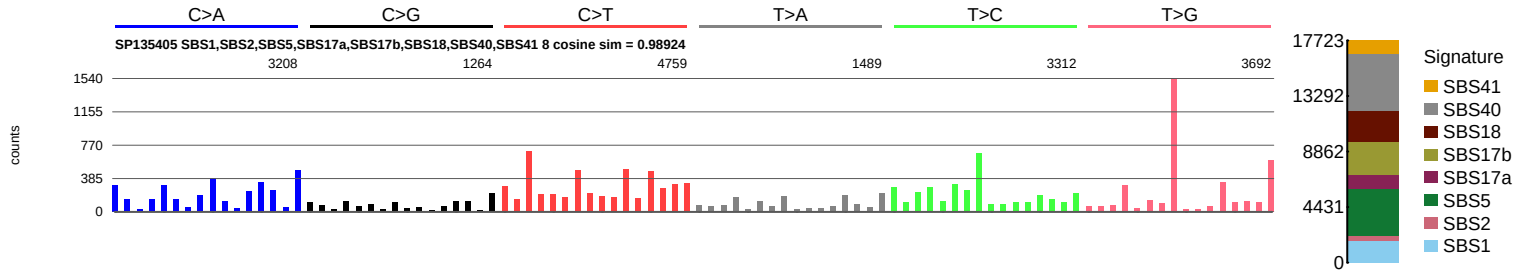
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98937
3205
1260
4730
1477
3369
3684
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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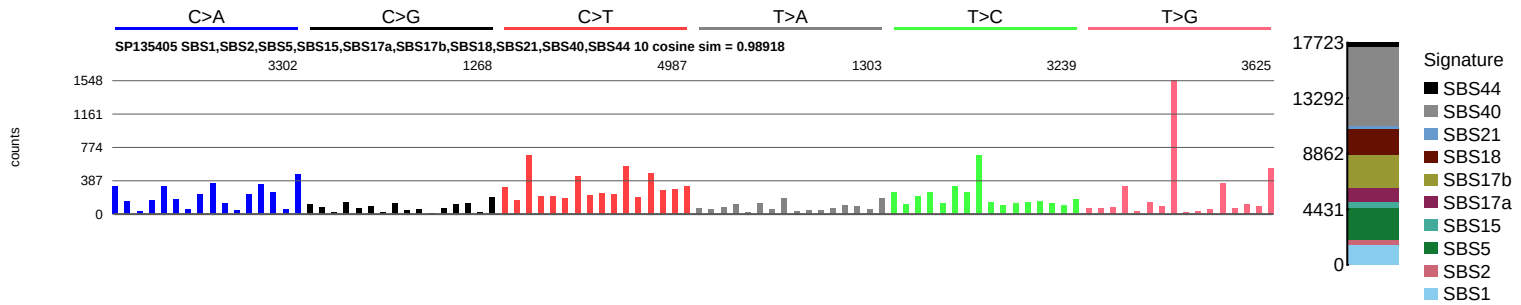
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98924
3208
1264
4759
1489
3312
3692
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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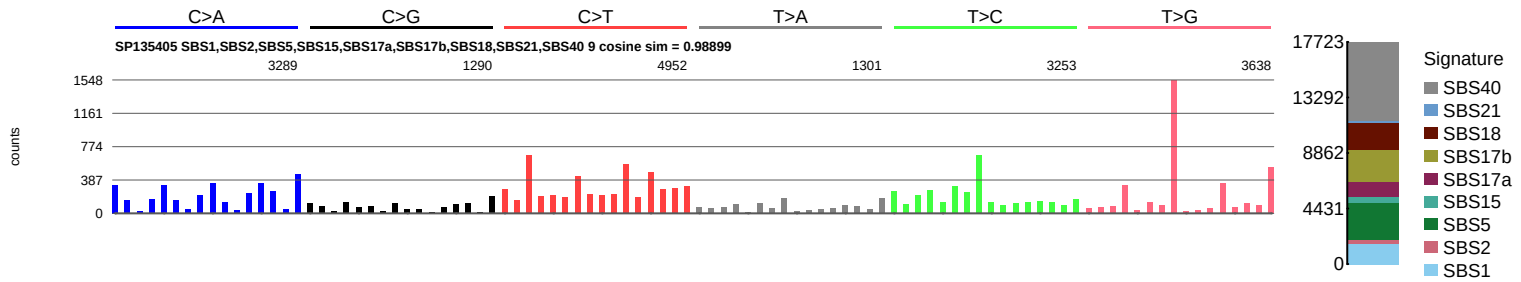
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98918
3302
1268
4987
1303
3239
3625
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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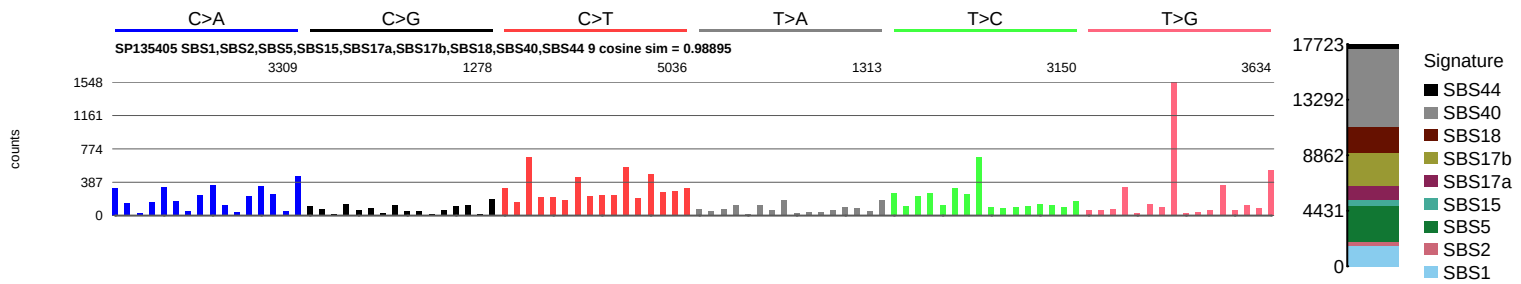
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98899
3289
1290
4952
1301
3253
3638
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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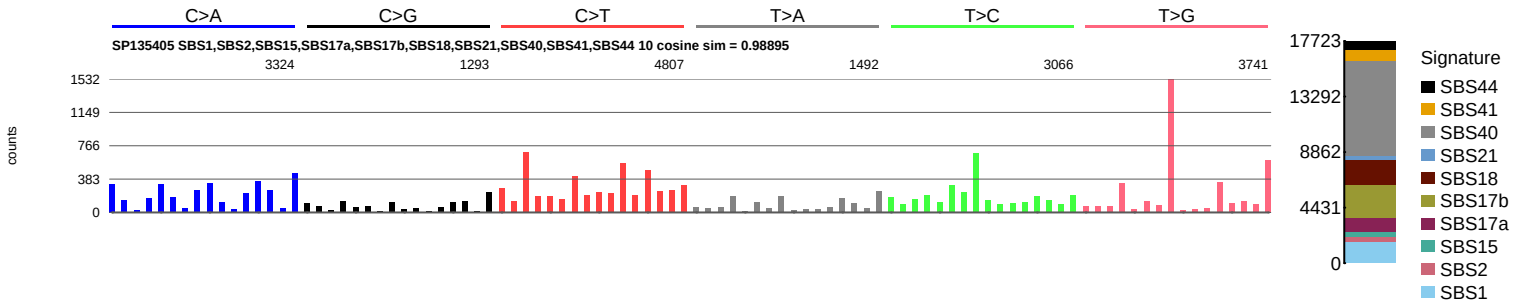
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98895
3309
1278
5036
1313
3150
3634
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98895
3324
1293
4807
1492
3066
3741
counts
1532
1149
766
383
0
17723
13292
8862
4431
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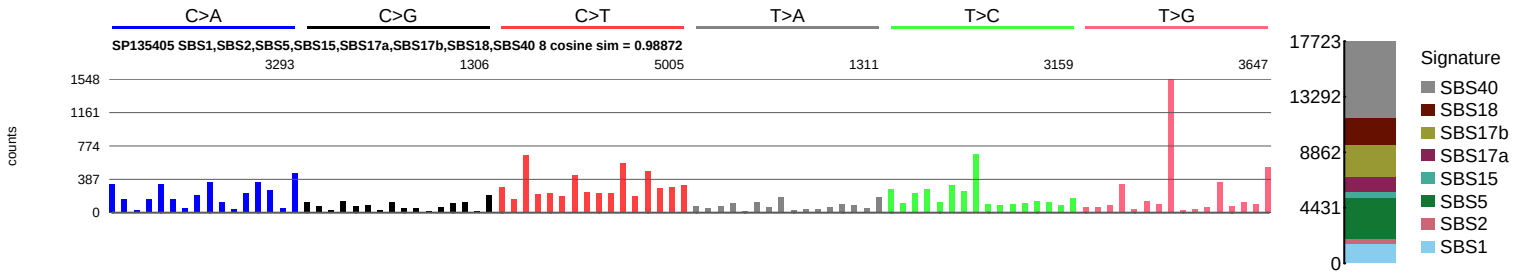
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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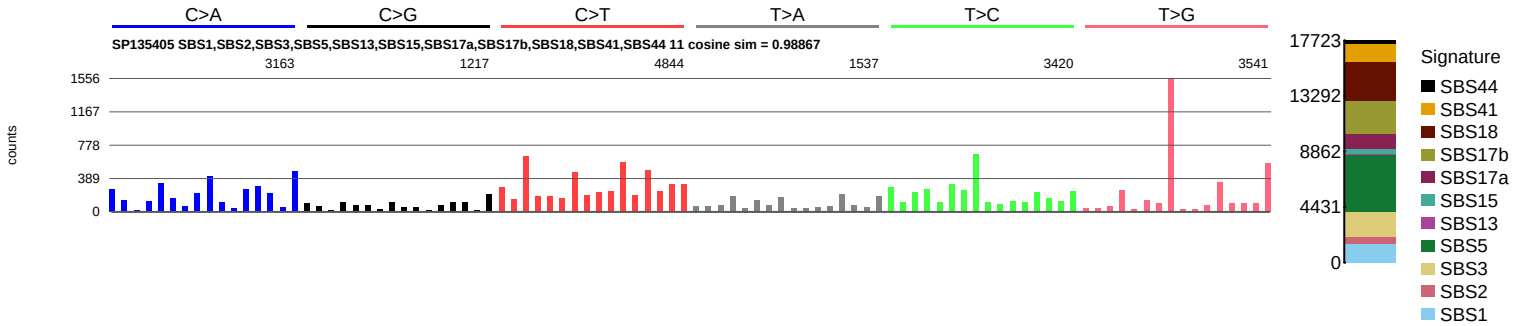
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98872
3293
1306
5005
1311
3159
3647
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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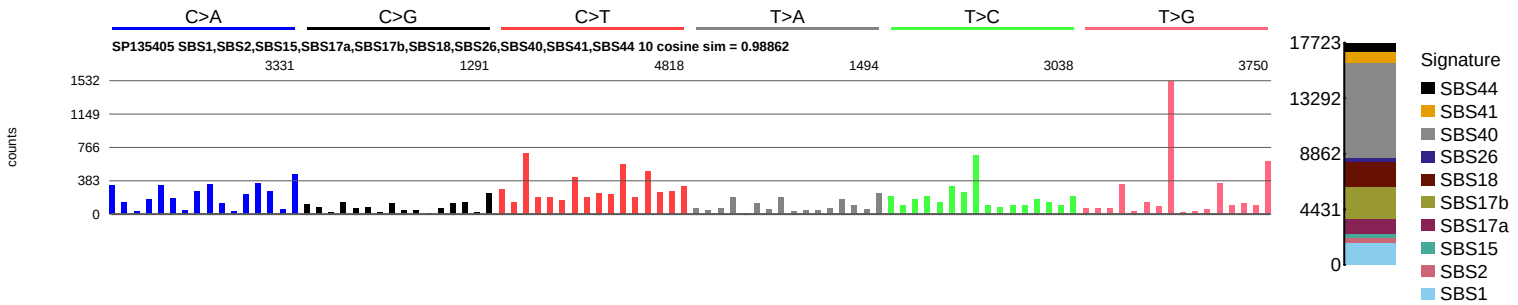
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98867
3163
1217
4844
1537
3420
3541
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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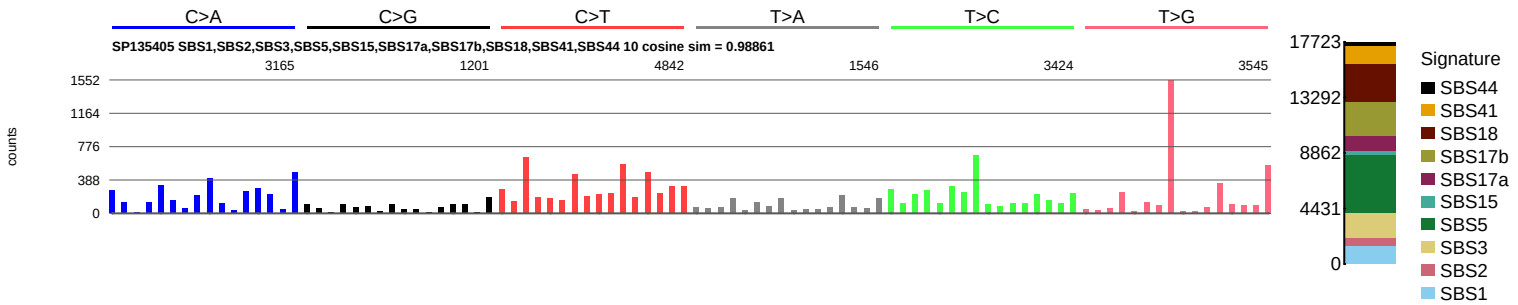
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98862
3331
1291
4818
1494
3038
3750
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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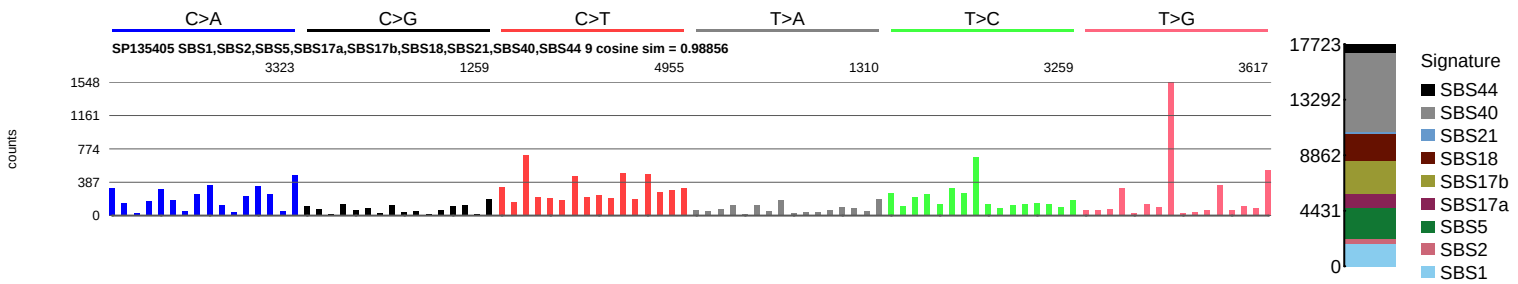
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98861
3165
1201
4842
1546
3424
3545
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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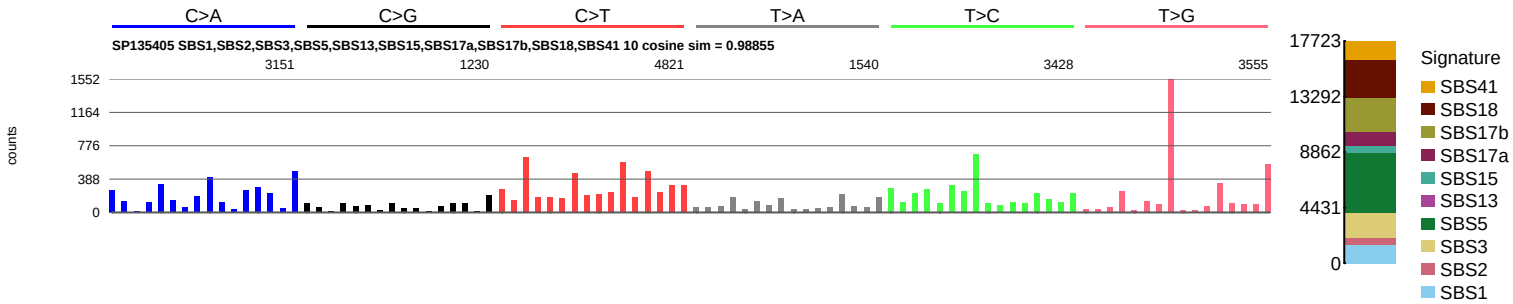
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98856
3323
1259
4955
1310
3259
3617
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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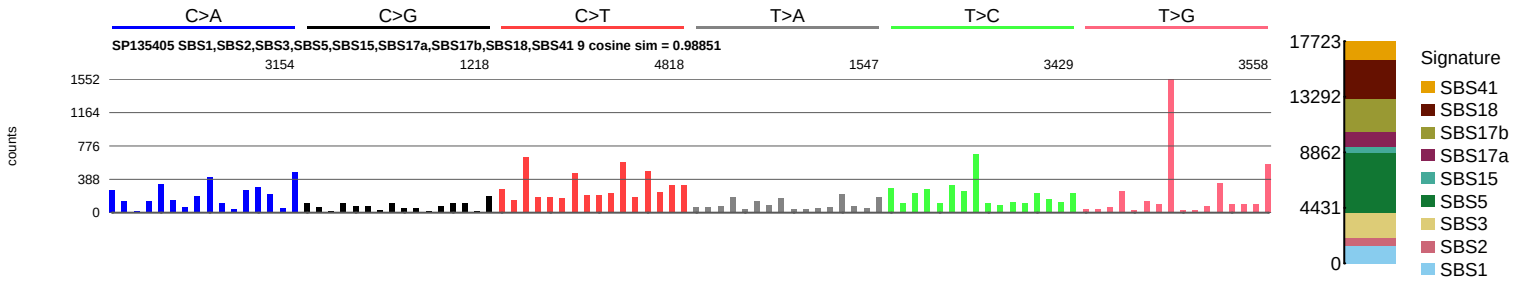
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98855
3151
1230
4821
1540
3428
3555
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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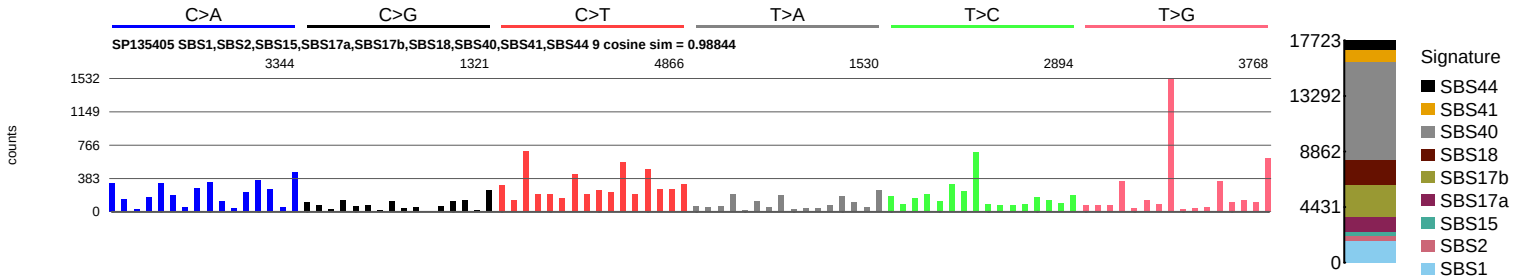
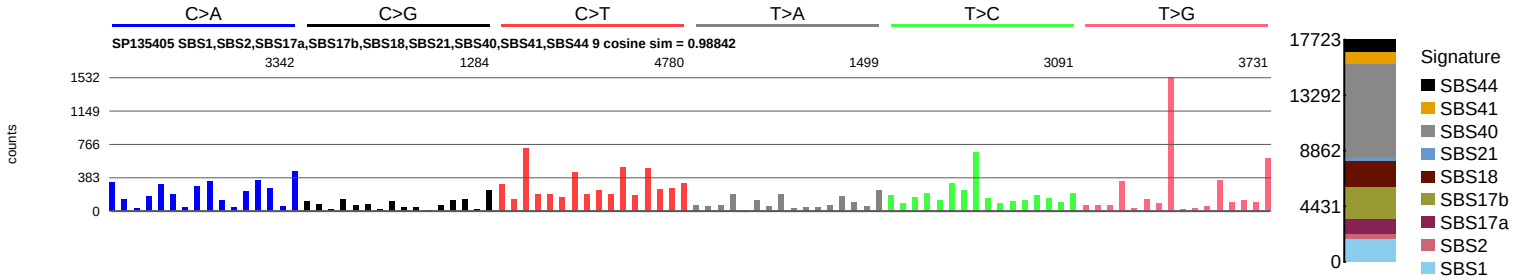
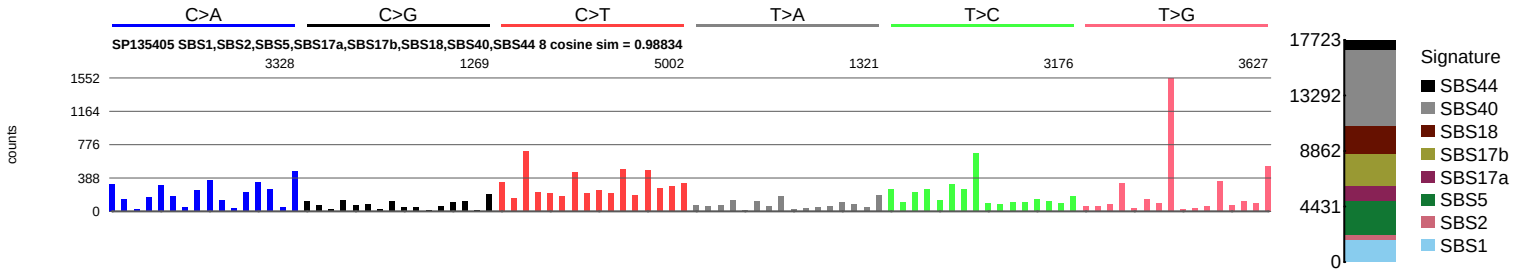
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98851
3154
1218
4818
1547
3429
3558
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98844
3344
1321
4866
1530
2894
3768
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98842
3342
1284
4780
1499
3091
3731
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98834
3328
1269
5002
1321
3176
3627
counts
1552
1164
776
388
0
17723
13292
8862
4431
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SBS44
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SBS18
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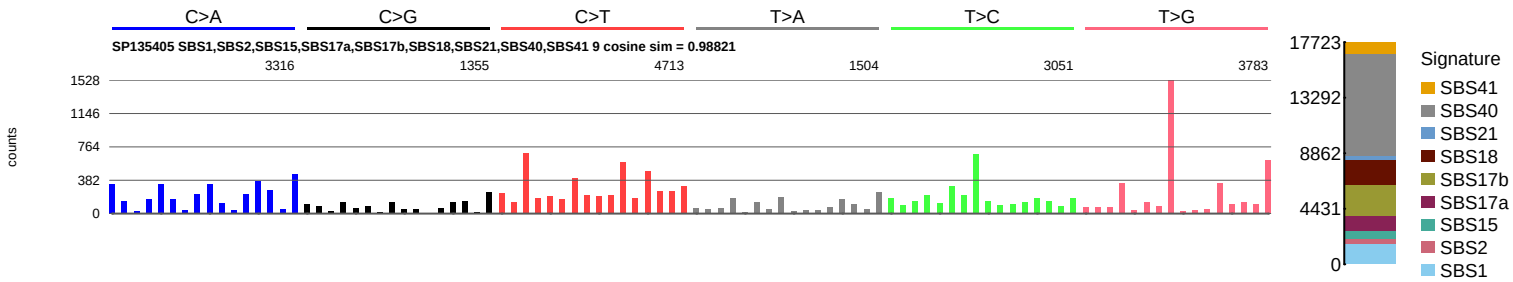
SBS2
SBS1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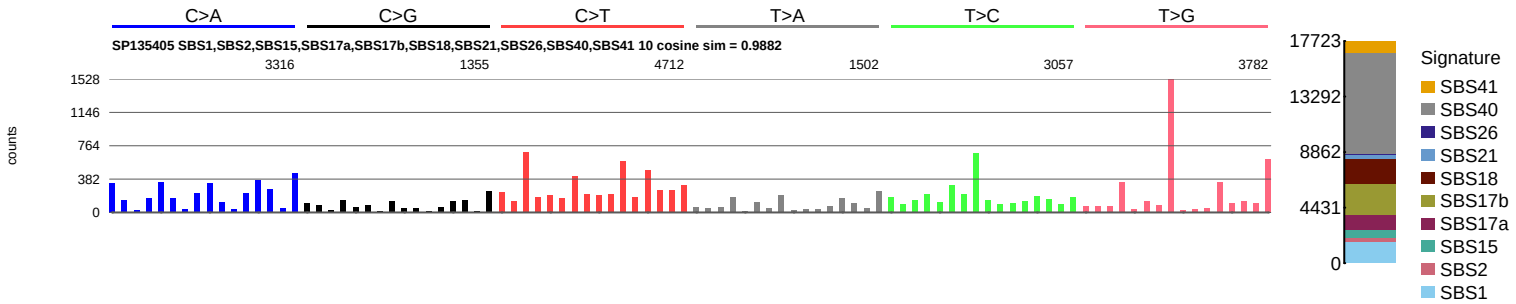
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98821
3316
1355
4713
1504
3051
3783
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9882
3316
1355
4712
1502
3057
3782
counts
1528
1146
764
382
0
17723
13292
8862
4431
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SBS21
SBS18
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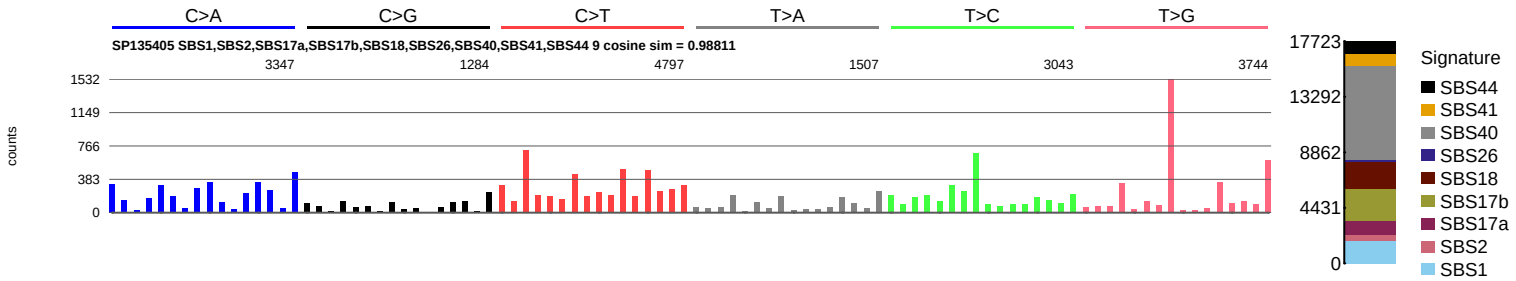
SBS2
SBS1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SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98811
3347
1284
4797
1507
3043
3744
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
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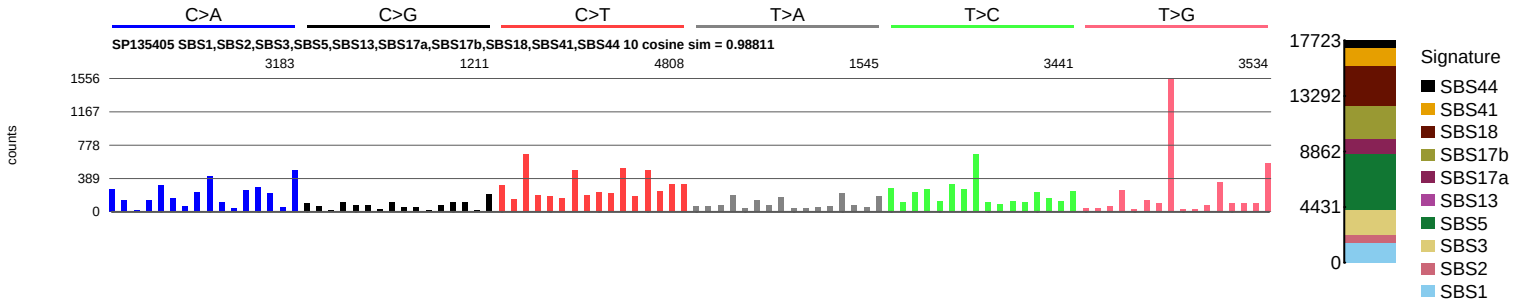
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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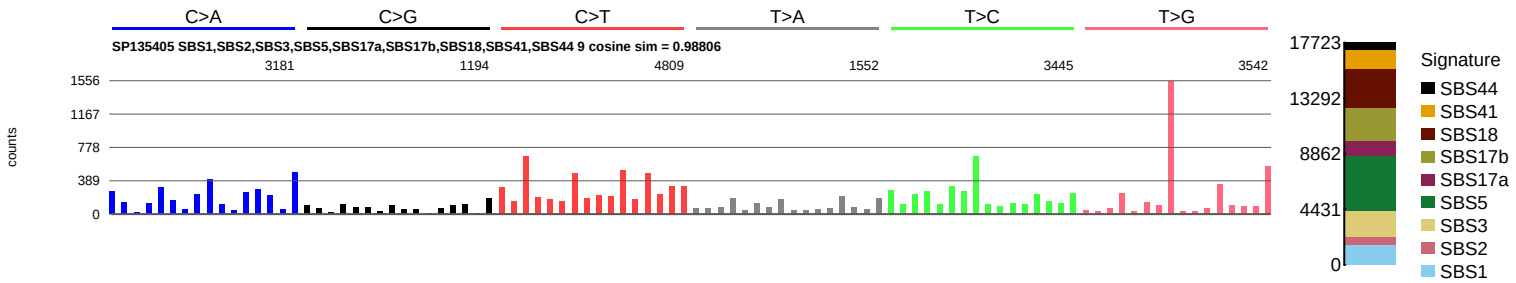
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98811
3183
1211
4808
1545
3441
3534
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98806
3181
1194
4809
1552
3445
3542
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
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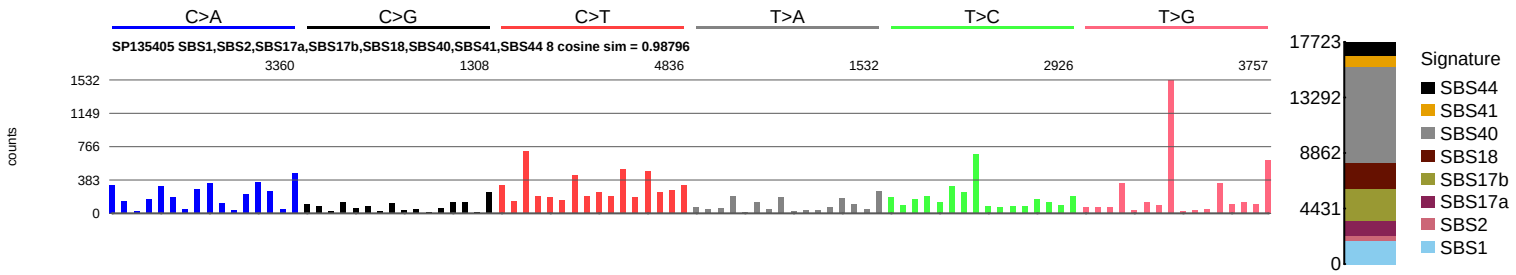
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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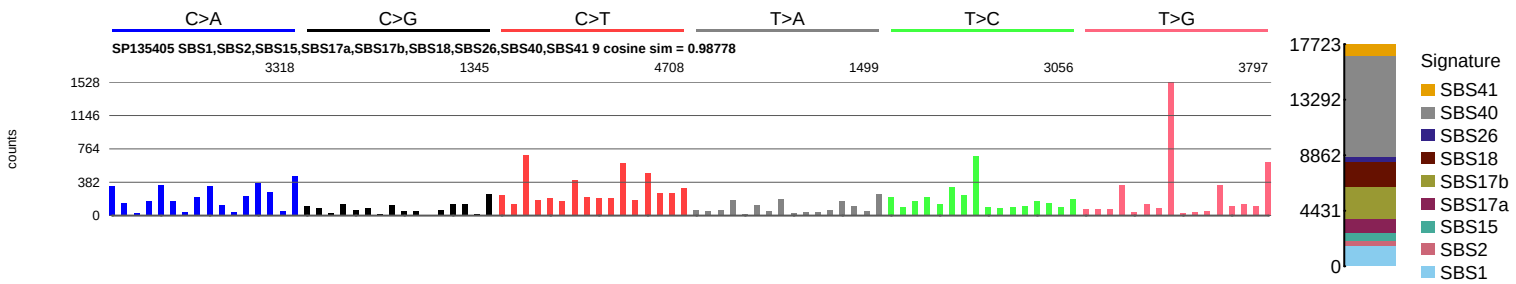
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98796
3360
1308
4836
1532
2926
3757
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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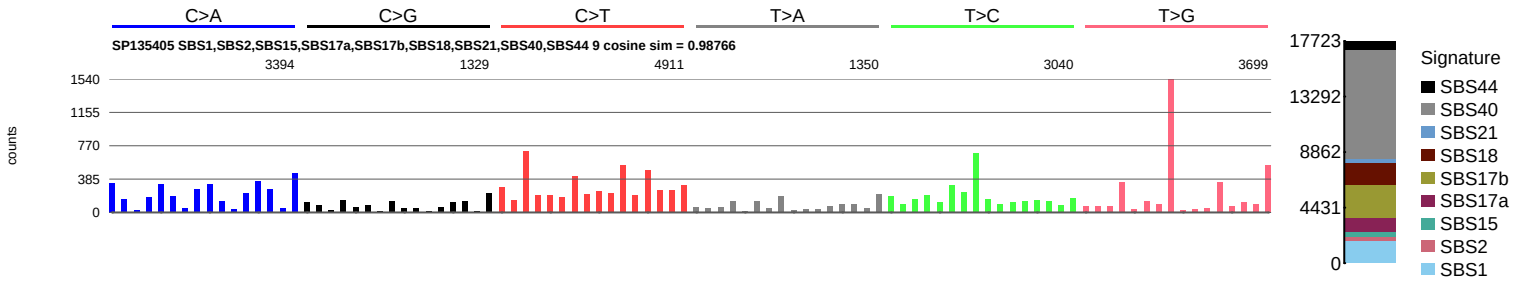
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98778
3318
1345
4708
1499
3056
3797
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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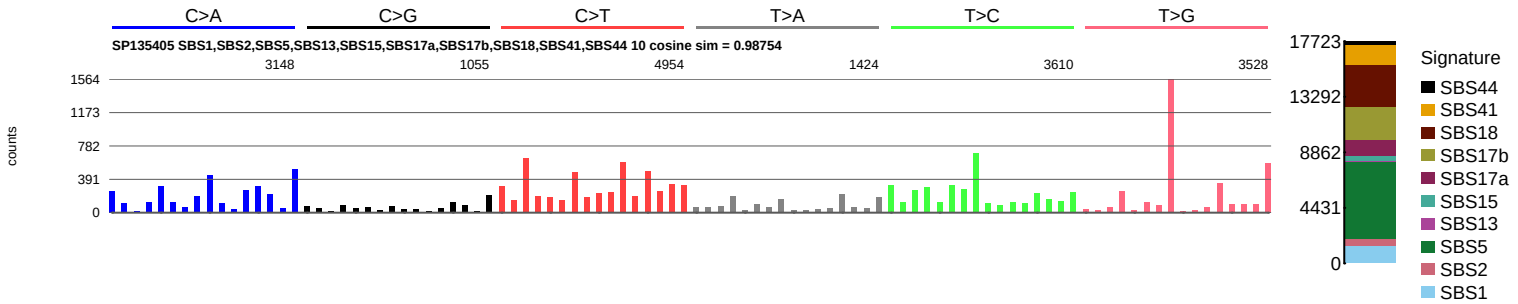
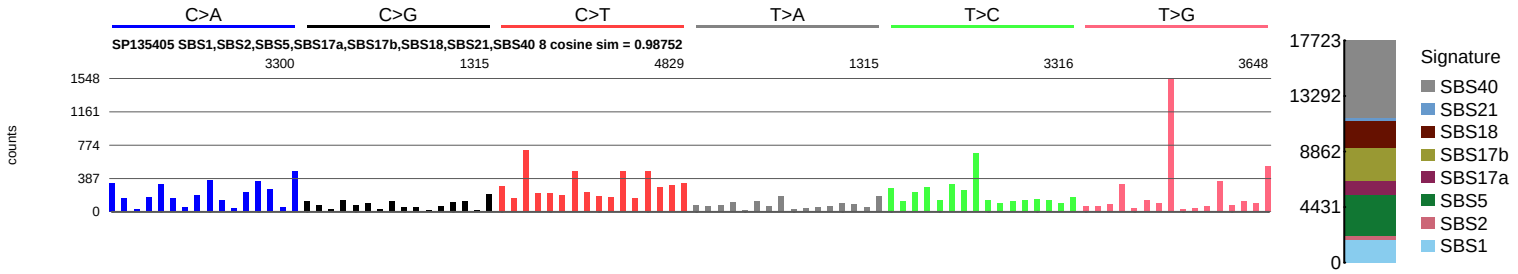
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98766
3394
1329
4911
1350
3040
3699
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98754
3148
1055
4954
1424
3610
3528
counts
1564
1173
782
391
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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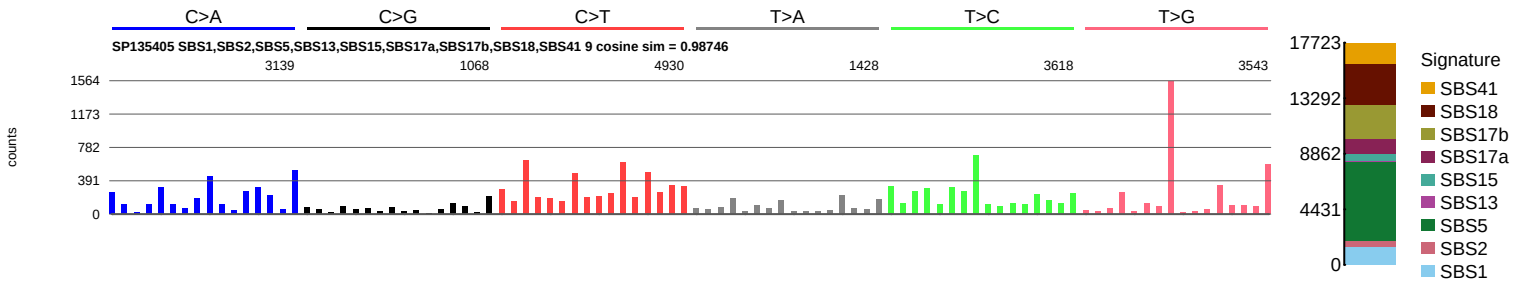
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98752
3300
1315
4829
1315
3316
3648
counts
1548
1161
774
387
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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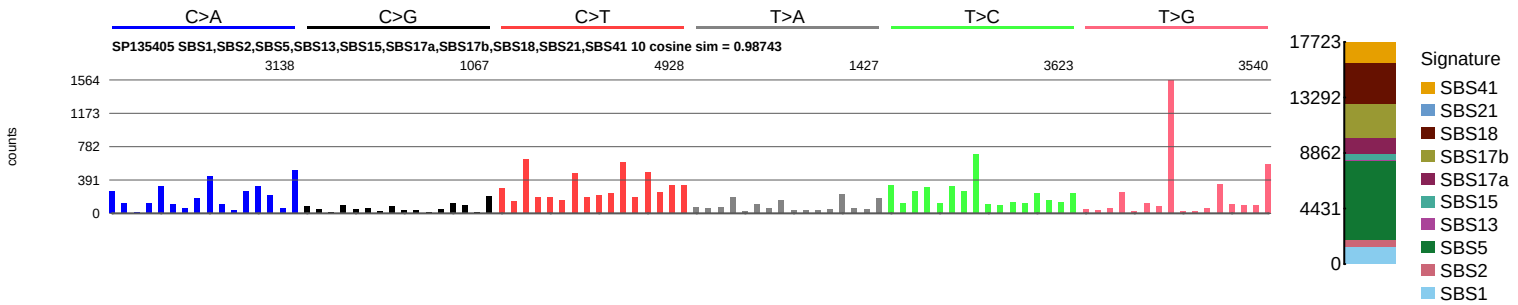
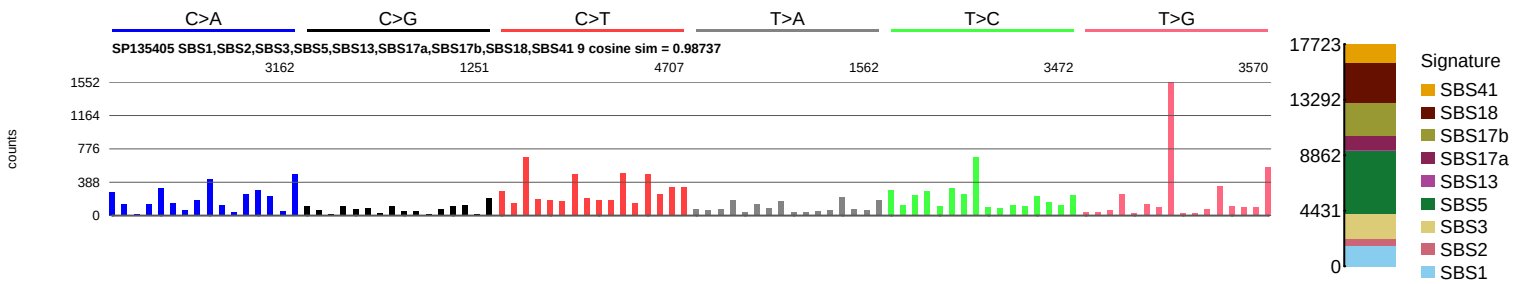
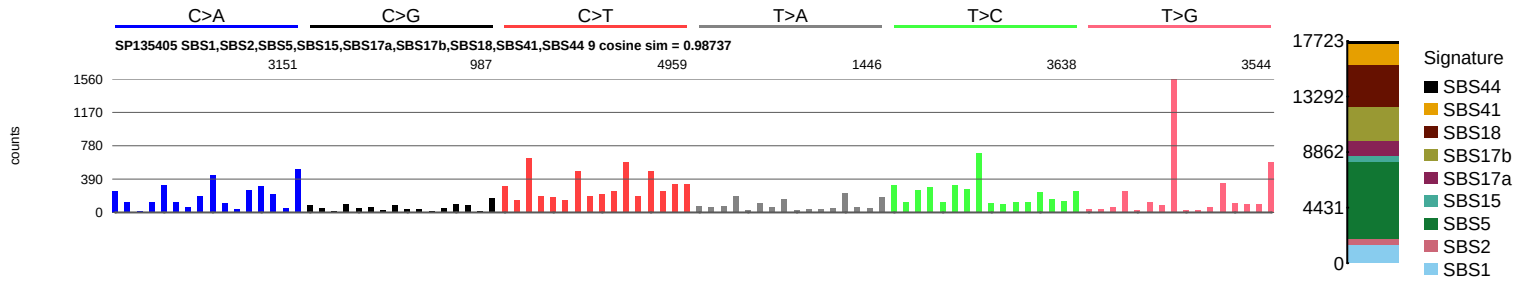
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98746
3139
1068
4930
1428
3618
3543
counts
1564
1173
782
391
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98743
3138
1067
4928
1427
3623
3540
counts
1564
1173
782
391
0
17723
13292
8862
4431
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98737
3162
1251
4707
1562
3472
3570
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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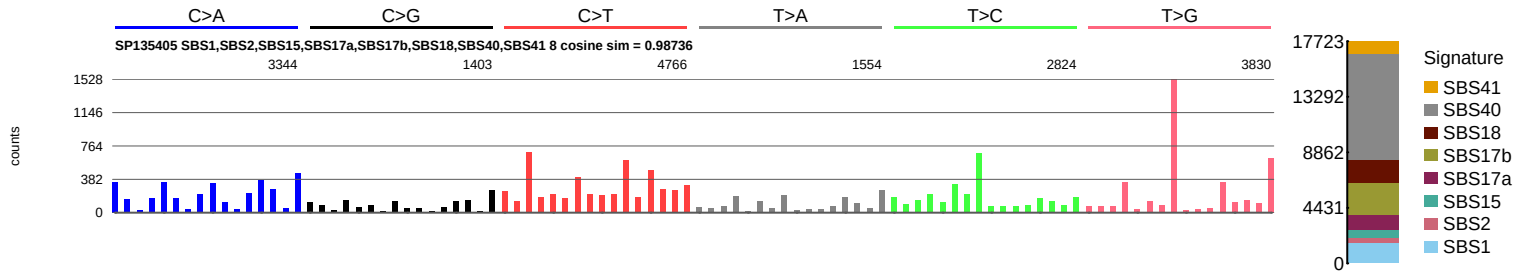
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98737
3151
987
4959
1446
3638
3544
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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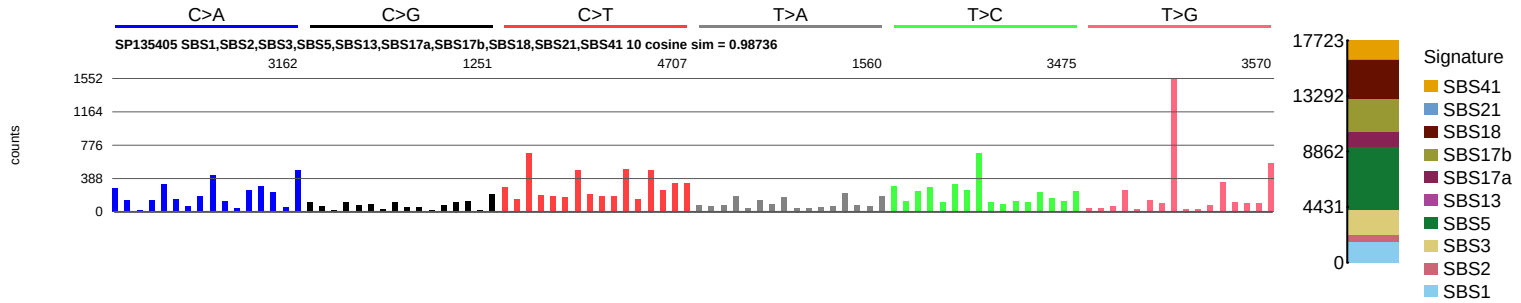
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98736
3344
1403
4766
1554
2824
3830
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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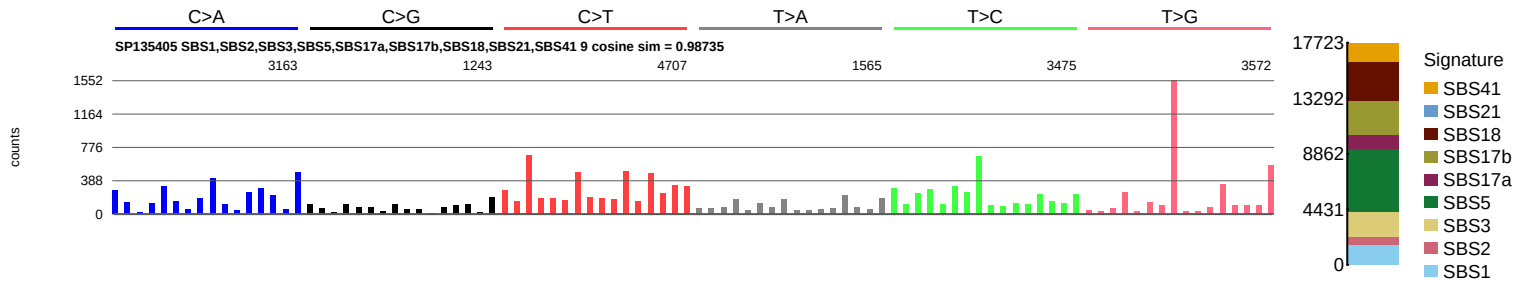
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98736
3162
1251
4707
1560
3475
3570
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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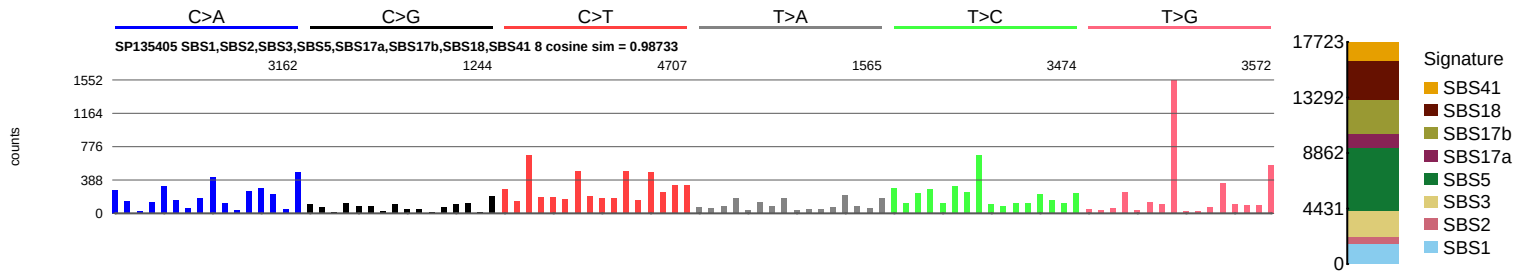
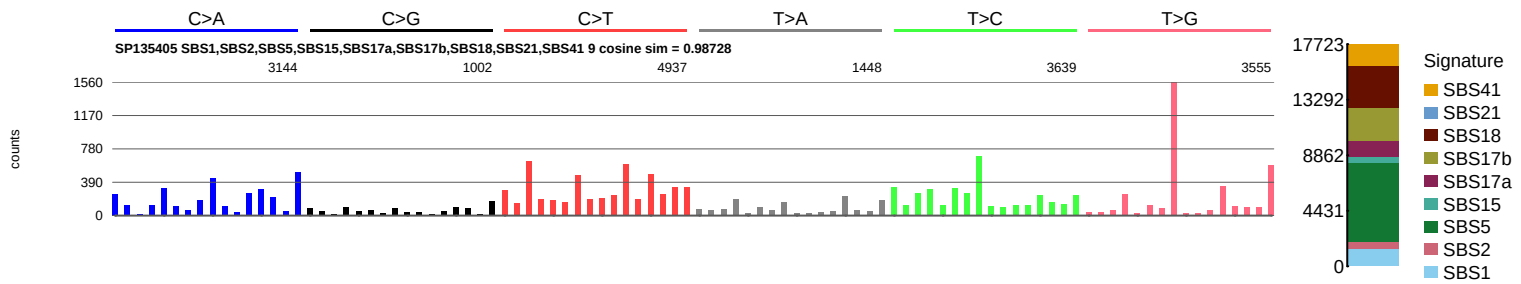
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98735
3163
1243
4707
1565
3475
3572
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98733
3162
1244
4707
1565
3474
3572
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98728
3144
1002
4937
1448
3639
3555
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
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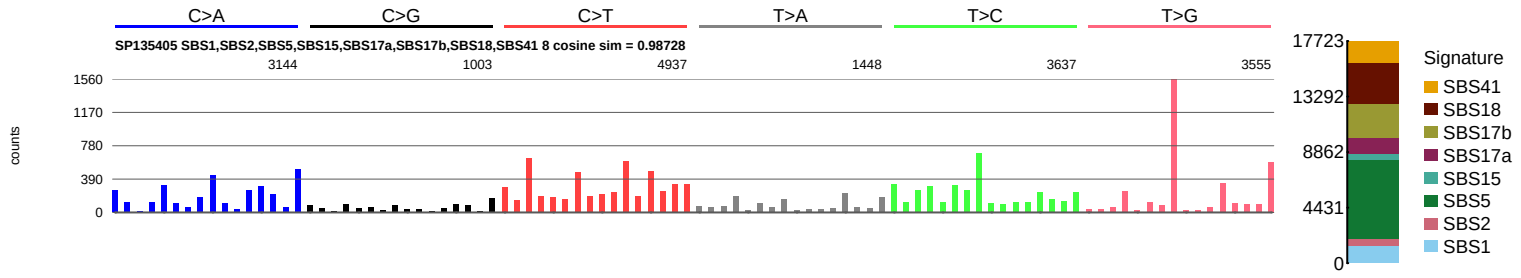
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98728
3144
1003
4937
1448
3637
3555
counts
1560
1170
780
390
0
17723
13292
8862
4431
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Signature
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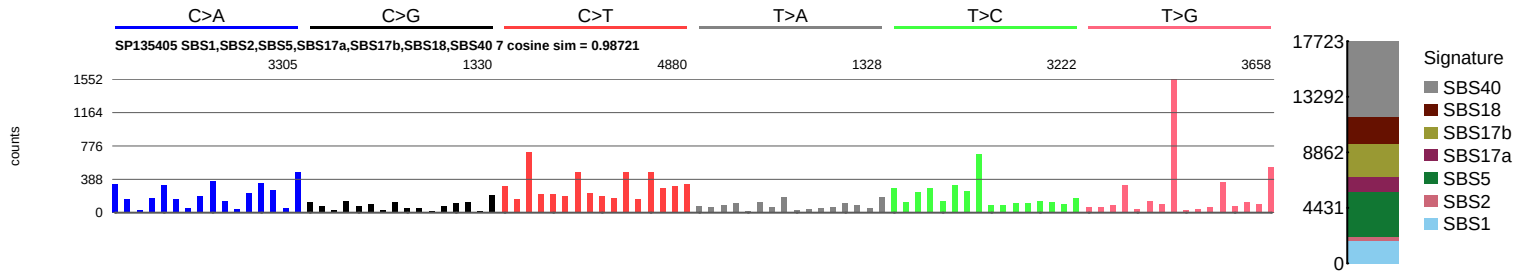
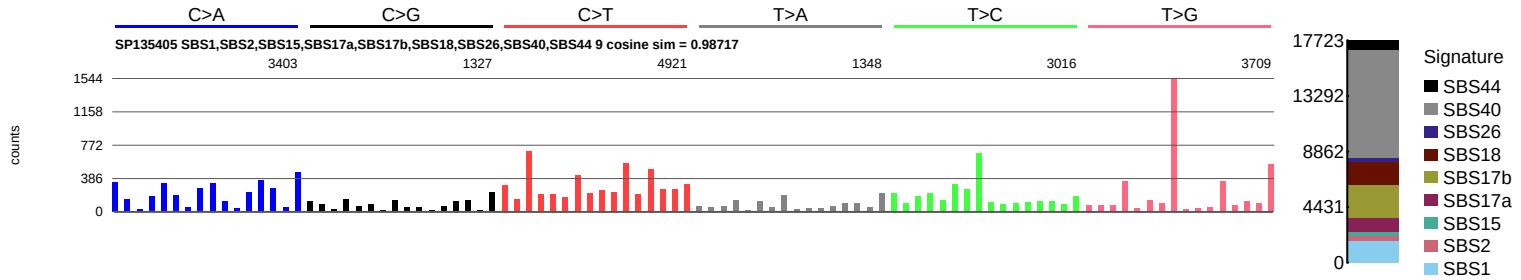
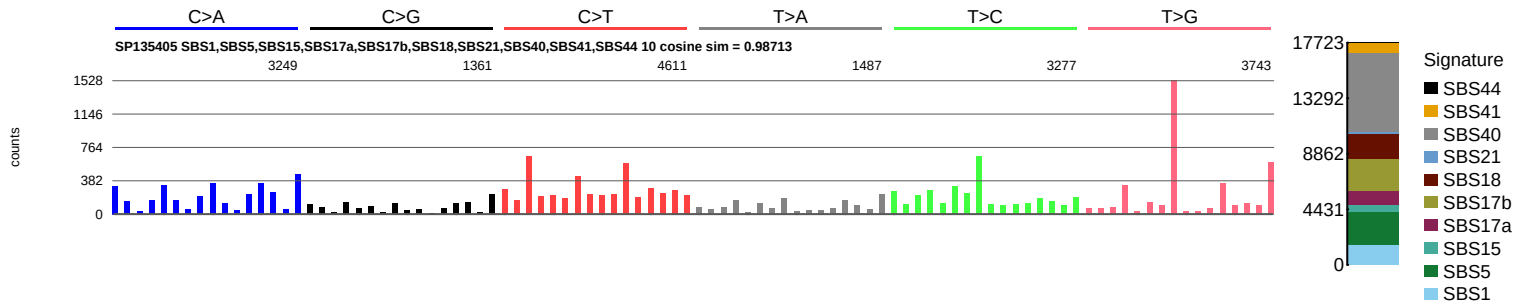
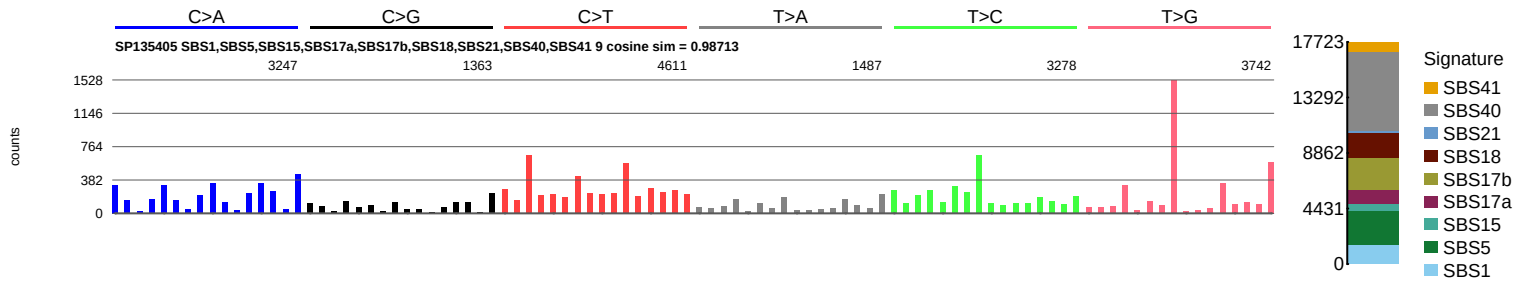
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98721
3305
1330
4880
1328
3222
3658
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98717
3403
1327
4921
1348
3016
3709
counts
1544
1158
772
386
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98713
3249
1361
4611
1487
3277
3743
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98713
3247
1363
4611
1487
3278
3742
counts
1528
1146
764
382
0
17723
13292
8862
4431
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Signature
SBS41
SBS40
SBS21
SBS18
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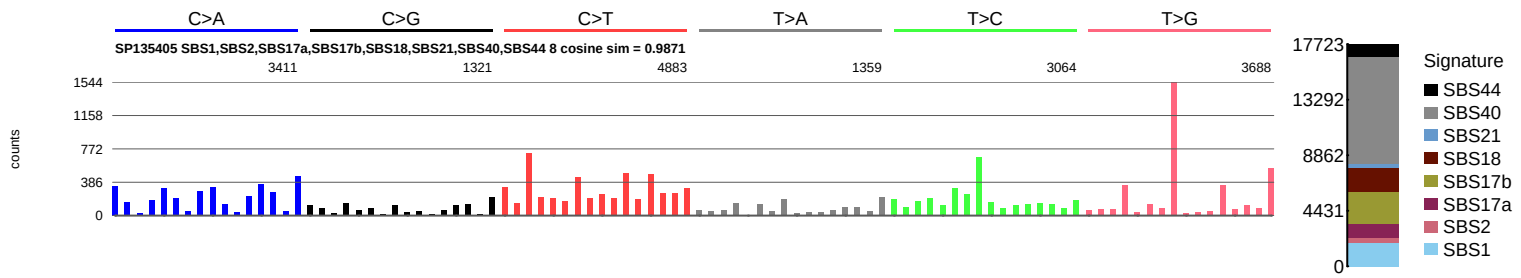
SBS5
SBS1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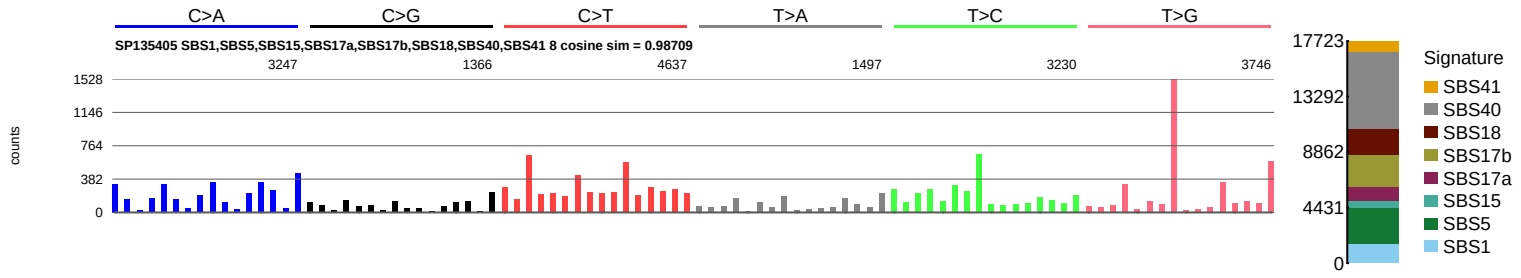
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9871
3411
1321
4883
1359
3064
3688
counts
1544
1158
772
386
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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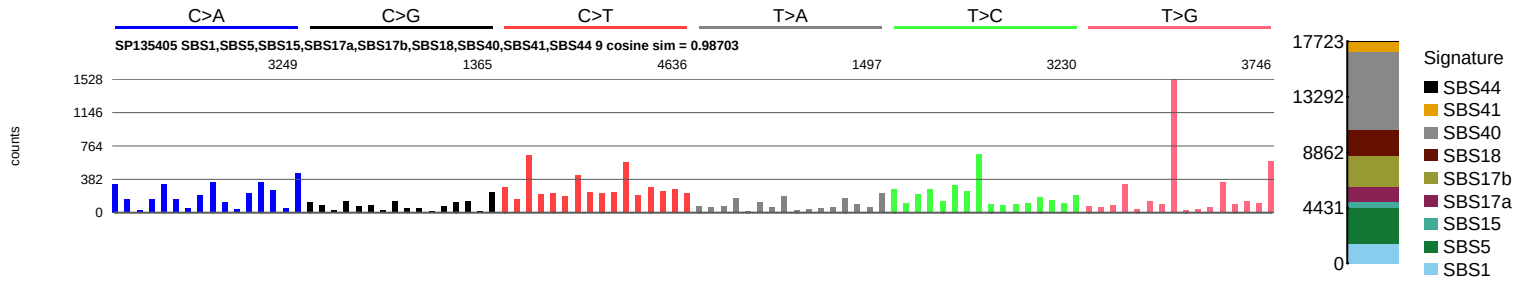
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98709
3247
1366
4637
1497
3230
3746
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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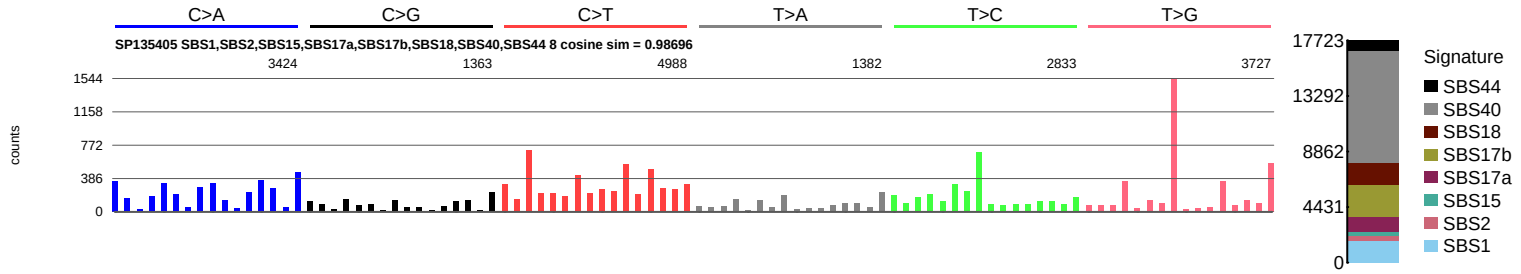
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98703
3249
1365
4636
1497
3230
3746
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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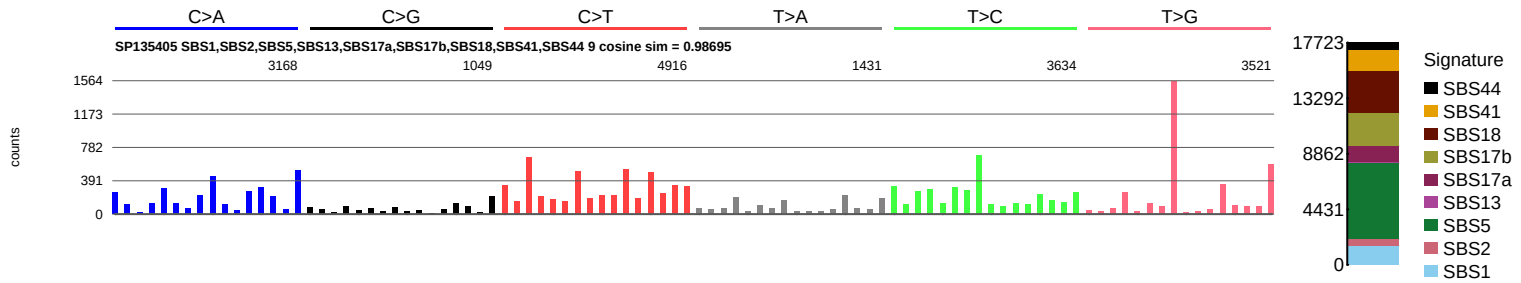
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98696
3424
1363
4988
1382
2833
3727
counts
1544
1158
772
386
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98695
3168
1049
4916
1431
3634
3521
counts
1564
1173
782
391
0
17723
13292
8862
4431
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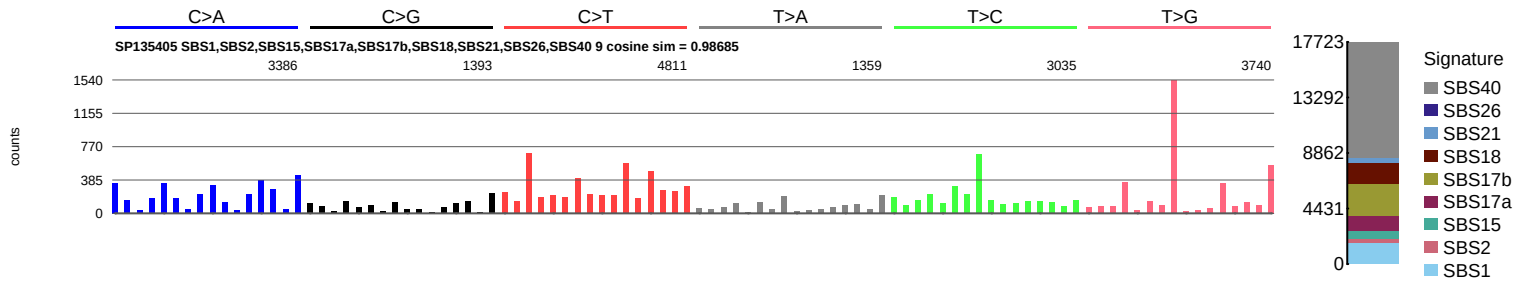
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
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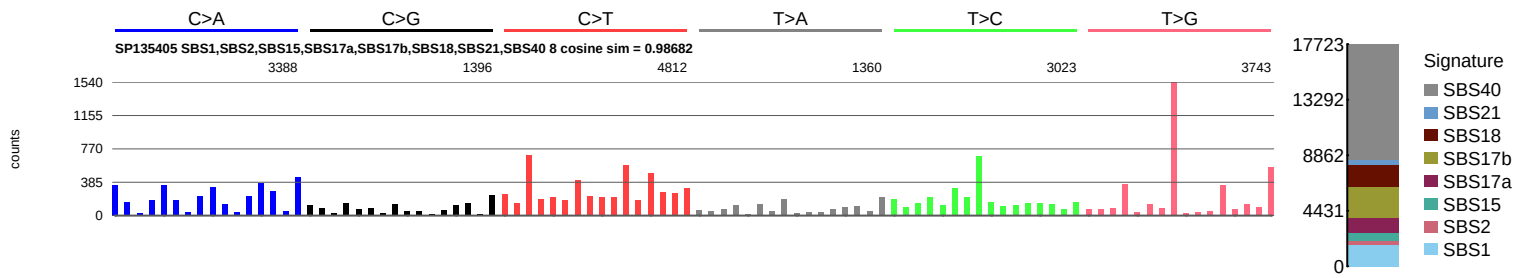
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98685
3386
1393
4811
1359
3035
3740
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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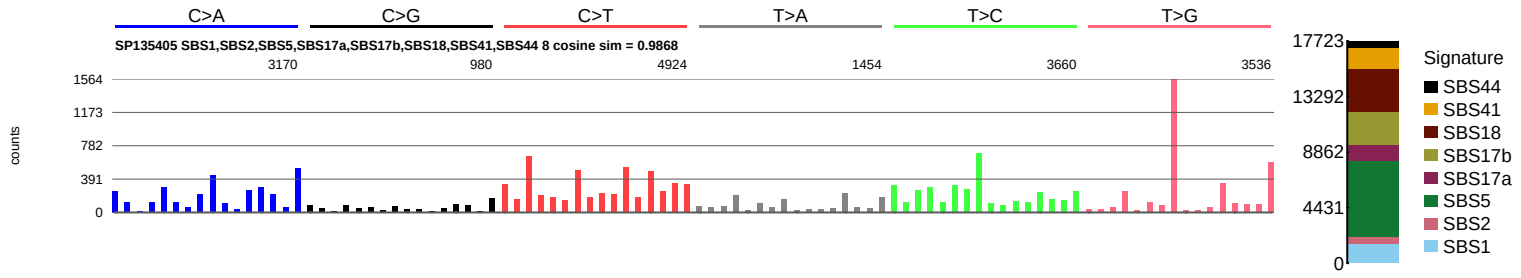
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98682
3388
1396
4812
1360
3023
3743
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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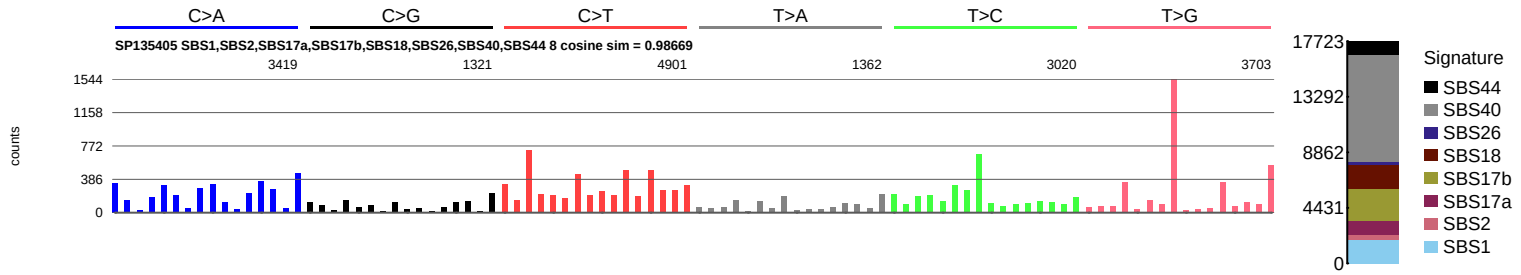
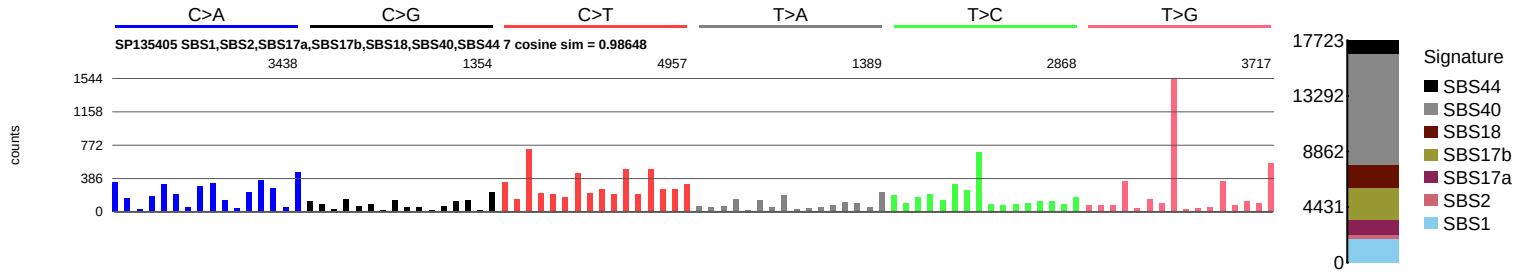
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9868
3170
980
4924
1454
3660
3536
counts
1564
1173
782
391
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98669
3419
1321
4901
1362
3020
3703
counts
1544
1158
772
386
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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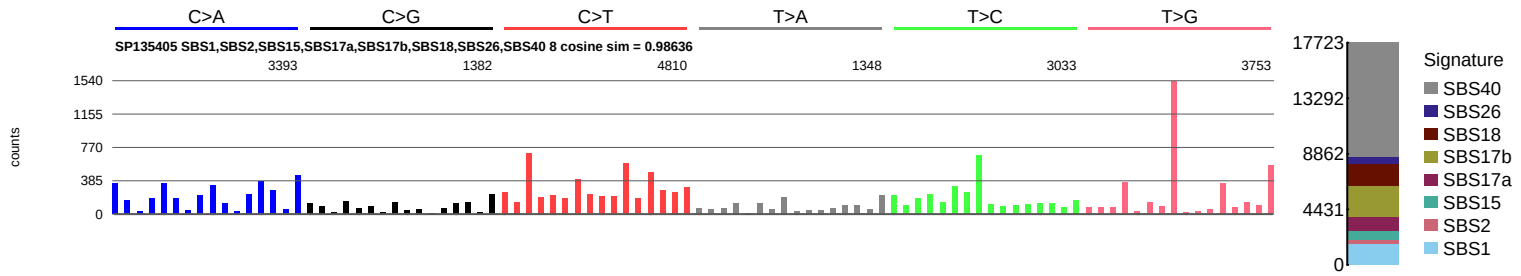
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98648
3438
1354
4957
1389
2868
3717
counts
1544
1158
772
386
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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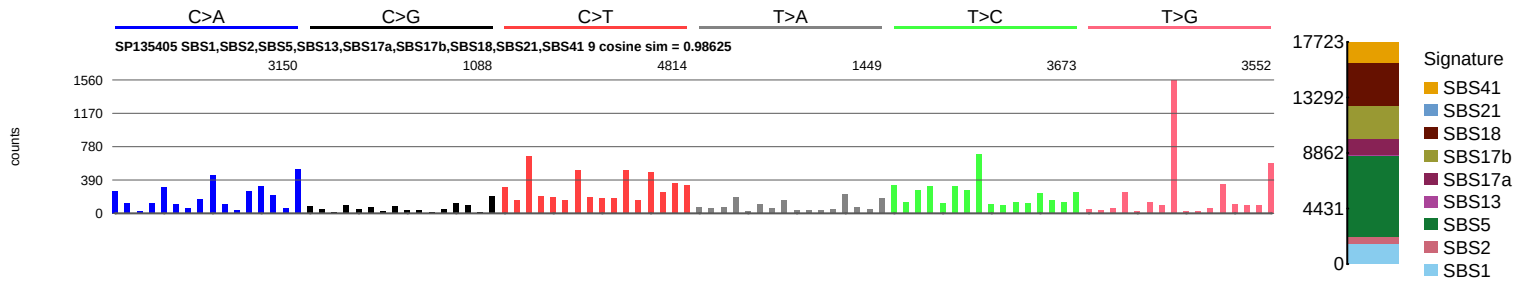
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98636
3393
1382
4810
1348
3033
3753
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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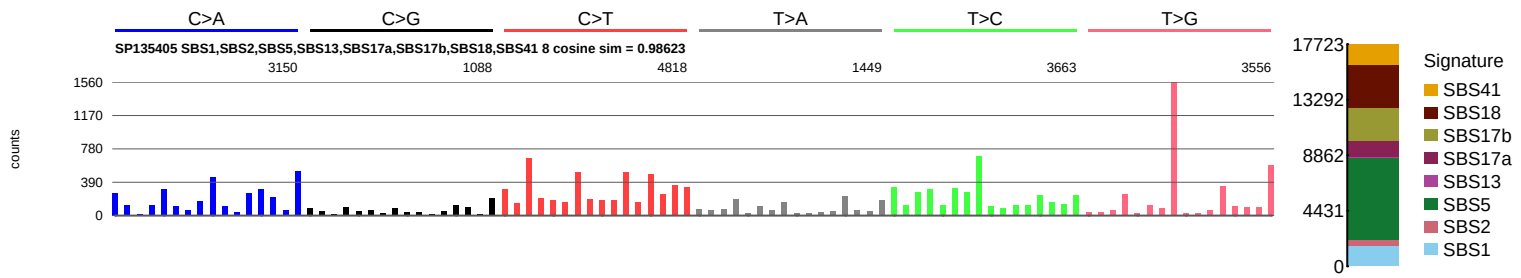
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98625
3150
1088
4814
1449
3673
3552
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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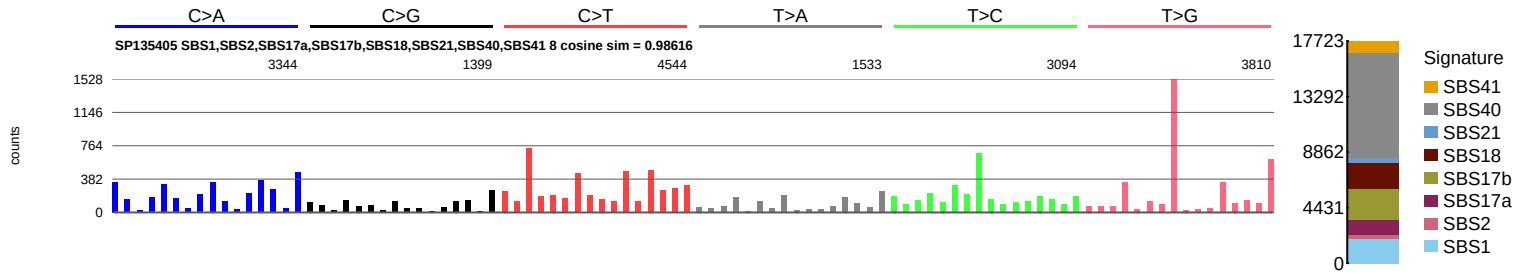
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98623
3150
1088
4818
1449
3663
3556
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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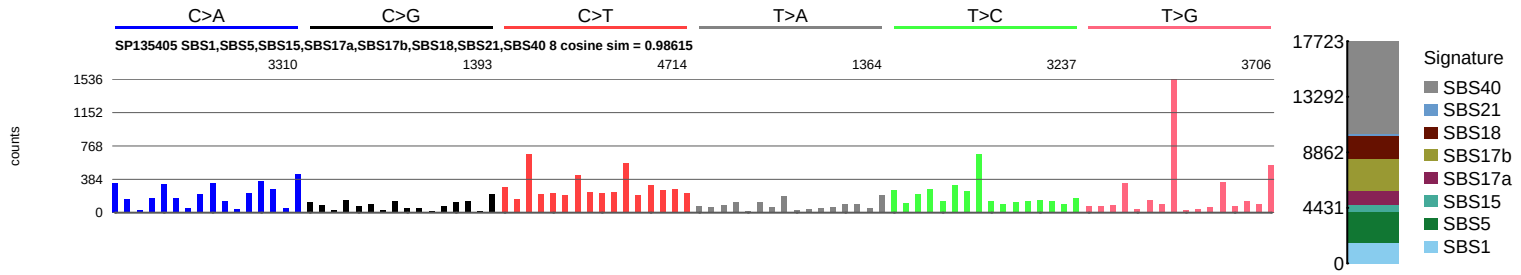
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98616
3344
1399
4544
1533
3094
3810
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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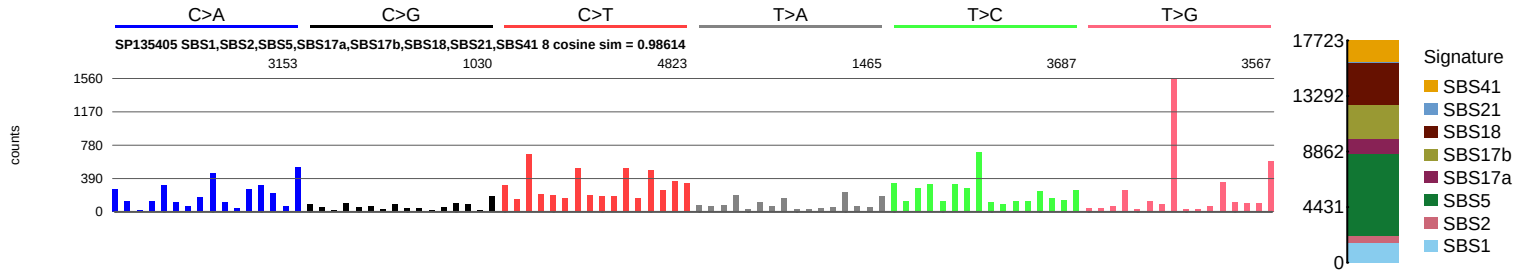
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98615
3310
1393
4714
1364
3237
3706
counts
1536
1152
768
384
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>C
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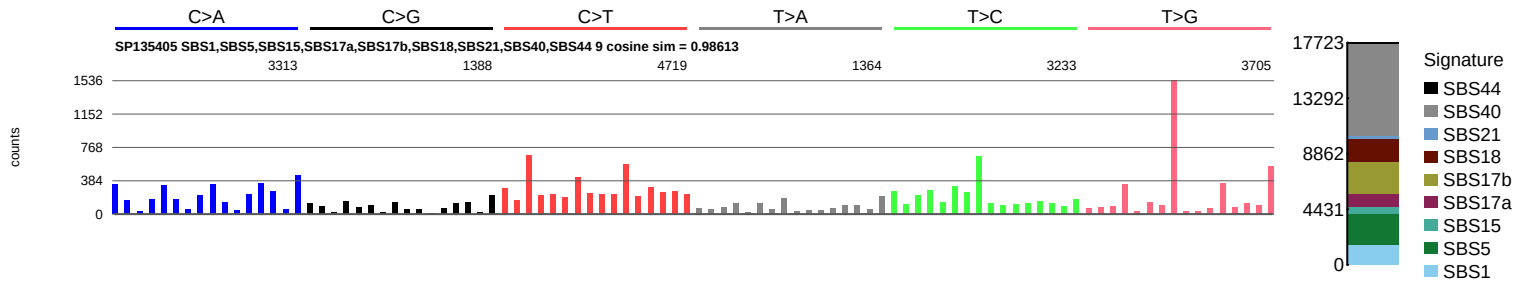
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98614
3153
1030
4823
1465
3687
3567
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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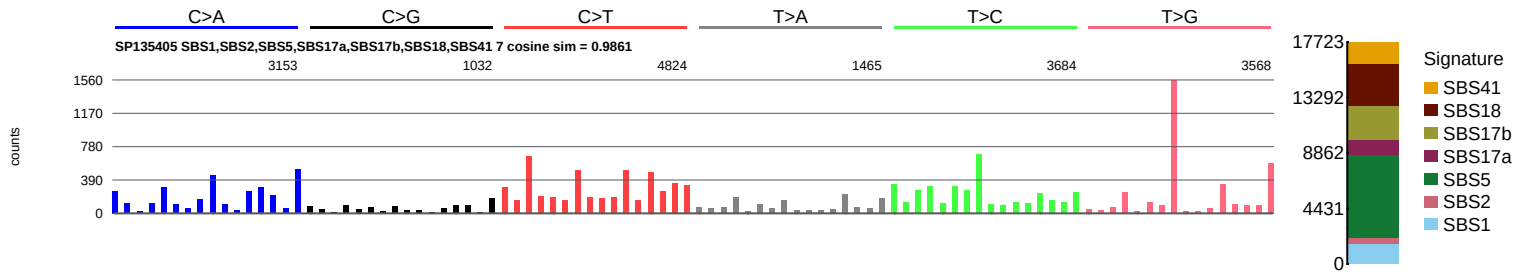
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98613
3313
1388
4719
1364
3233
3705
counts
1536
1152
768
384
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9861
3153
1032
4824
1465
3684
3568
counts
1560
1170
780
390
0
17723
13292
8862
4431
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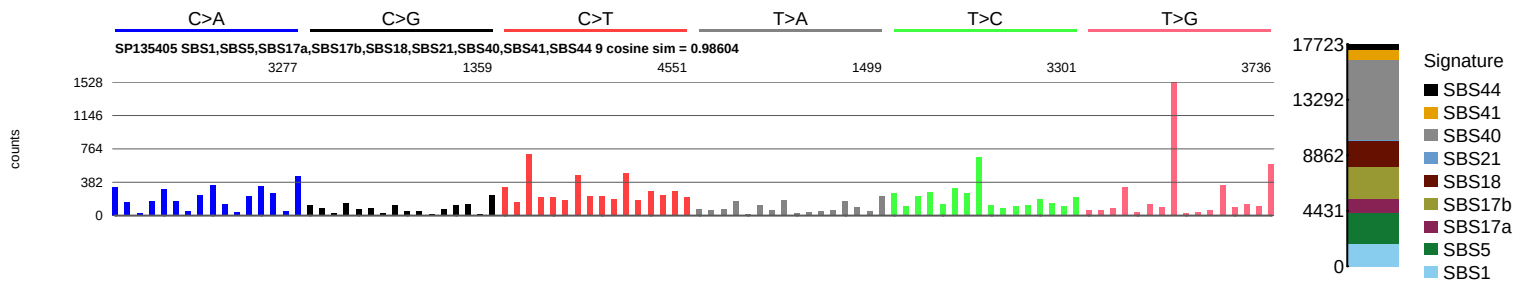
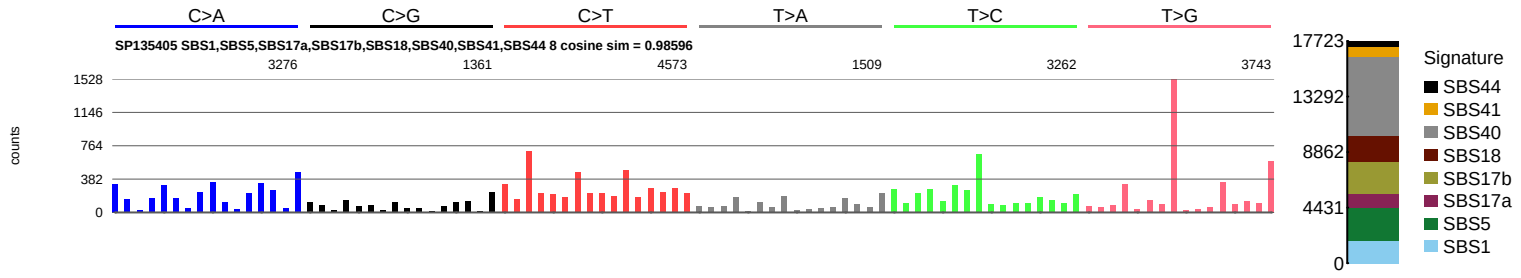
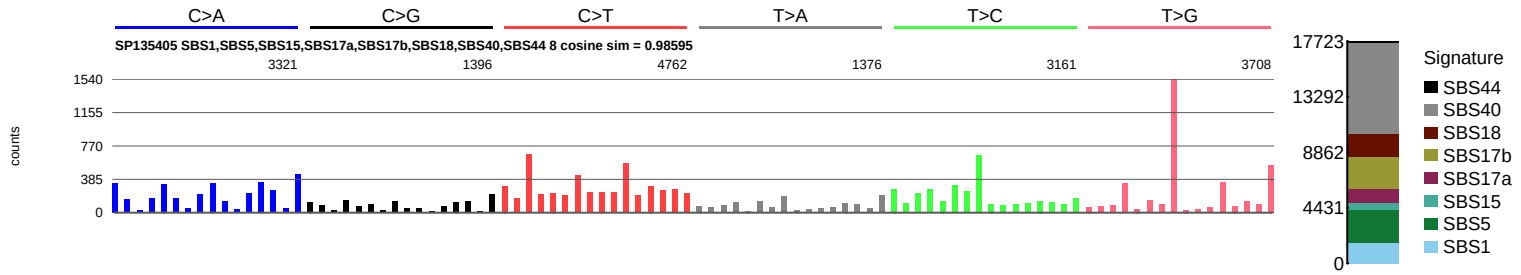
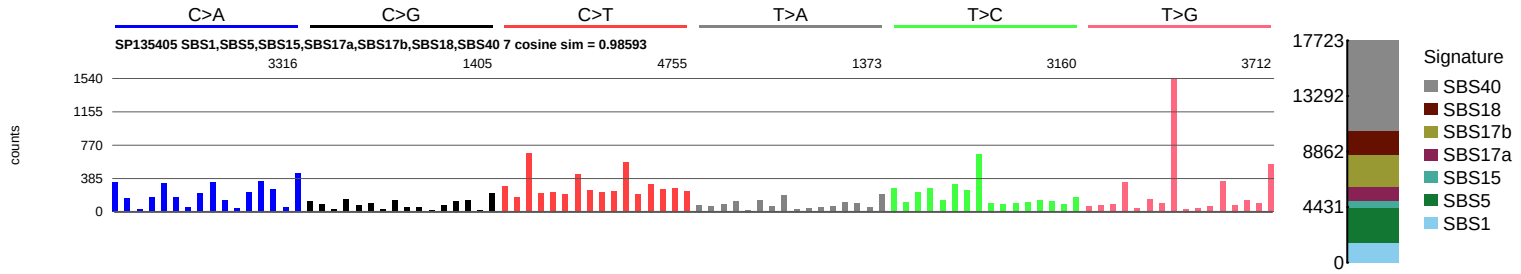
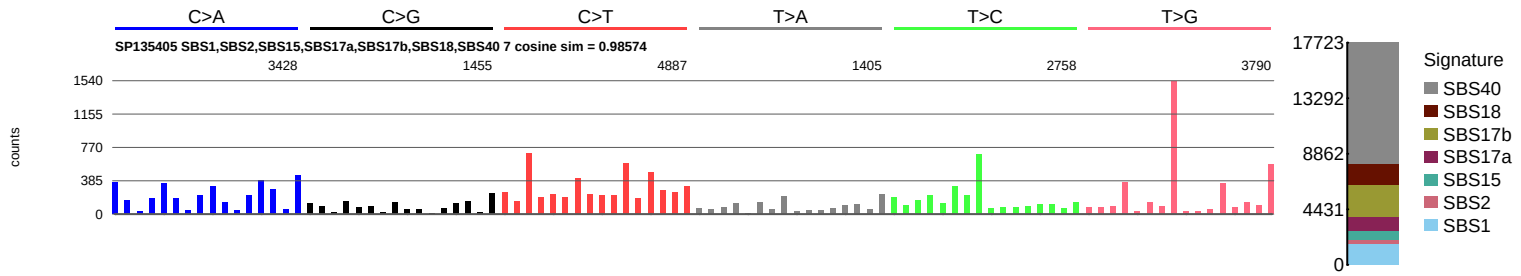
SBS2
SBS1

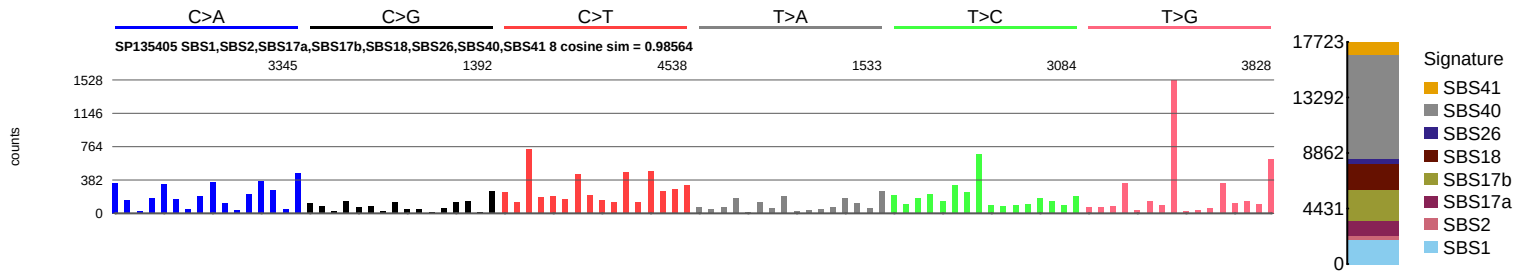

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98604
3277
1359
4551
1499
3301
3736
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

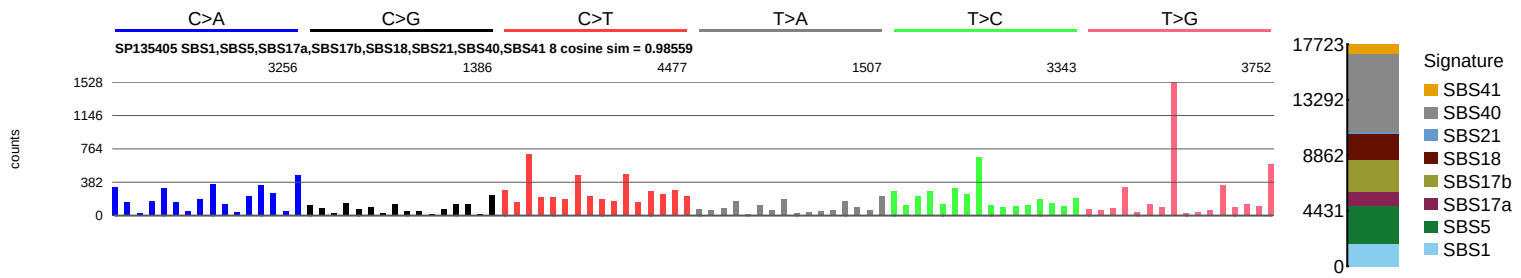

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98596
3276
1361
4573
1509
3262
3743
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98595
3321
1396
4762
1376
3161
3708
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98593
3316
1405
4755
1373
3160
3712
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98574
3428
1455
4887
1405
2758
3790
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98564
3345
1392
4538
1533
3084
3828
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98559
3256
1386
4477
1507
3343
3752
counts
1528
1146
764
382
0
17723
13292
8862
4431
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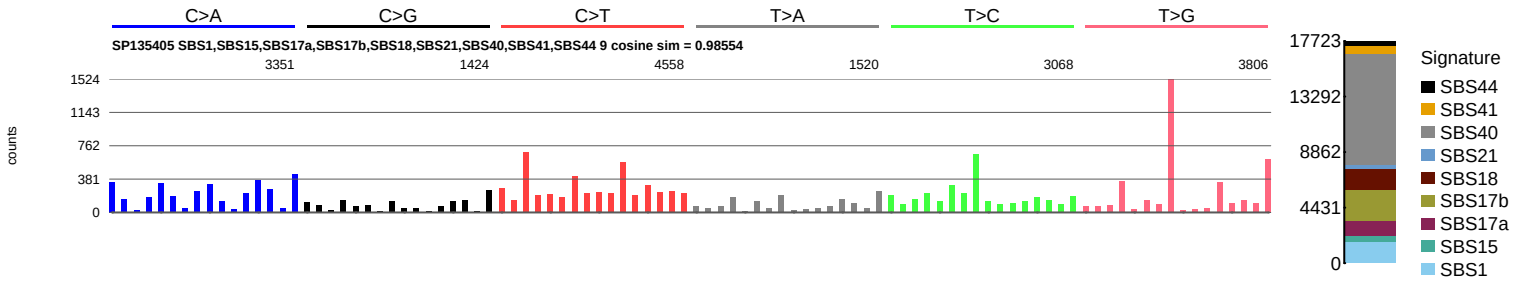
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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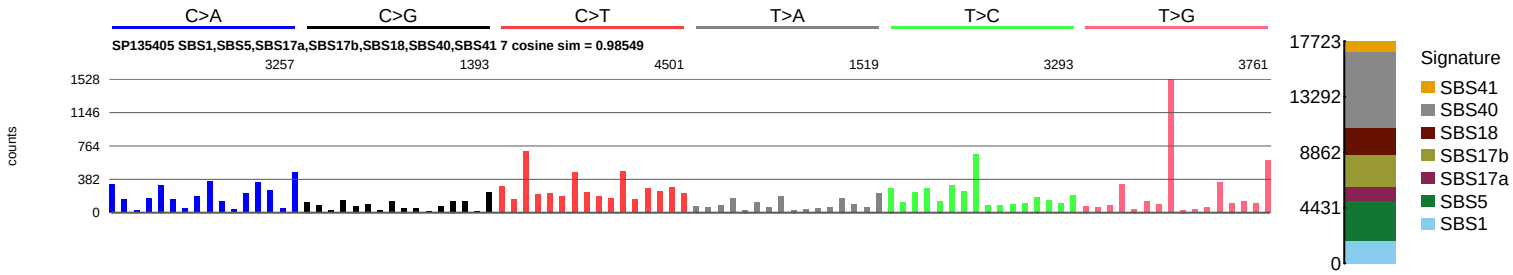
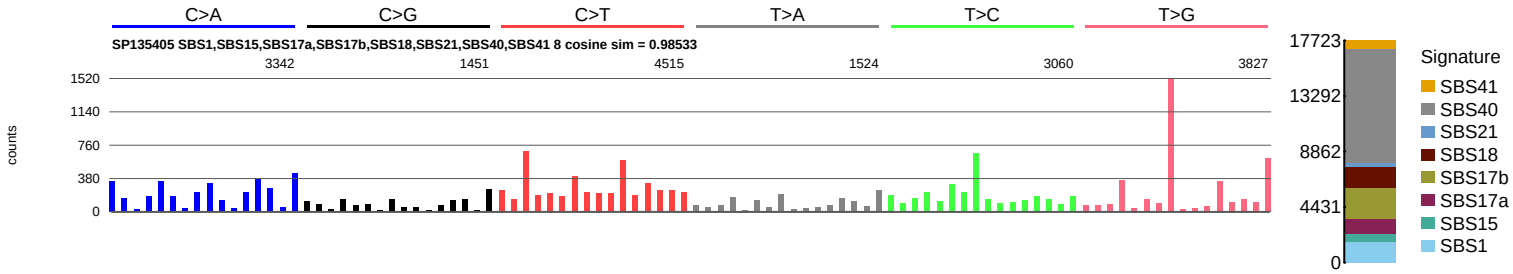
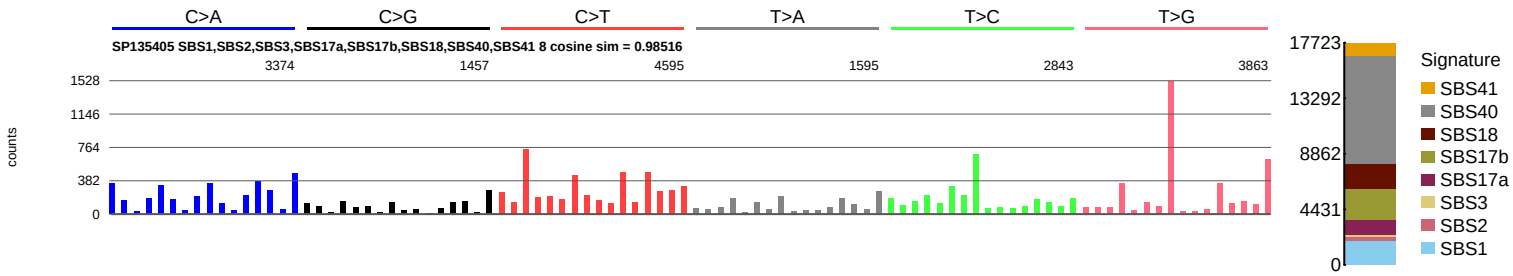
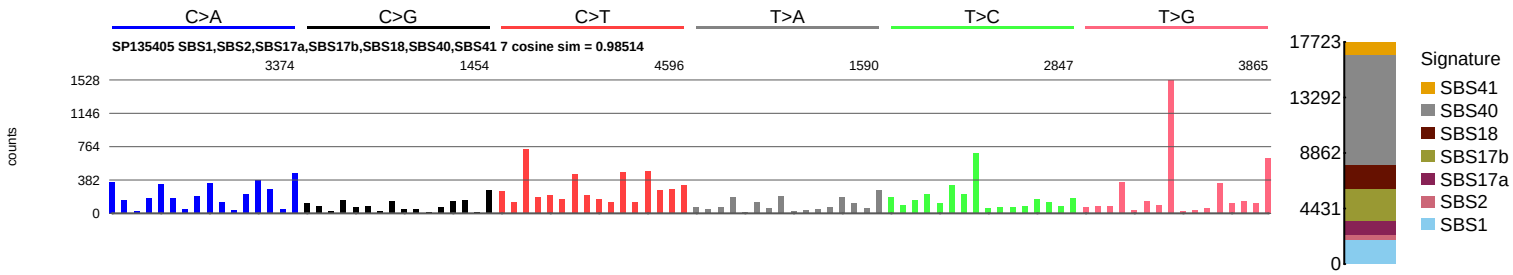
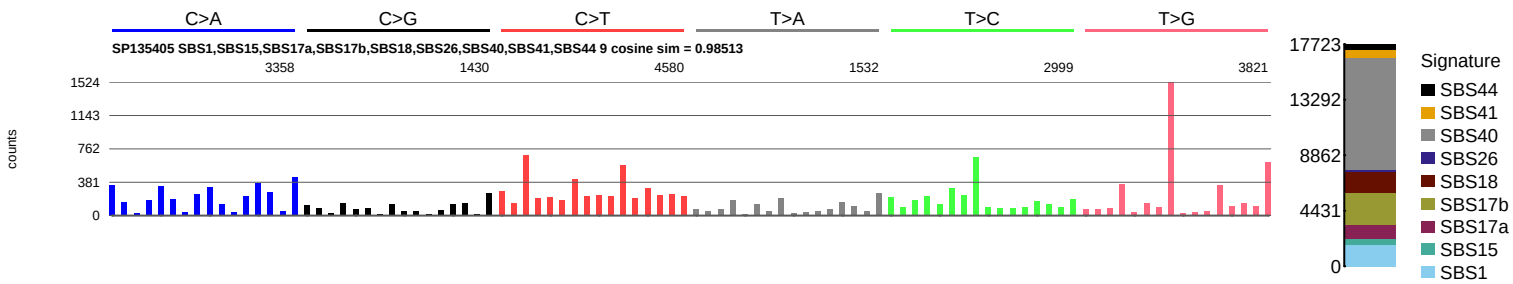
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98554
3351
1424
4558
1520
3068
3806
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

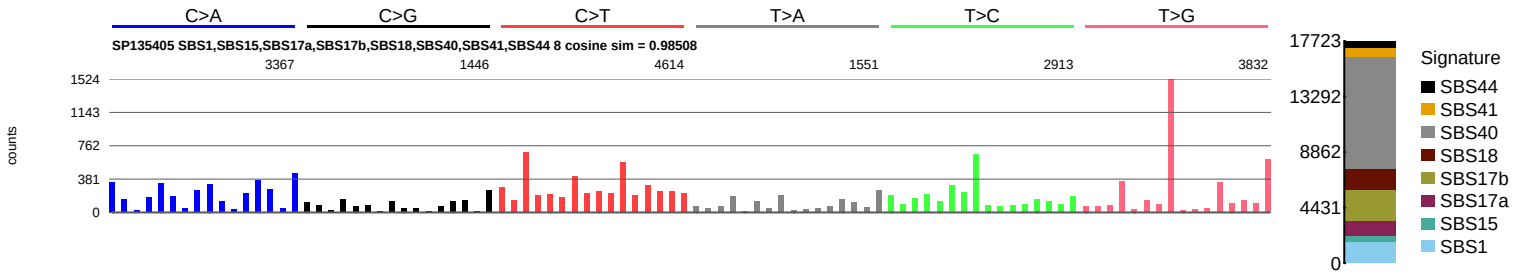

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98549
3257
1393
4501
1519
3293
3761
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

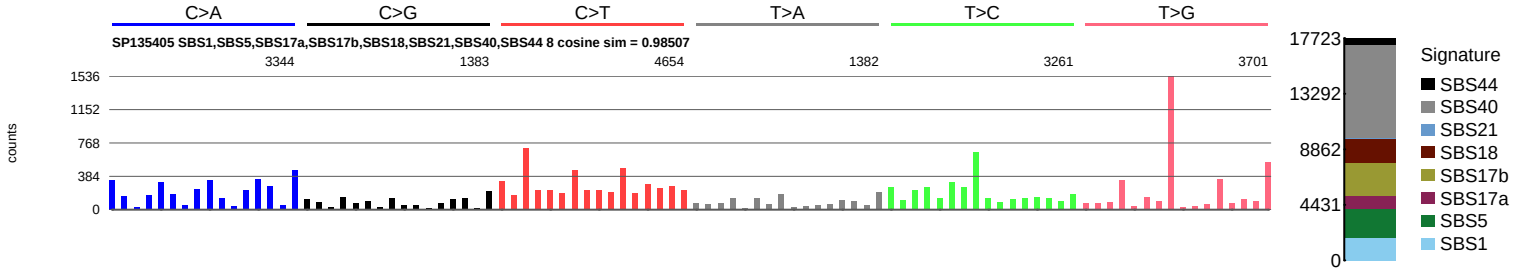

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98533
3342
1451
4515
1524
3060
3827
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

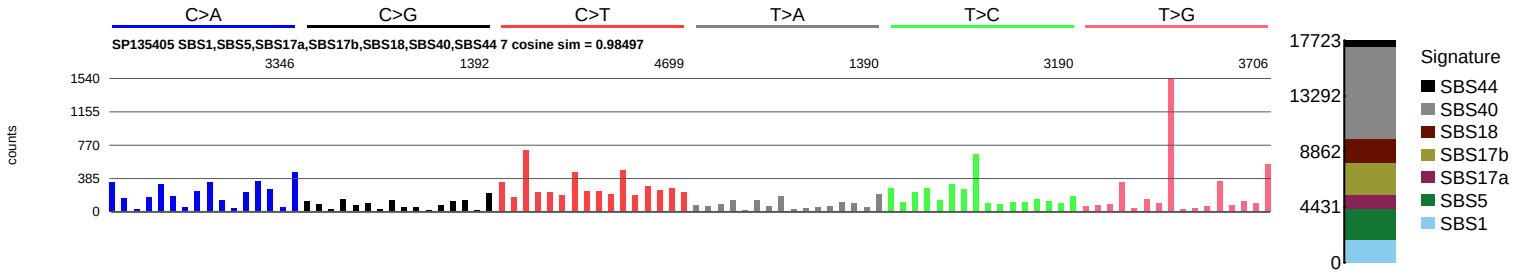

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98516
3374
1457
4595
1595
2843
3863
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

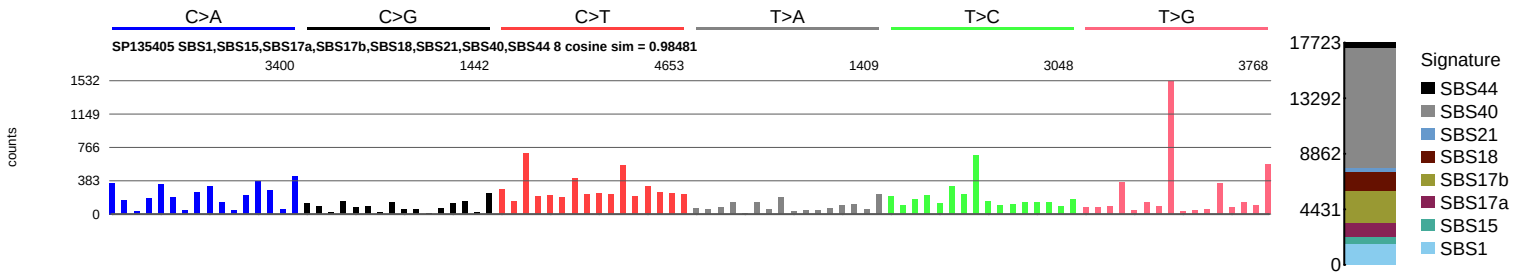

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98514
3374
1454
4596
1590
2847
3865
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

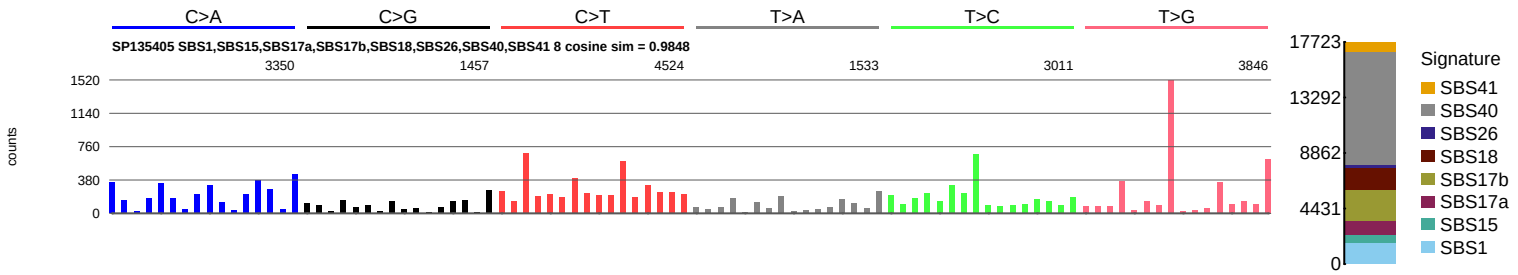

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98513
3358
1430
4580
1532
2999
3821
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

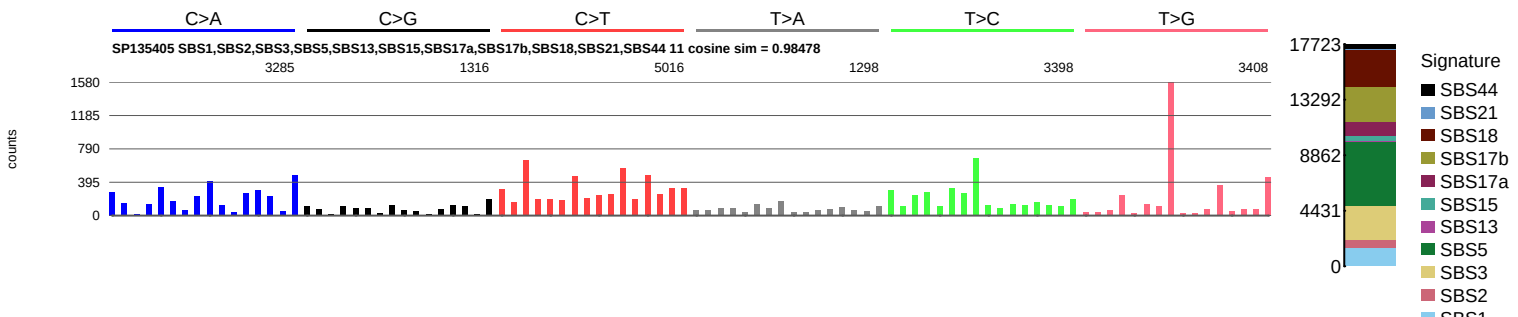

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98508
3367
1446
4614
1551
2913
3832
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

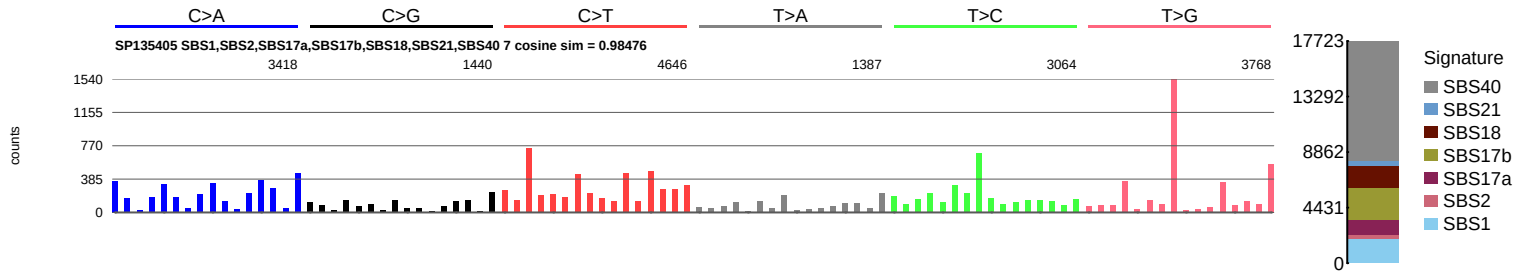

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98507
3344
1383
4654
1382
3261
3701
counts
1536
1152
768
384
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

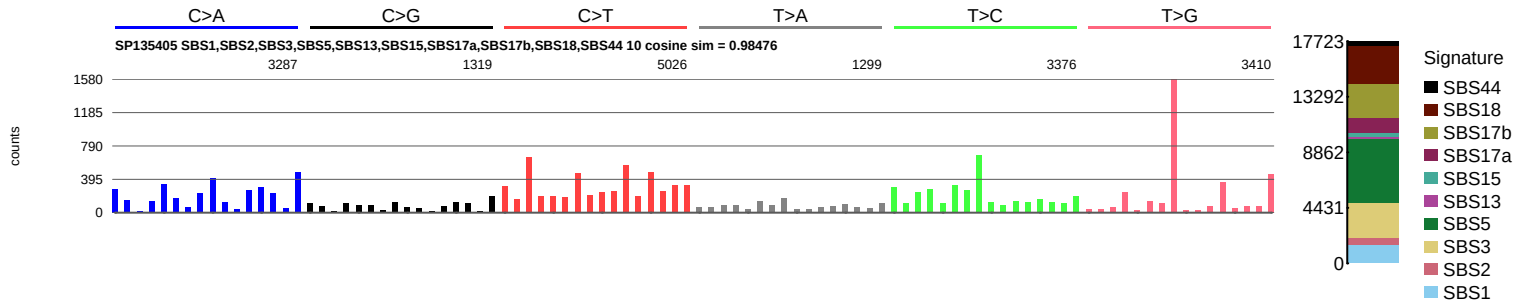

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98497
3346
1392
4699
1390
3190
3706
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

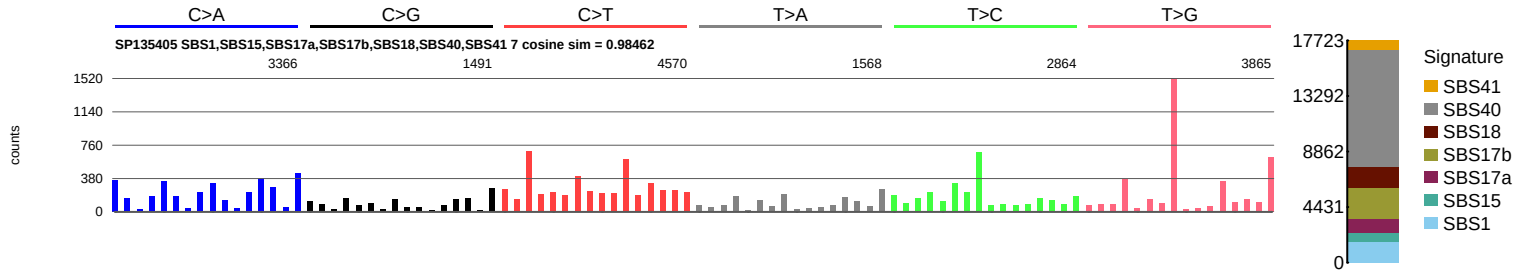

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98481
3400
1442
4653
1409
3048
3768
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

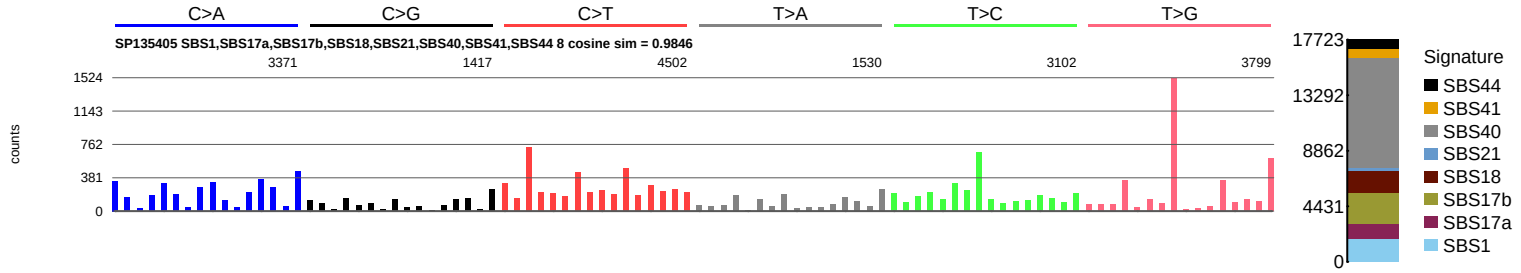

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9848
3350
1457
4524
1533
3011
3846
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

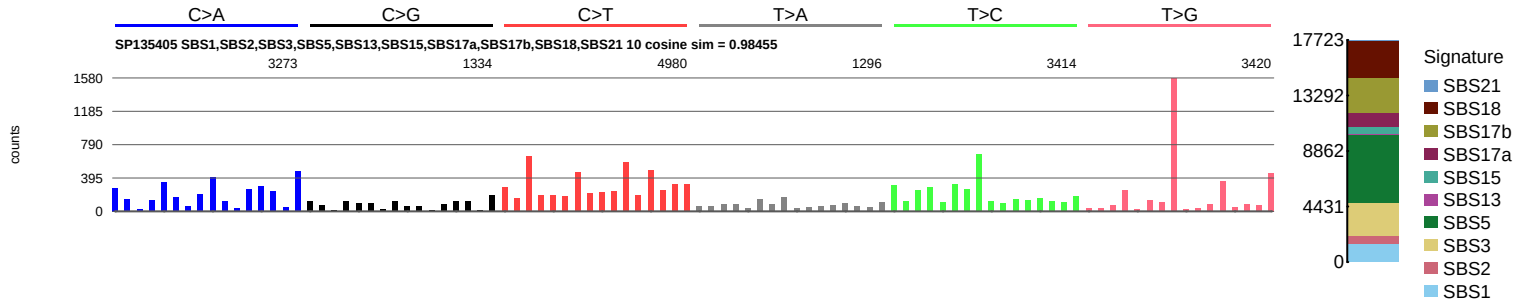

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98478
3285
1316
5016
1298
3398
3408
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

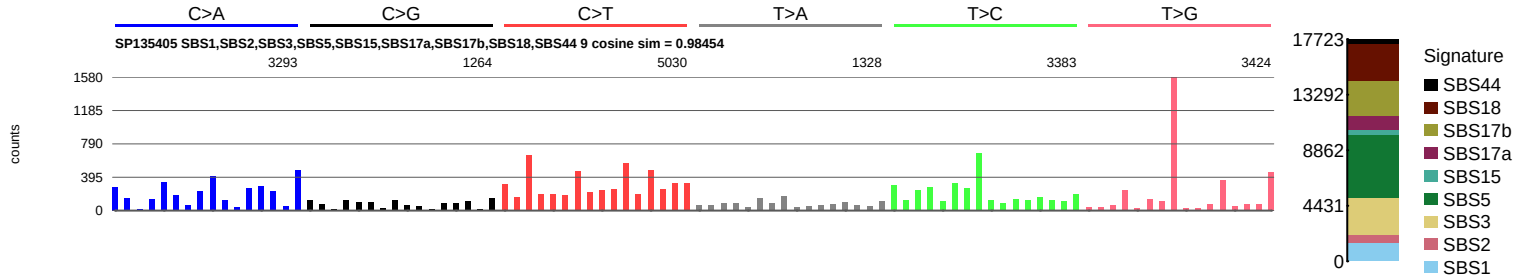

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98476
3418
1440
4646
1387
3064
3768
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

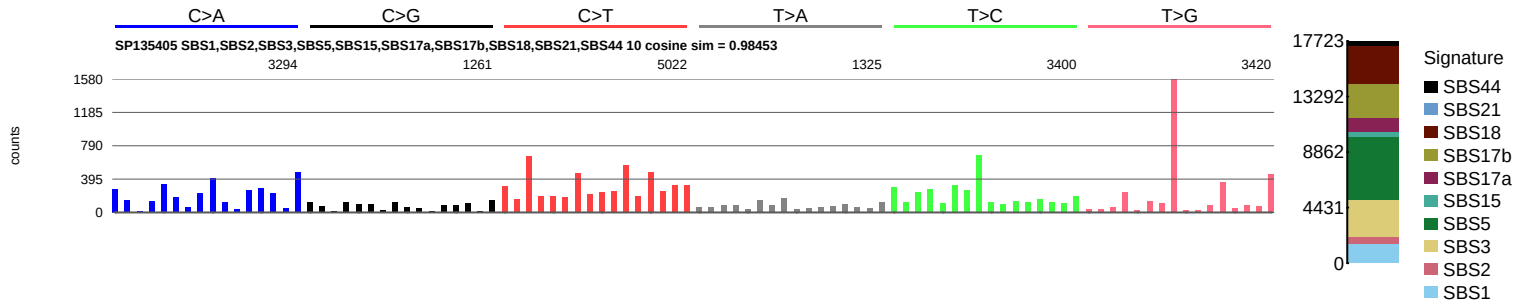

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98476
3287
1319
5026
1299
3376
3410
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

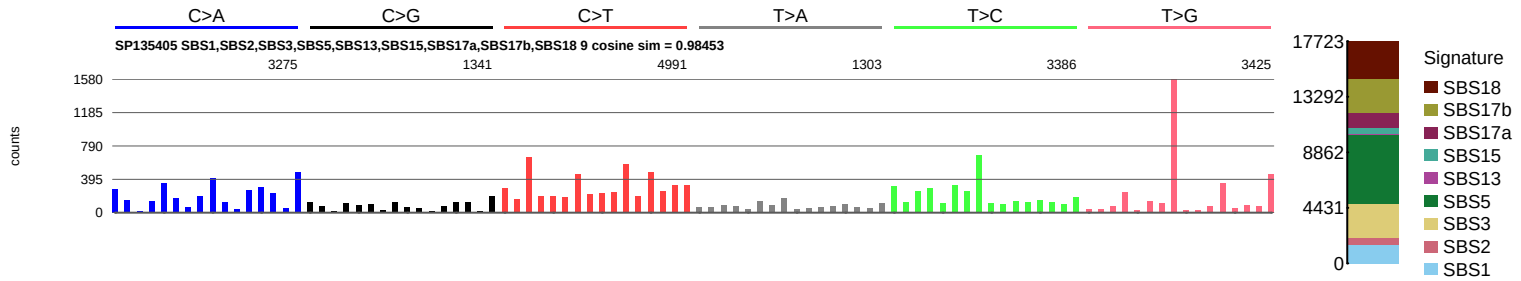

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98462
3366
1491
4570
1568
2864
3865
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

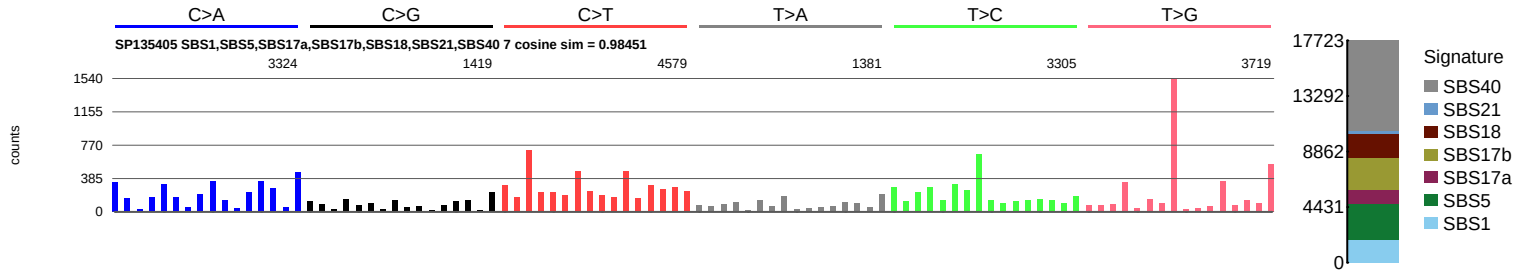

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9846
3371
1417
4502
1530
3102
3799
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

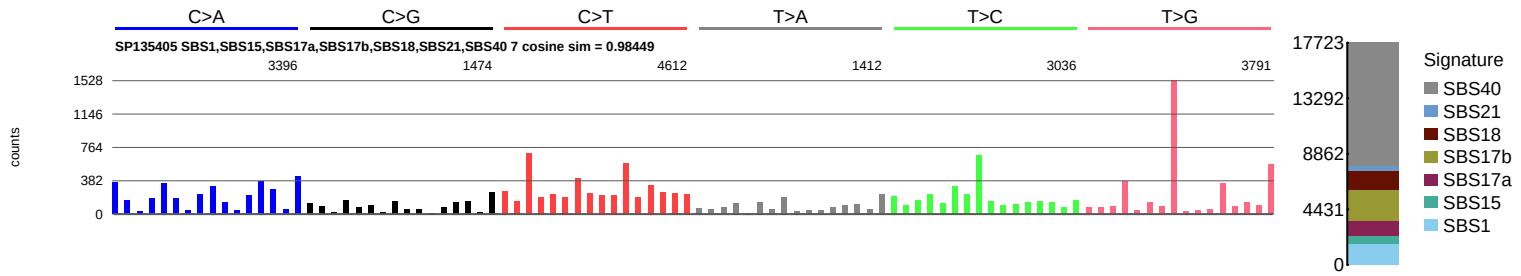

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98455
3273
1334
4980
1296
3414
3420
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

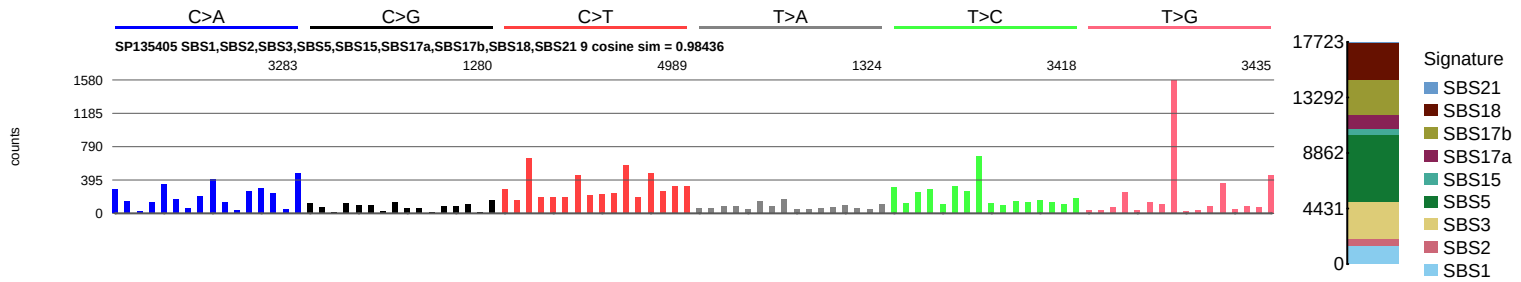

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98454
3293
1264
5030
1328
3383
3424
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

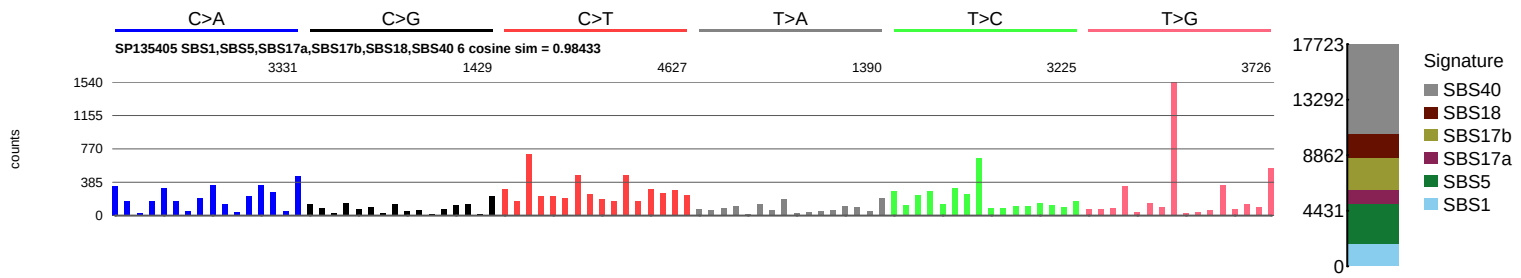

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98453
3294
1261
5022
1325
3400
3420
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98453
3275
1341
4991
1303
3386
3425
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98451
3324
1419
4579
1381
3305
3719
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98449
3396
1474
4612
1412
3036
3791
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98436
3283
1280
4989
1324
3418
3435
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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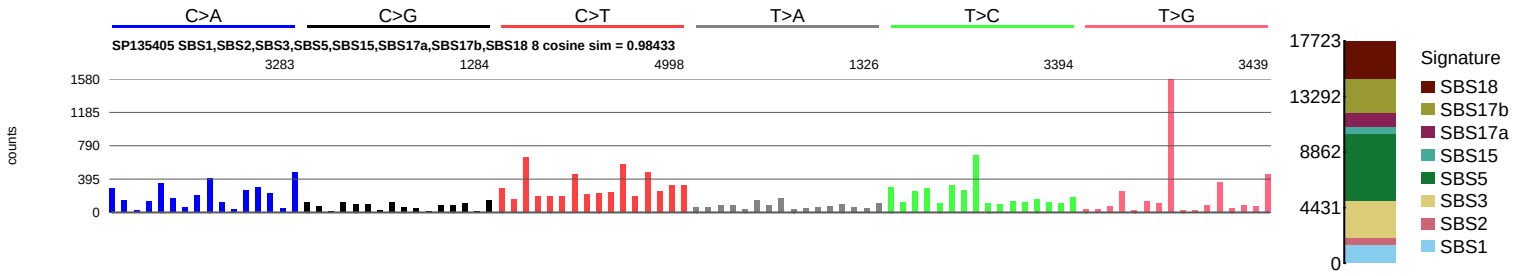
SP135405 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98433
3331
1429
4627
1390
3225
3726
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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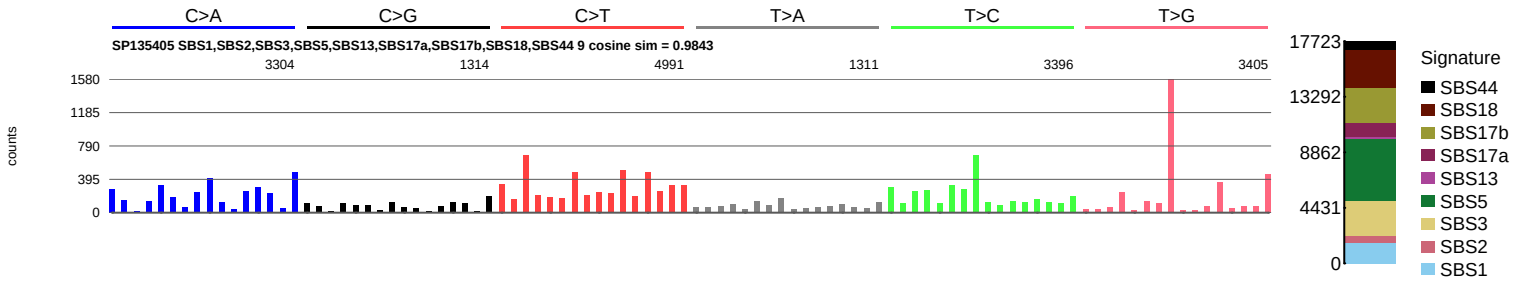
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98433
3283
1284
4998
1326
3394
3439
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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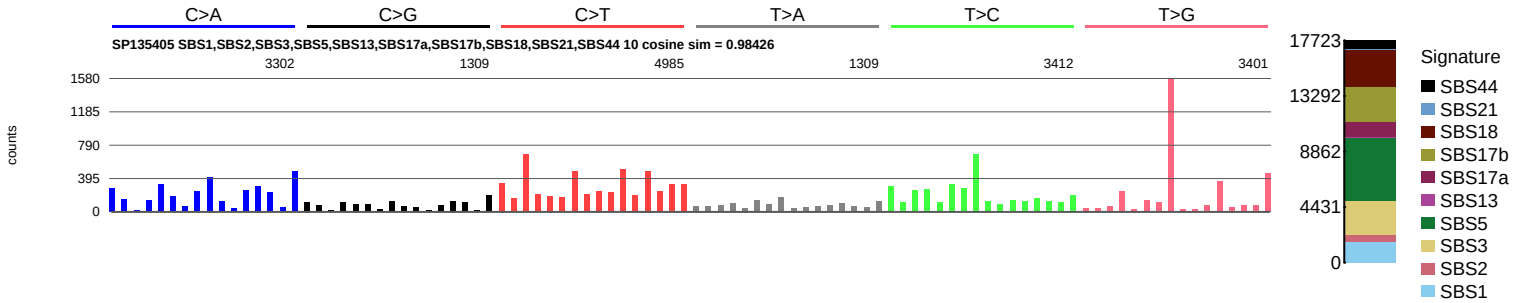
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.9843
3304
1314
4991
1311
3396
3405
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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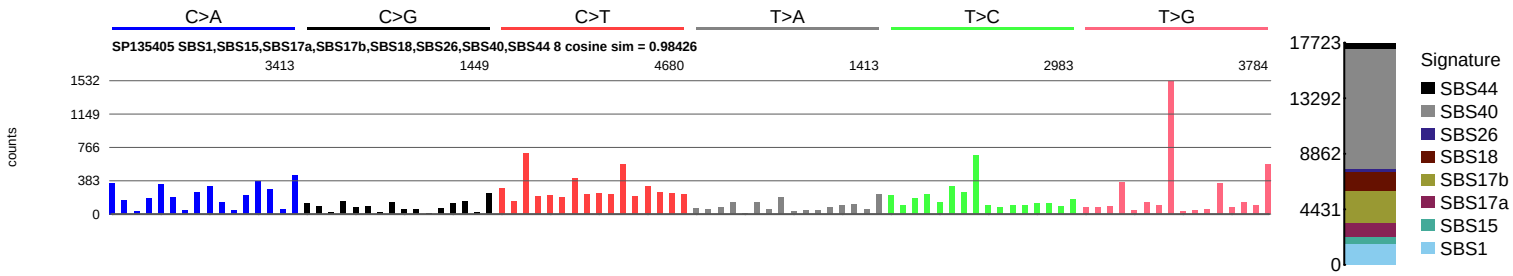
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98426
3302
1309
4985
1309
3412
3401
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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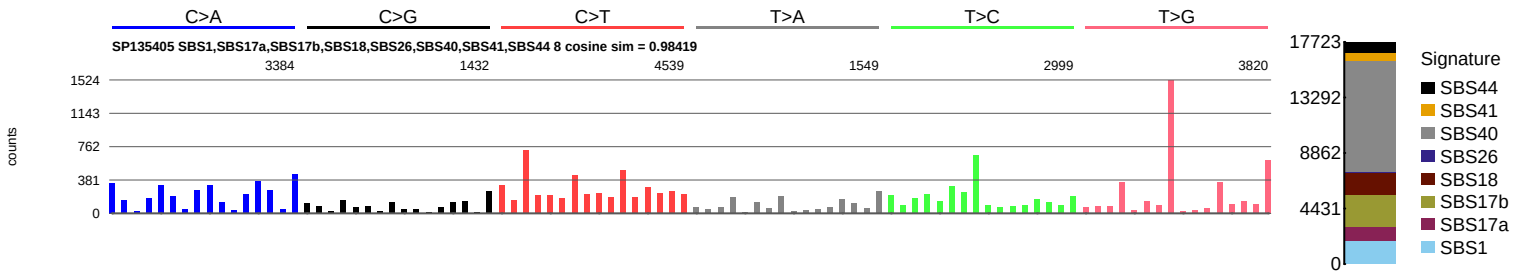
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98426
3413
1449
4680
1413
2983
3784
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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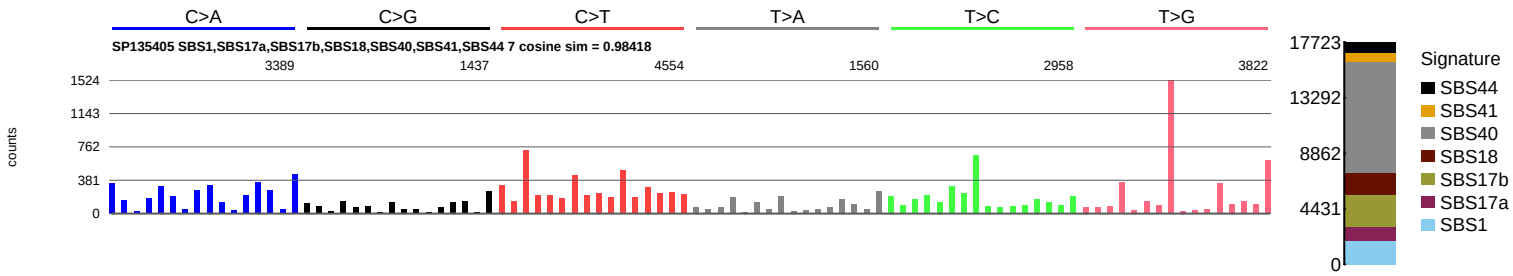
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98419
3384
1432
4539
1549
2999
3820
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
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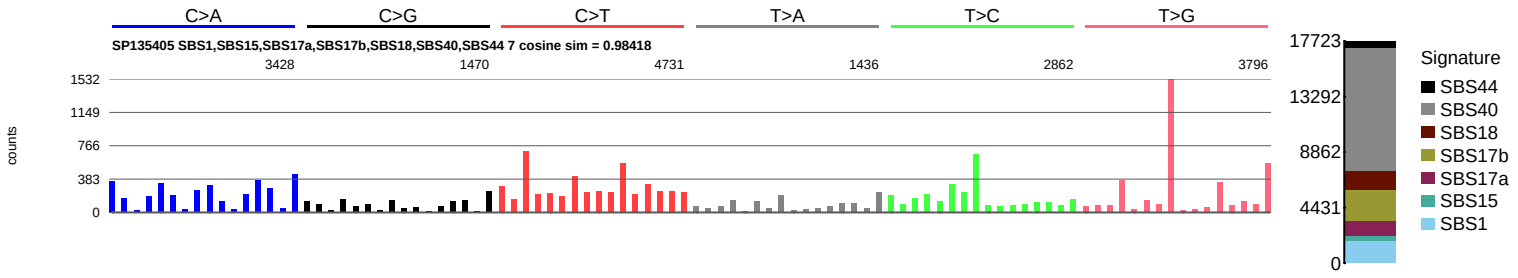
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98418
3389
1437
4554
1560
2958
3822
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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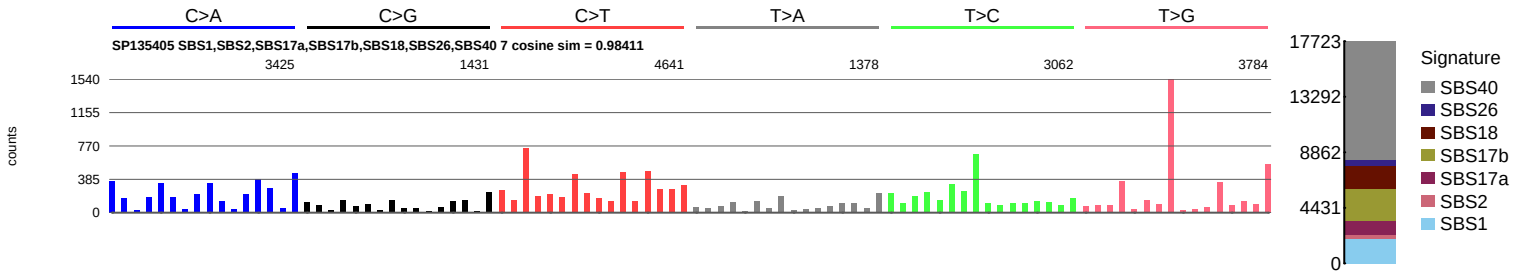
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98418
3428
1470
4731
1436
2862
3796
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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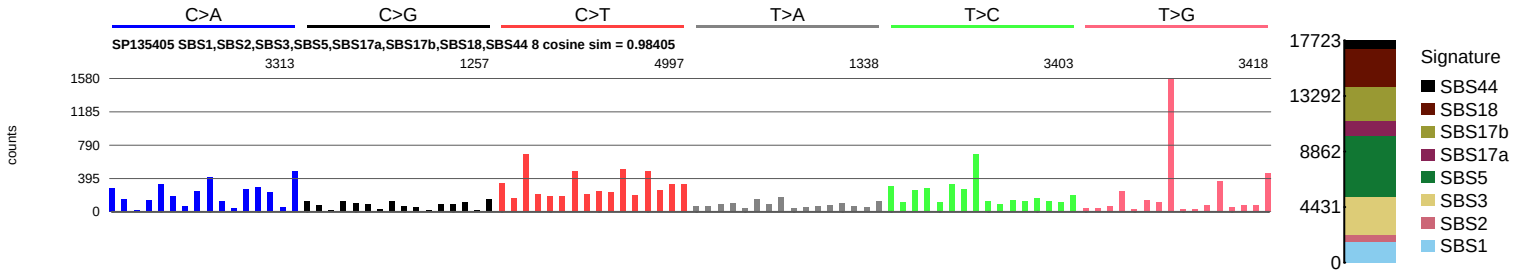
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98411
3425
1431
4641
1378
3062
3784
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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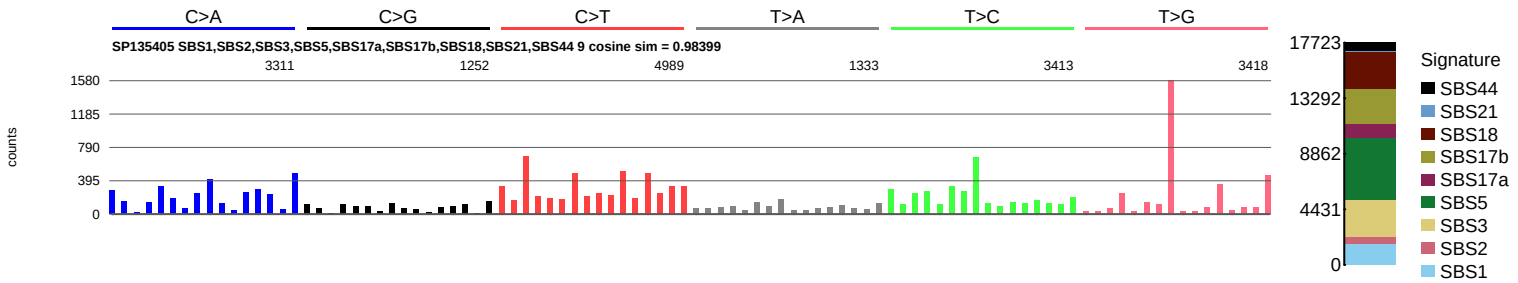
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98405
3313
1257
4997
1338
3403
3418
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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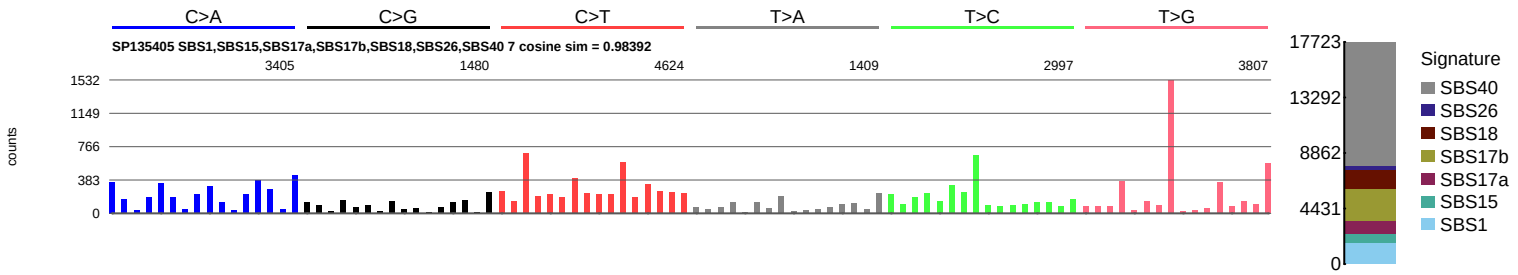
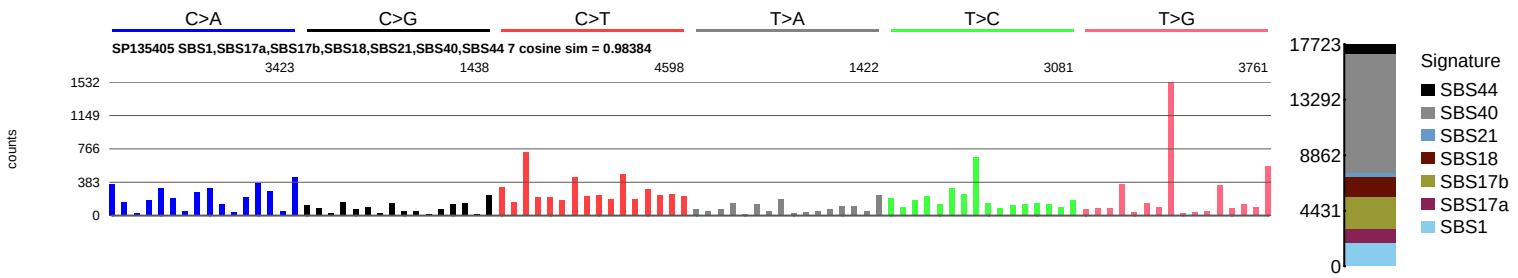
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98399
3311
1252
4989
1333
3413
3418
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98392
3405
1480
4624
1409
2997
3807
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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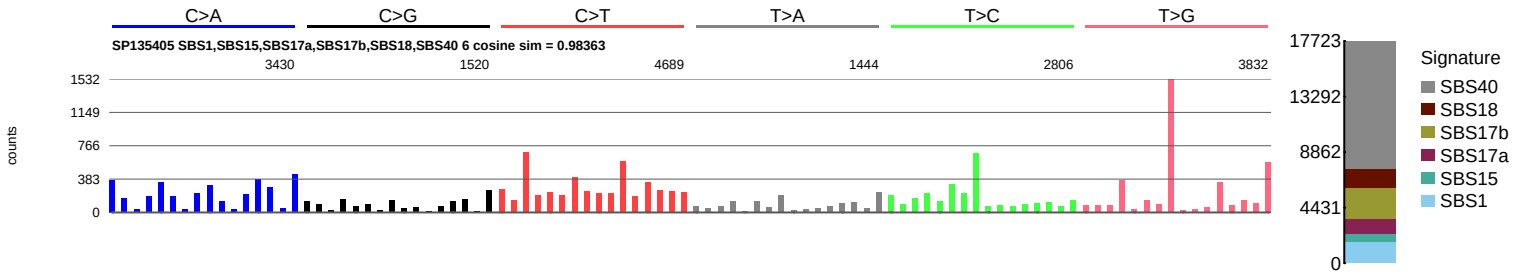
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98384
3423
1438
4598
1422
3081
3761
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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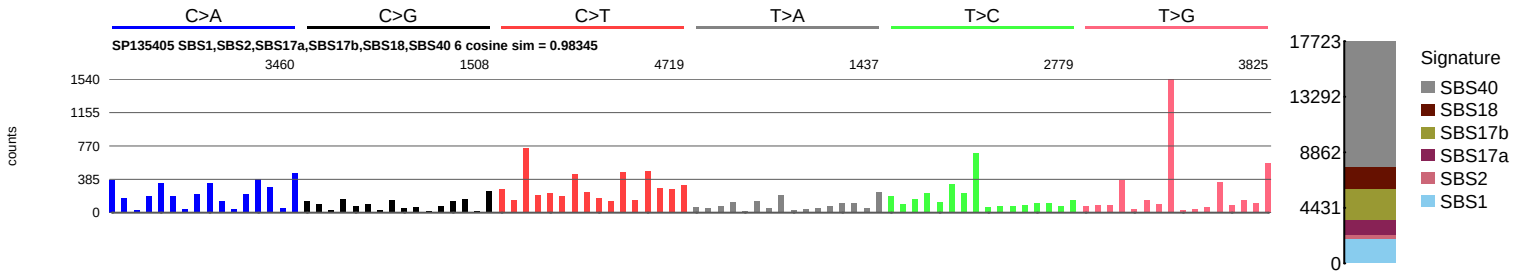
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98363
3430
1520
4689
1444
2806
3832
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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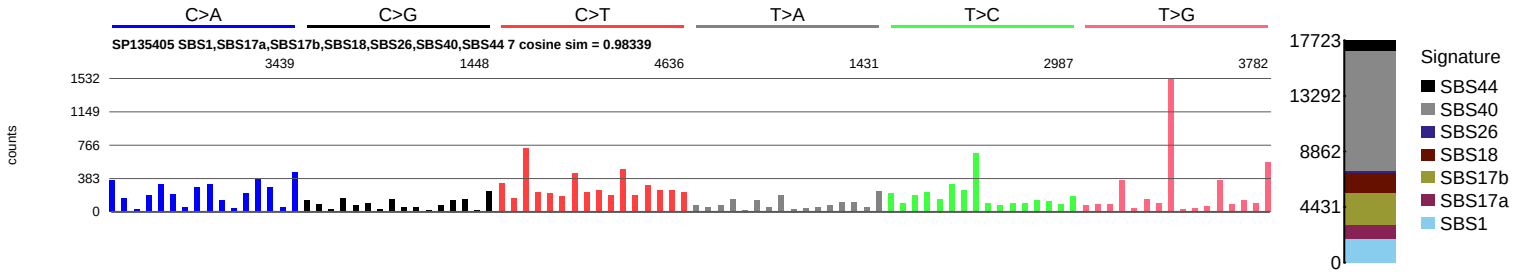
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98345
3460
1508
4719
1437
2779
3825
counts
1540
1155
770
385
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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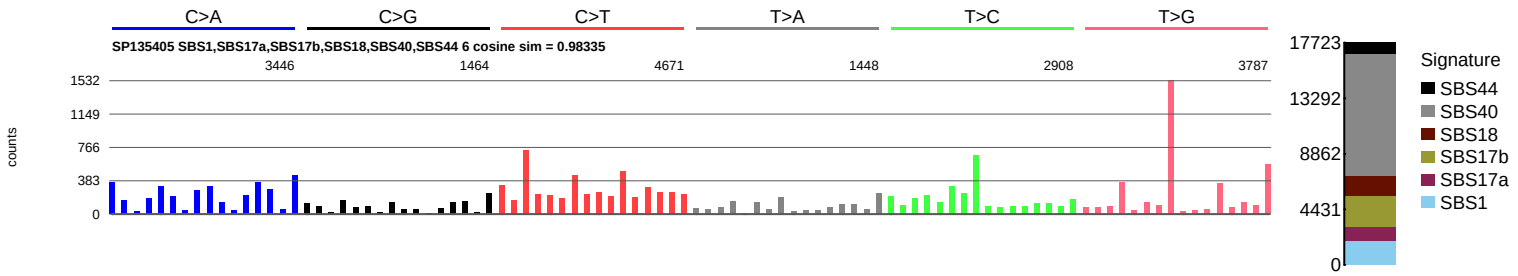
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98339
3439
1448
4636
1431
2987
3782
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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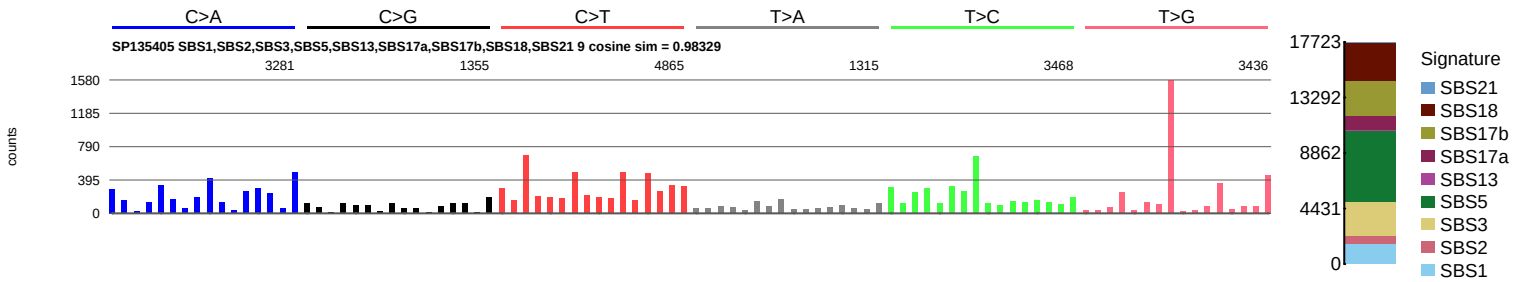
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98335
3446
1464
4671
1448
2908
3787
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1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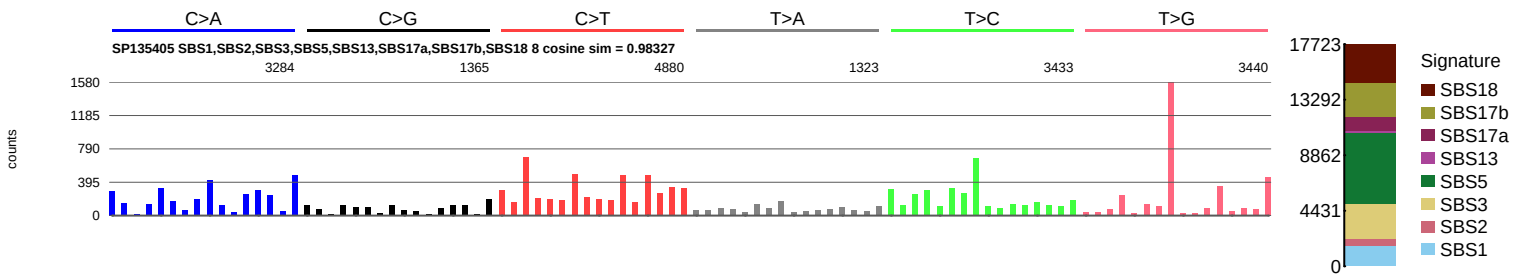
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98329
3281
1355
4865
1315
3468
3436
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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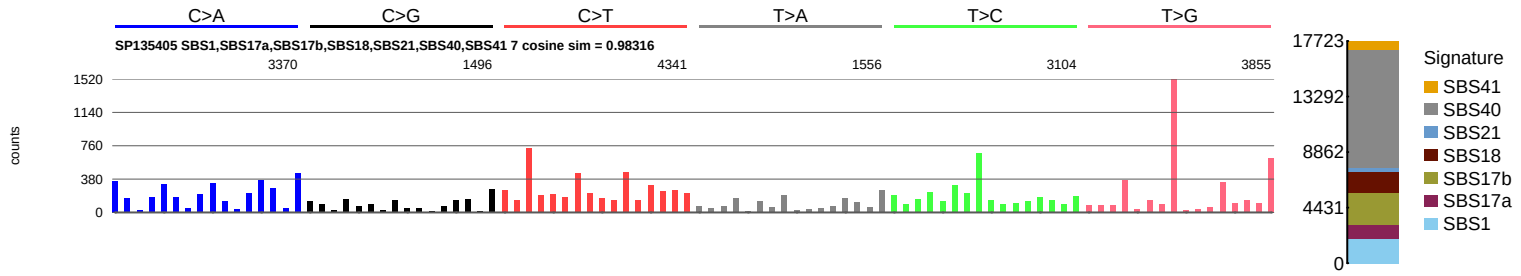
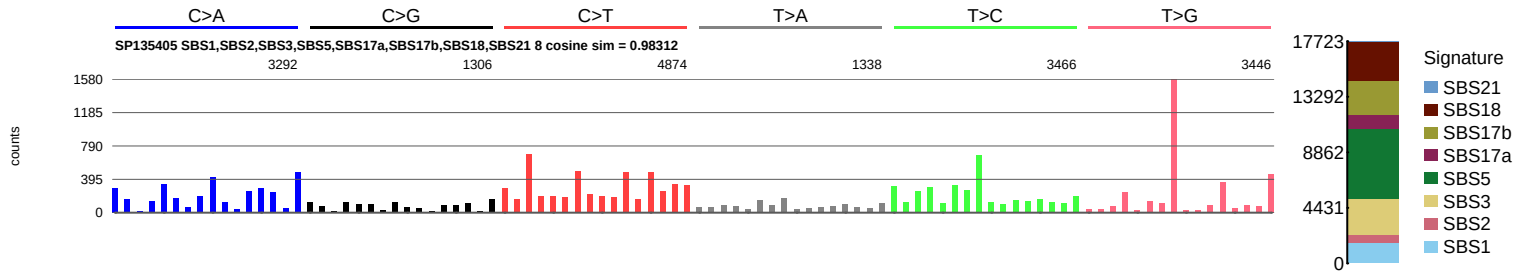
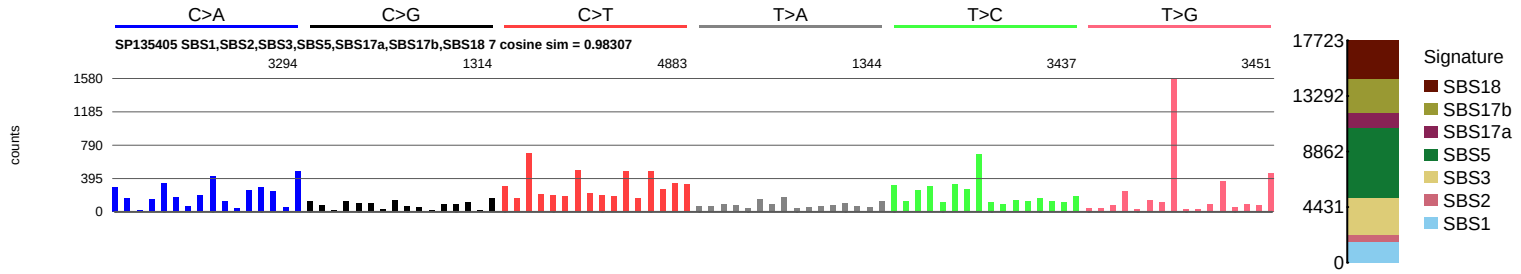
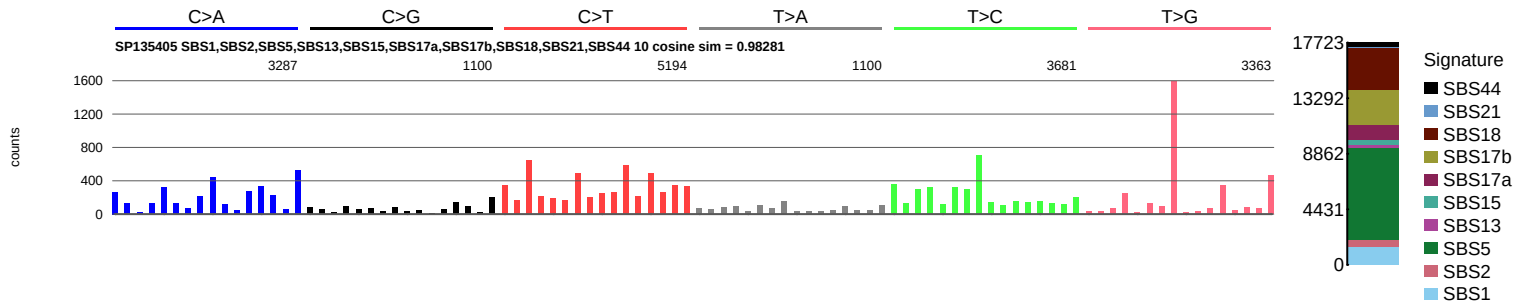
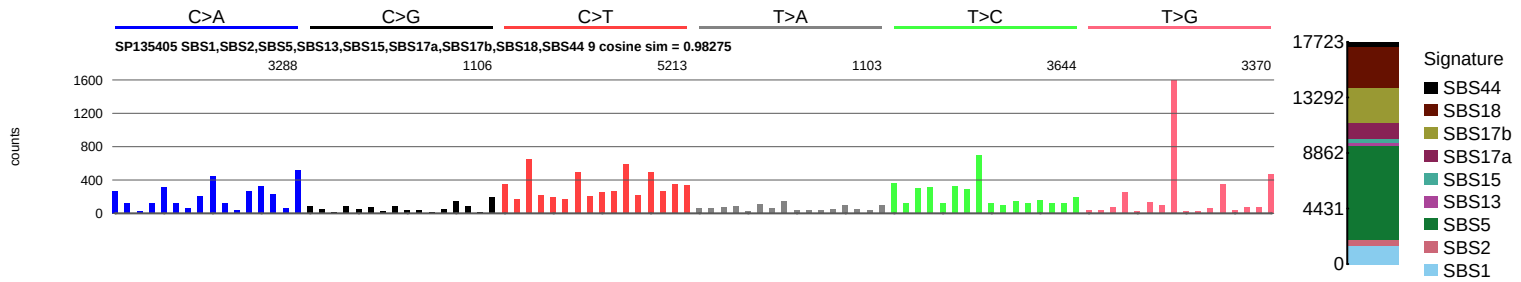
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98327
3284
1365
4880
1323
3433
3440
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

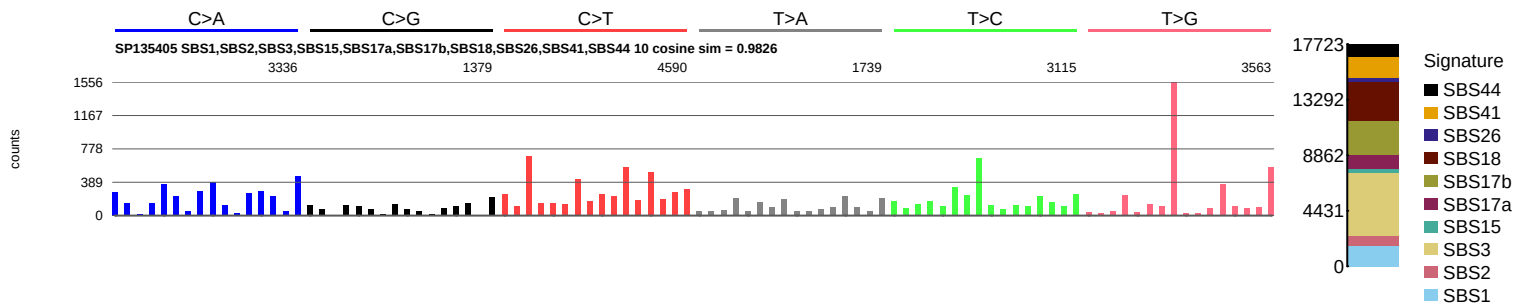

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98316
3370
1496
4341
1556
3104
3855
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

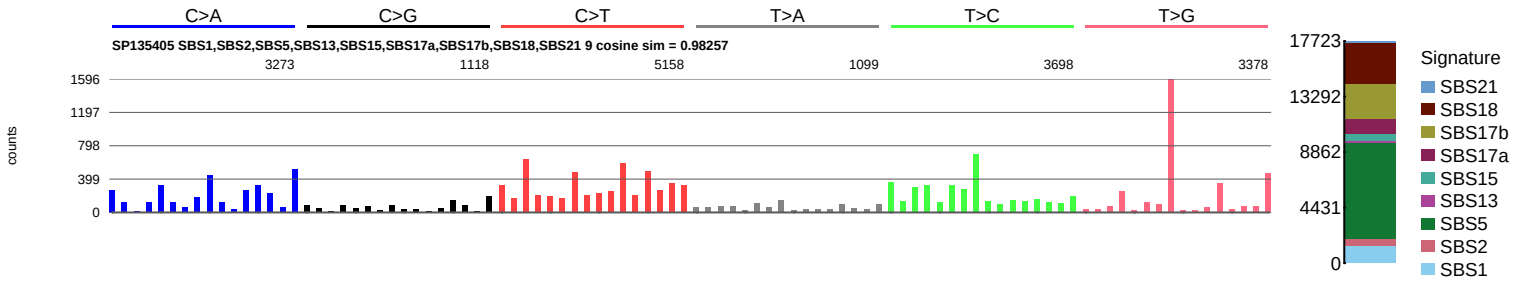

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98312
3292
1306
4874
1338
3466
3446
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98307
3294
1314
4883
1344
3437
3451
counts
1580
1185
790
395
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98281
3287
1100
5194
1100
3681
3363
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98275
3288
1106
5213
1103
3644
3370
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9826
3336
1379
4590
1739
3115
3563
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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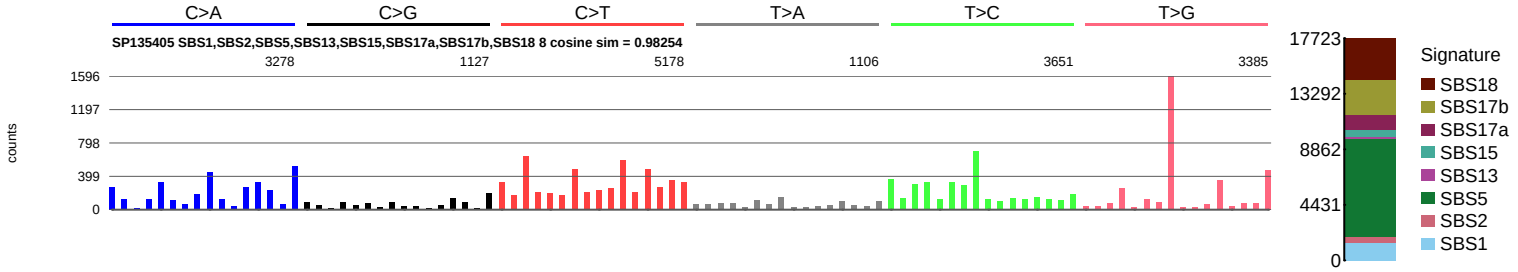
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98257
3273
1118
5158
1099
3698
3378
counts
1596
1197
798
399
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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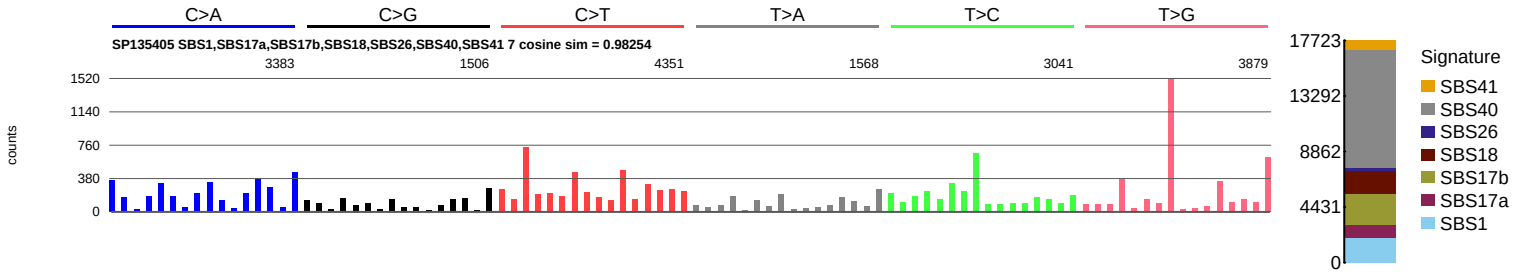
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98254
3278
1127
5178
1106
3651
3385
counts
1596
1197
798
399
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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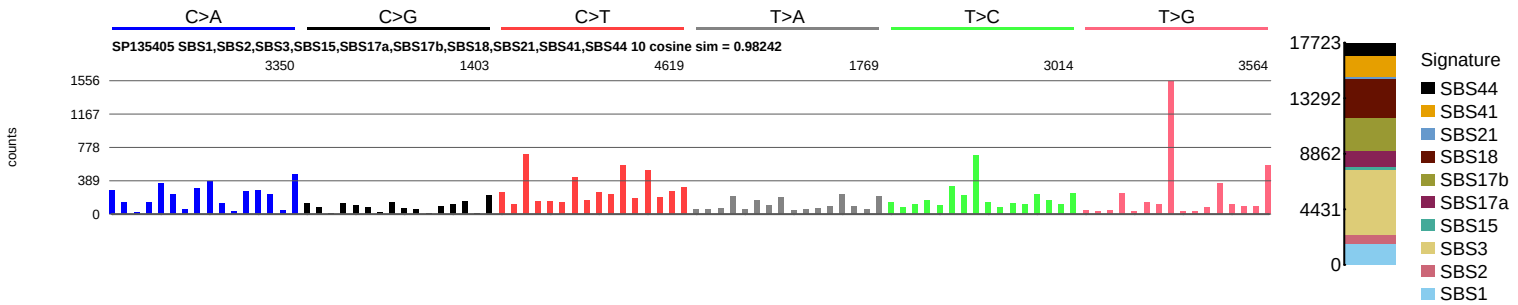
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98254
3383
1506
4351
1568
3041
3879
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
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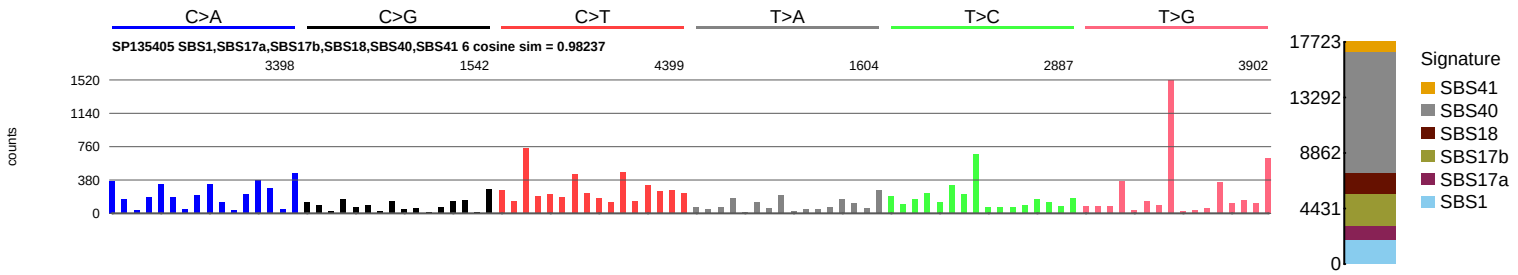
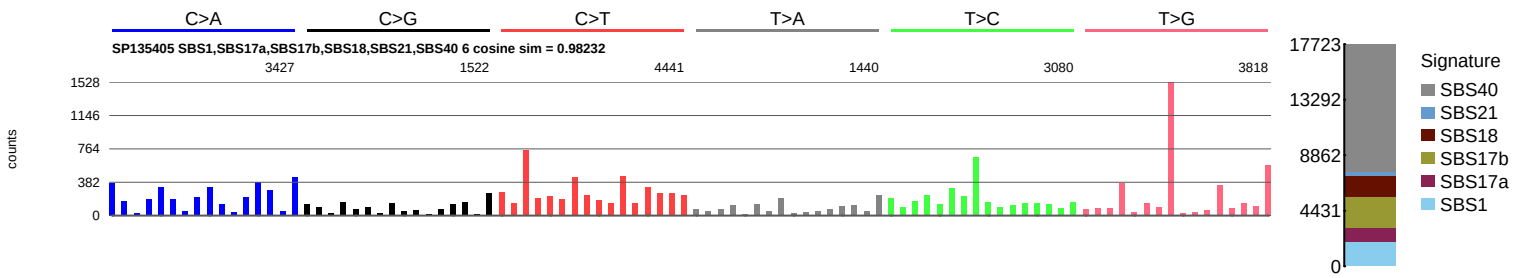
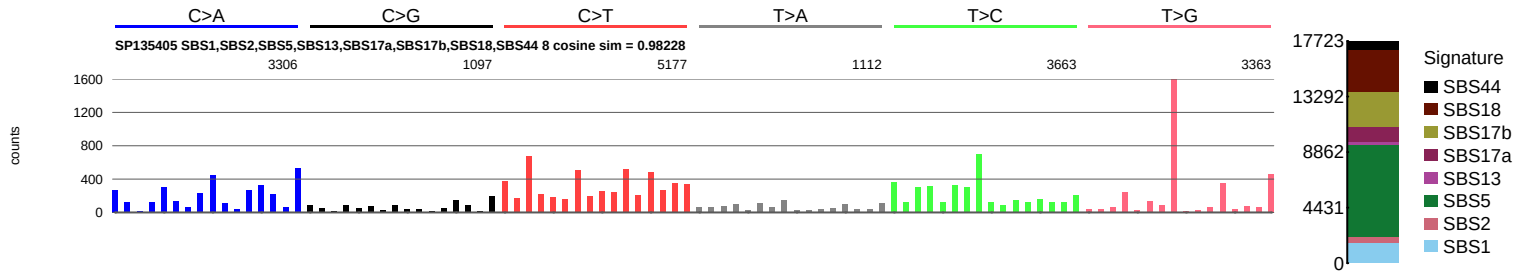
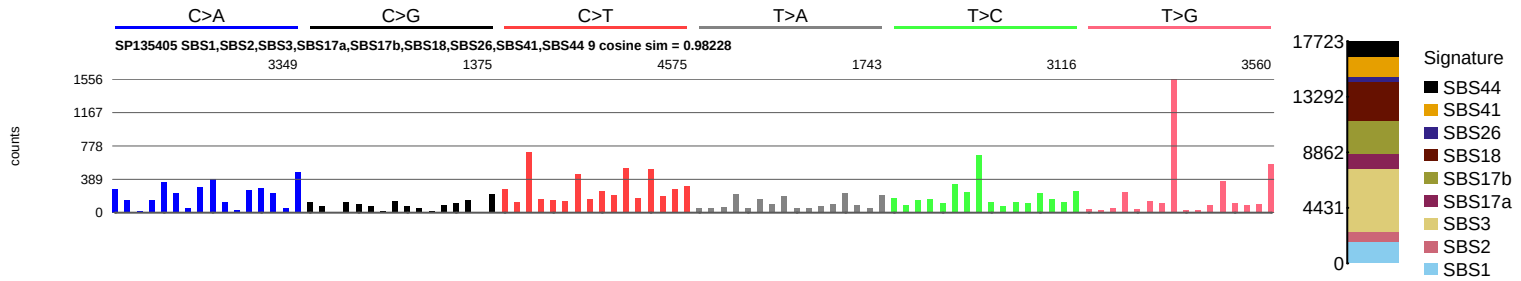
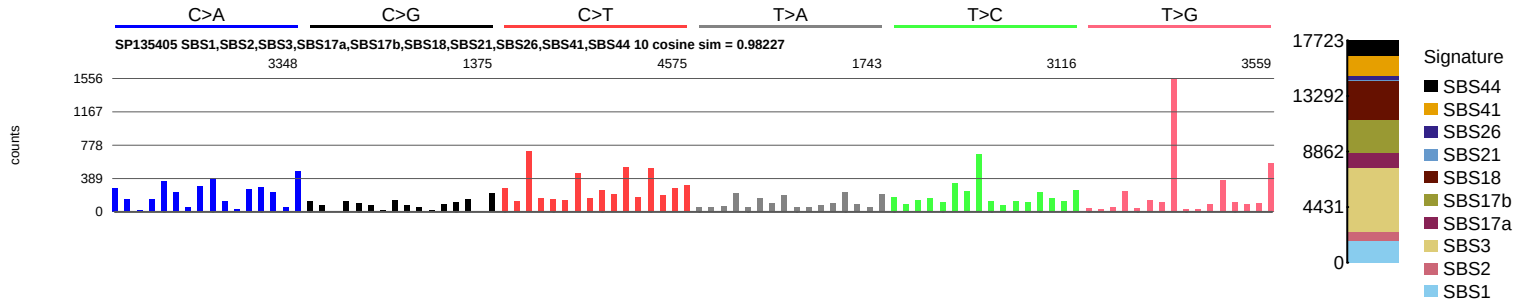
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98242
3350
1403
4619
1769
3014
3564
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

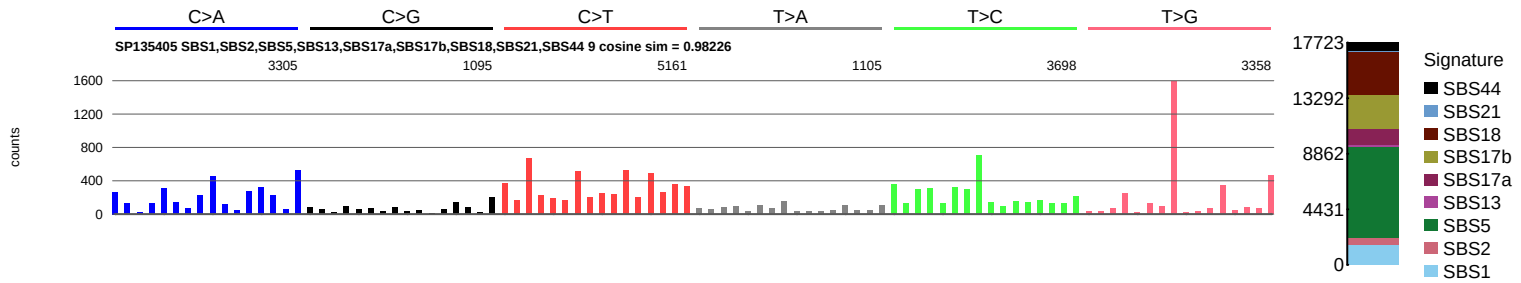

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98237
3398
1542
4399
1604
2887
3902
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

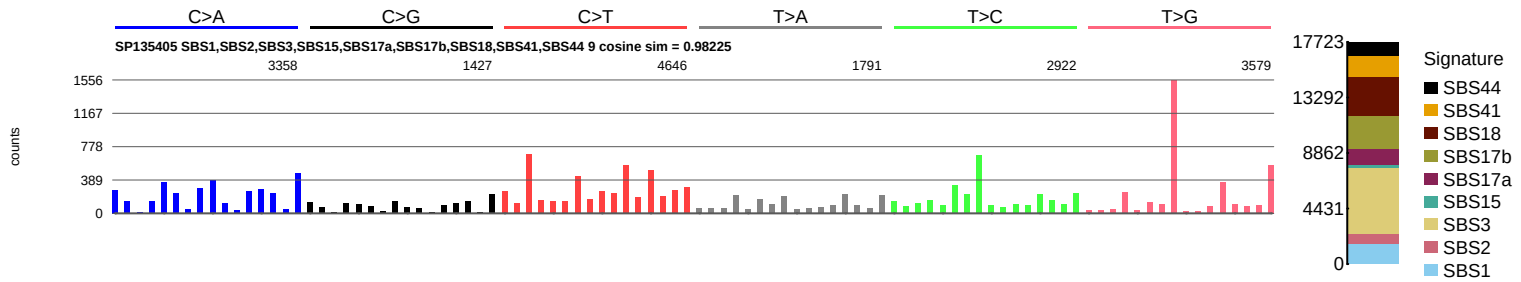

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98232
3427
1522
4441
1440
3080
3818
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

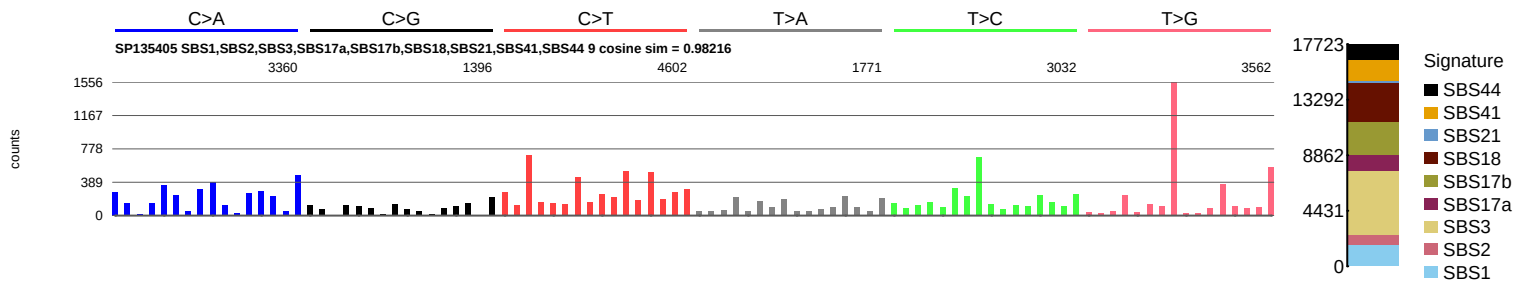

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98228
3306
1097
5177
1112
3663
3363
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

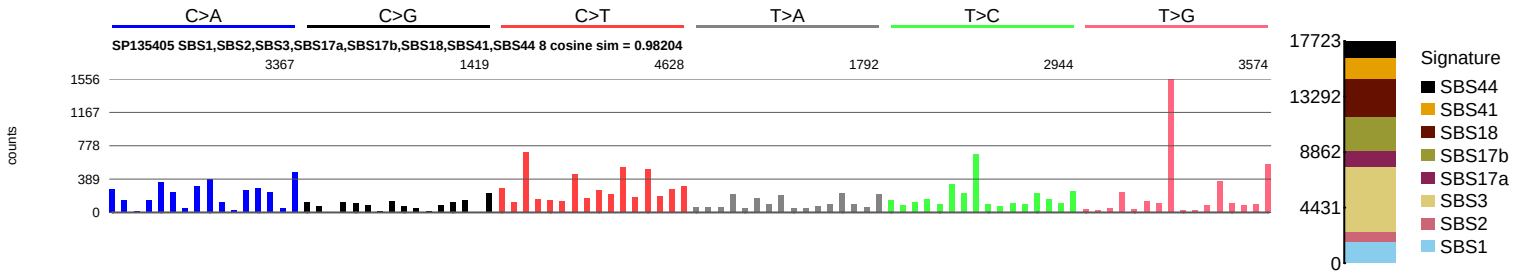

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98228
3349
1375
4575
1743
3116
3560
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

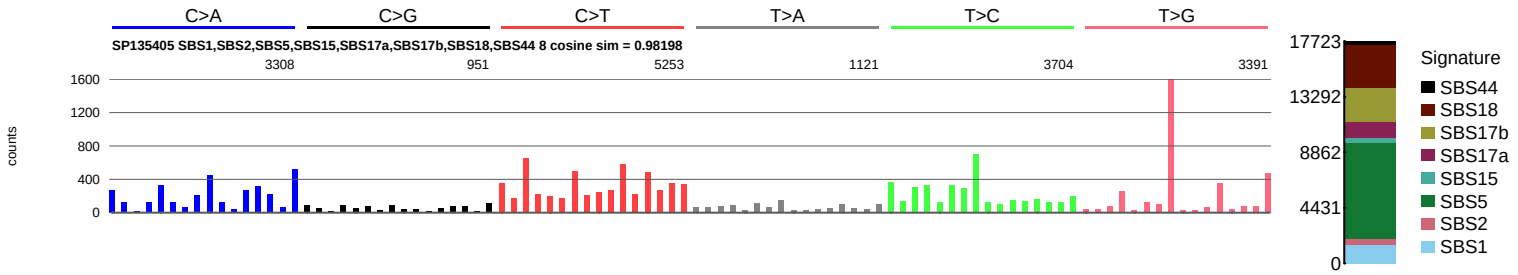

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98227
3348
1375
4575
1743
3116
3559
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

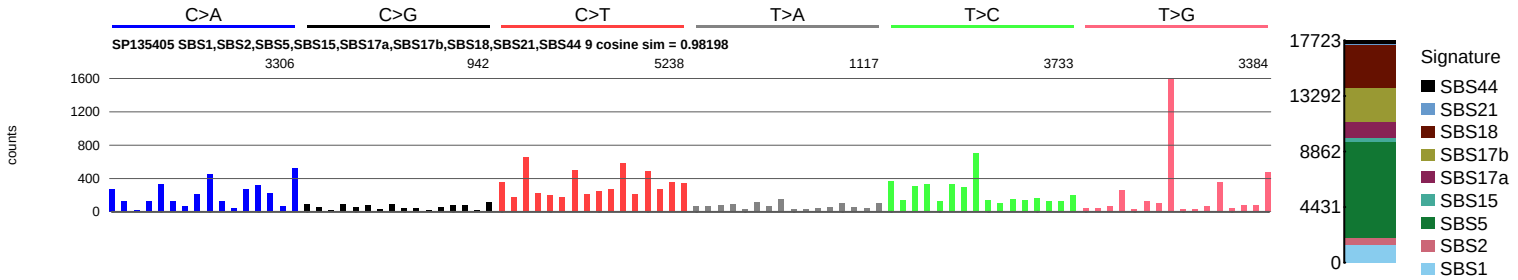

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98226
3305
1095
5161
1105
3698
3358
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

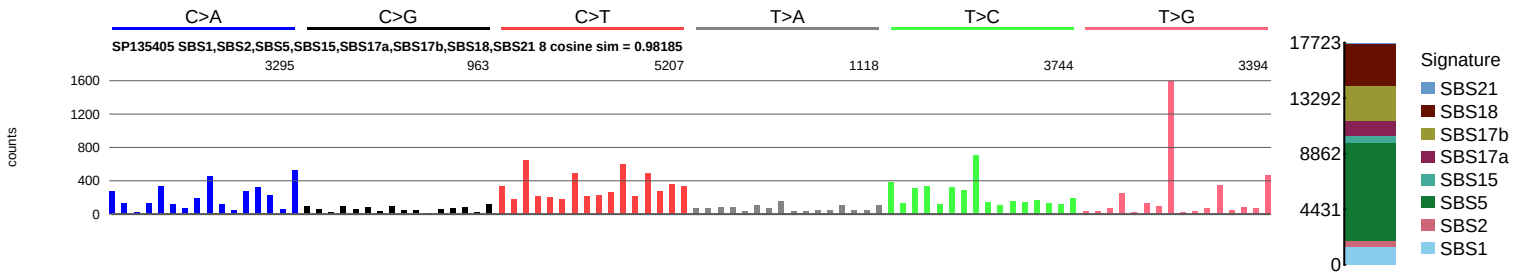

C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98225
3358
1427
4646
1791
2922
3579
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98216
3360
1396
4602
1771
3032
3562
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98204
3367
1419
4628
1792
2944
3574
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98198
3308
951
5253
1121
3704
3391
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98198
3306
942
5238
1117
3733
3384
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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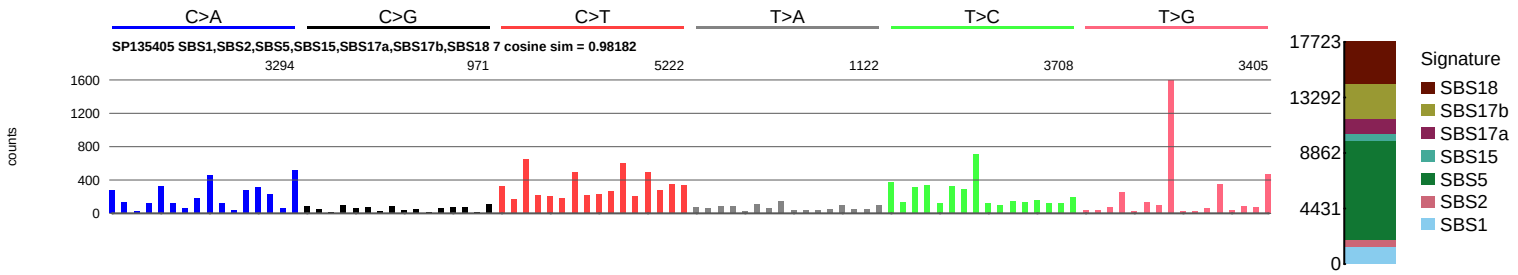
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98185
3295
963
5207
1118
3744
3394
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98182
3294
971
5222
1122
3708
3405
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
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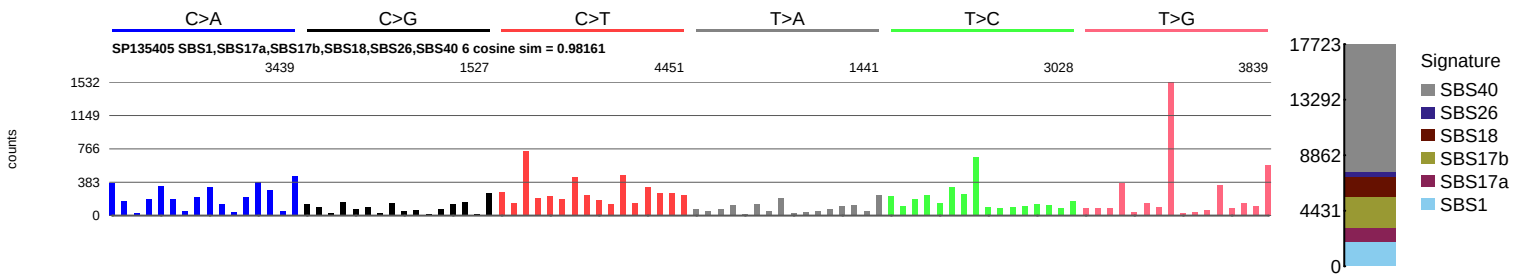
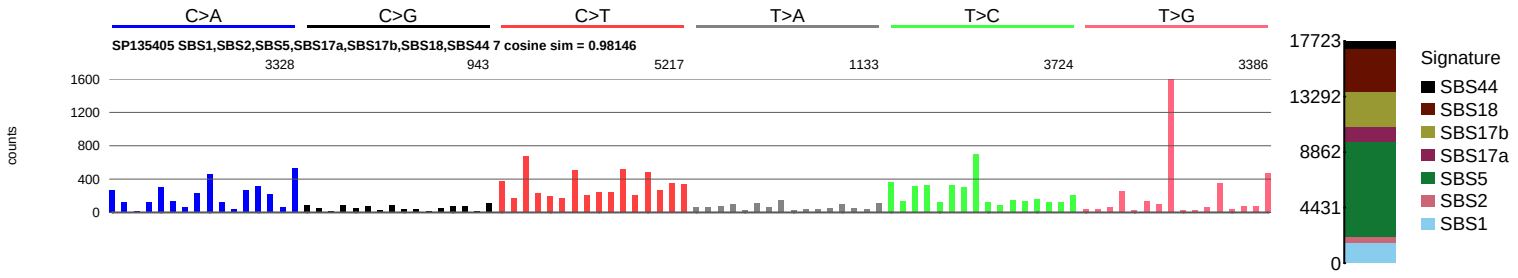
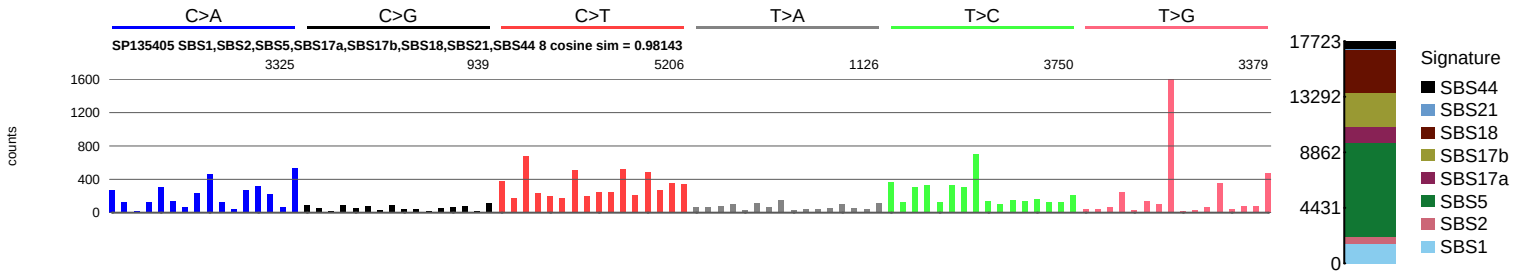
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98161
3439
1527
4451
1441
3028
3839
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98146
3328
943
5217
1133
3724
3386
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98143
3325
939
5206
1126
3750
3379
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS44
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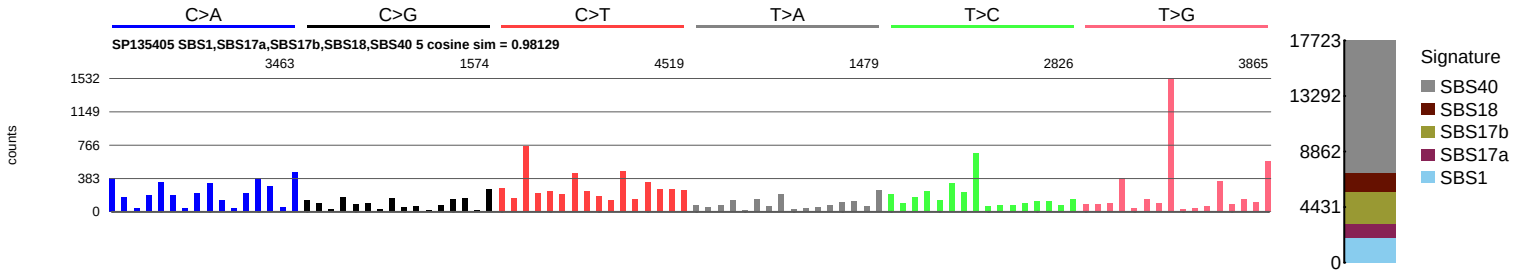
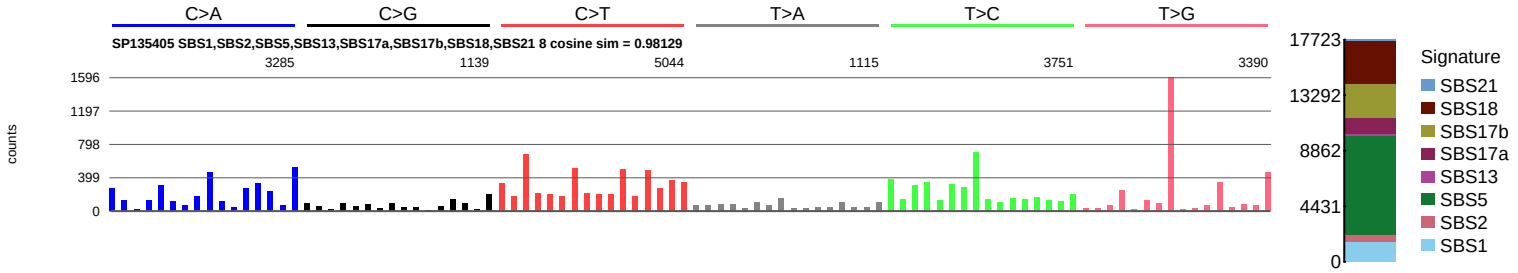
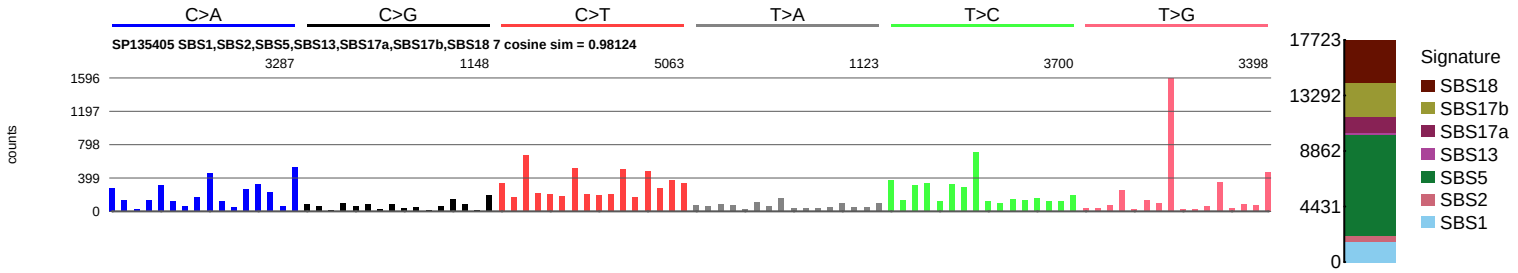
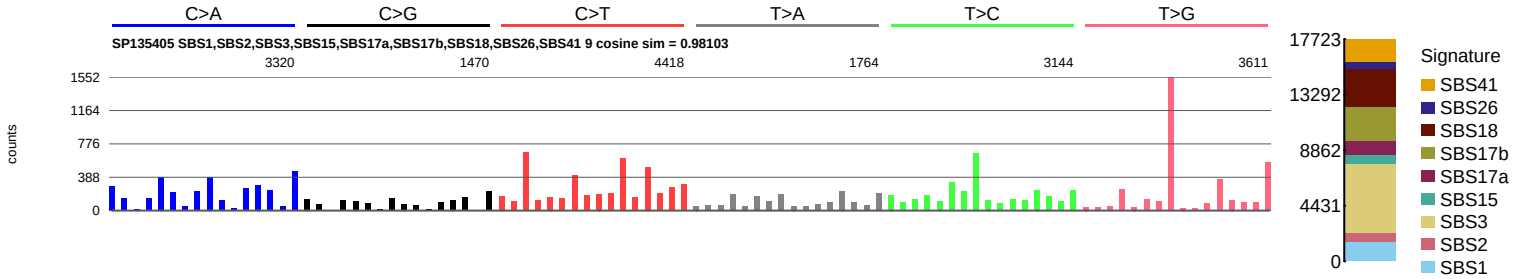
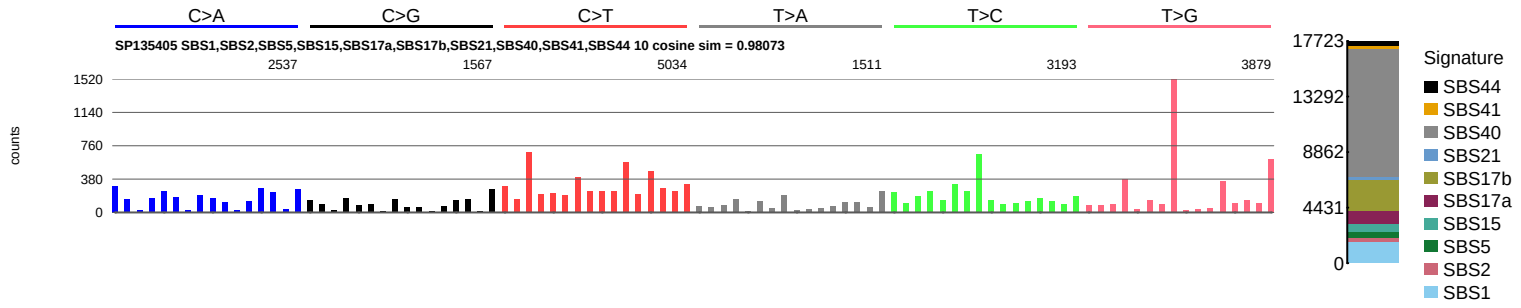
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98129
3463
1574
4519
1479
2826
3865
counts
1532
1149
766
383
0
17723
13292
8862
4431
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98129
3285
1139
5044
1115
3751
3390
counts
1596
1197
798
399
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98124
3287
1148
5063
1123
3700
3398
counts
1596
1197
798
399
0
17723
13292
8862
4431
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98103
3320
1470
4418
1764
3144
3611
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98073
2537
1567
5034
1511
3193
3879
counts
1520
1140
760
380
0
17723
13292
8862
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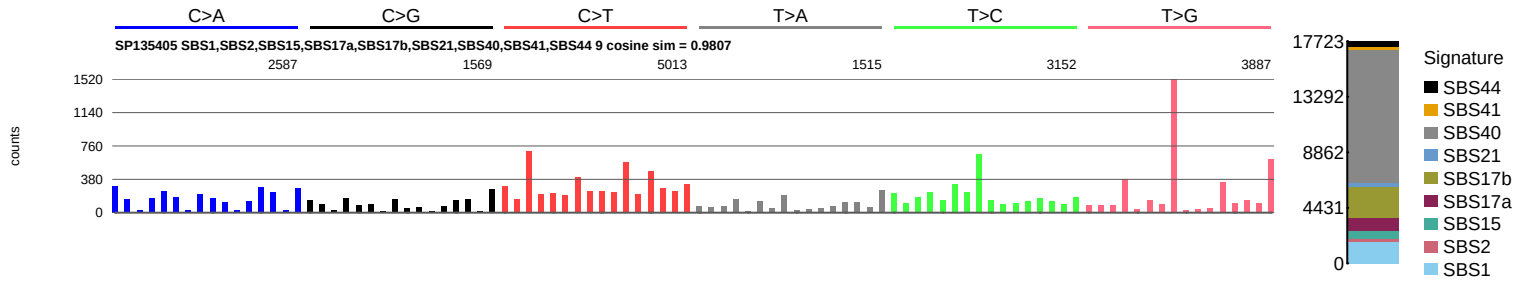
SBS1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SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9807
2587
1569
5013
1515
3152
3887
counts
1520
1140
760
380
0
17723
13292
8862
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Signature
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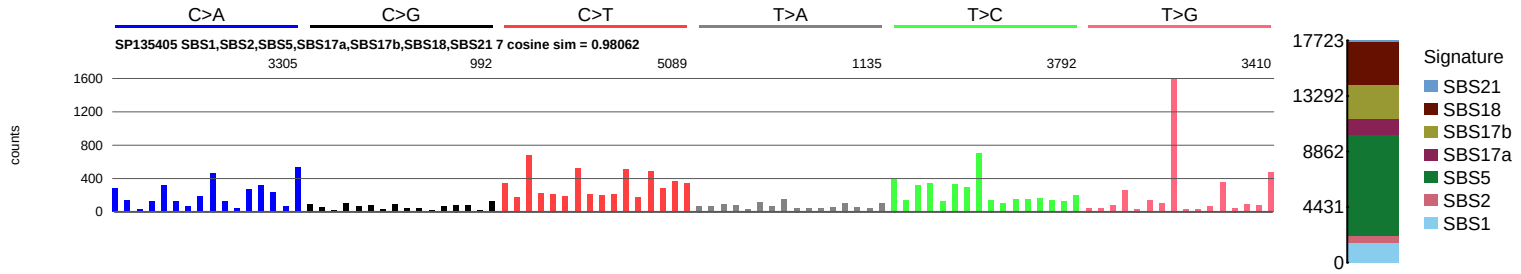
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C>G
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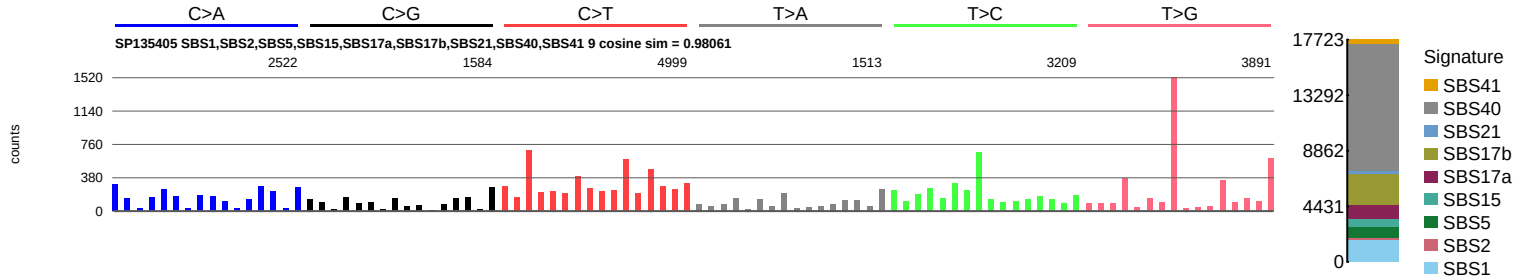
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98062
3305
992
5089
1135
3792
3410
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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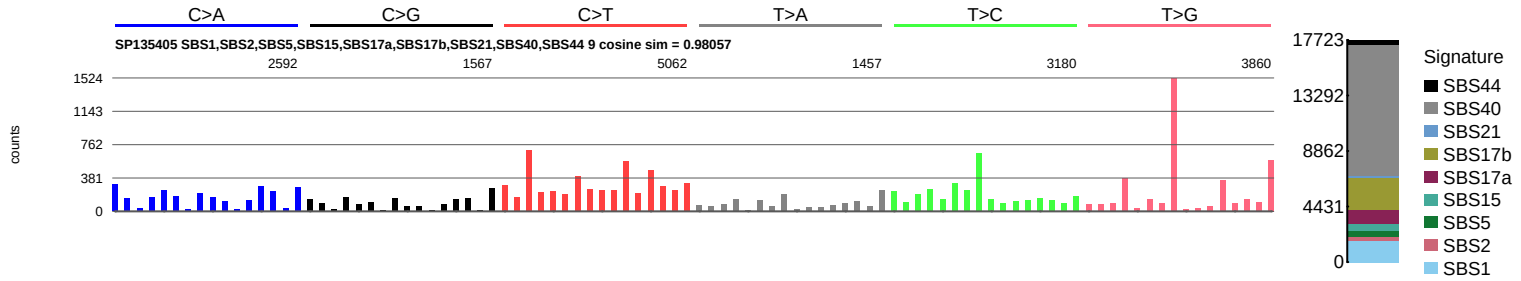
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98061
2522
1584
4999
1513
3209
3891
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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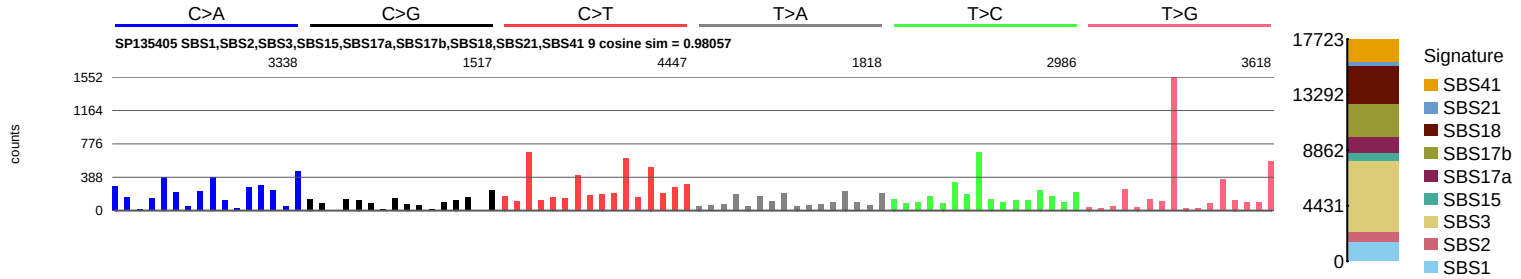
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98057
2592
1567
5062
1457
3180
3860
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98057
3338
1517
4447
1818
2986
3618
counts
1552
1164
776
388
0
17723
13292
8862
4431
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Signature
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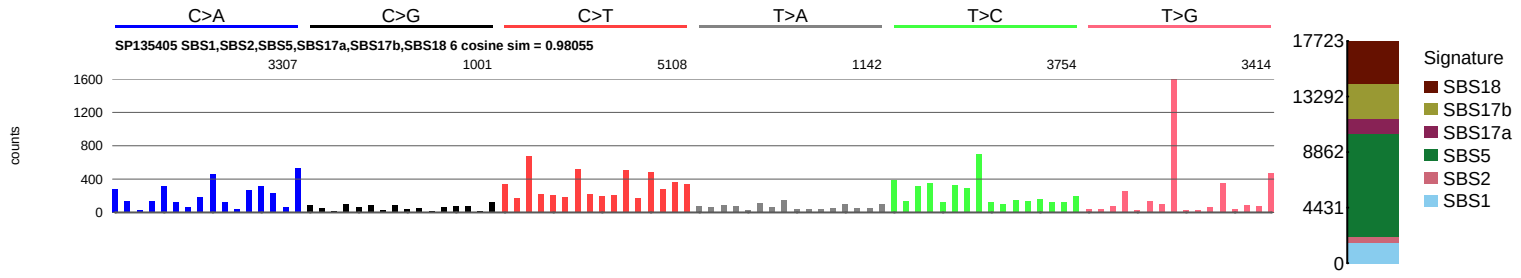
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98055
3307
1001
5108
1142
3754
3414
counts
1600
1200
800
400
0
17723
13292
8862
4431
0
Signature
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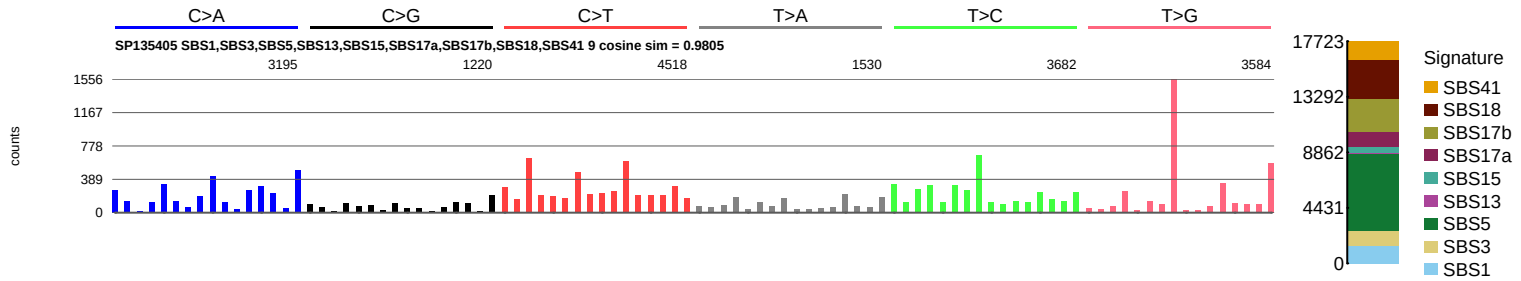
SBS2
SBS1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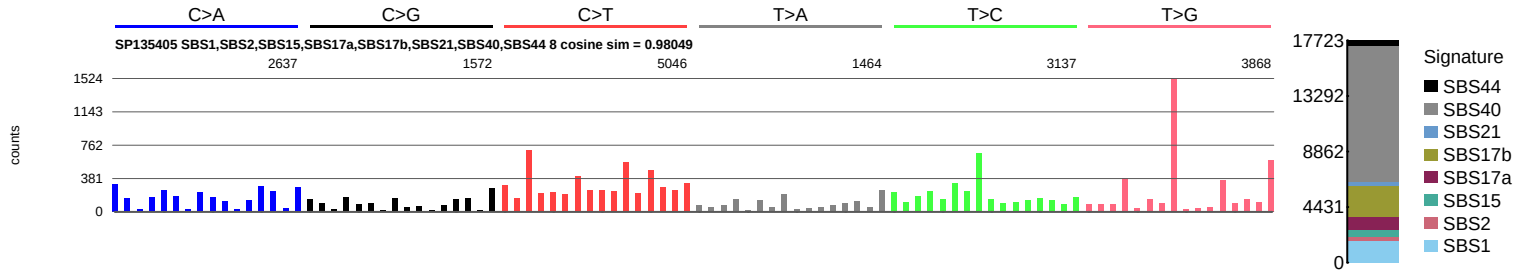
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9805
3195
1220
4518
1530
3682
3584
counts
1556
1167
778
389
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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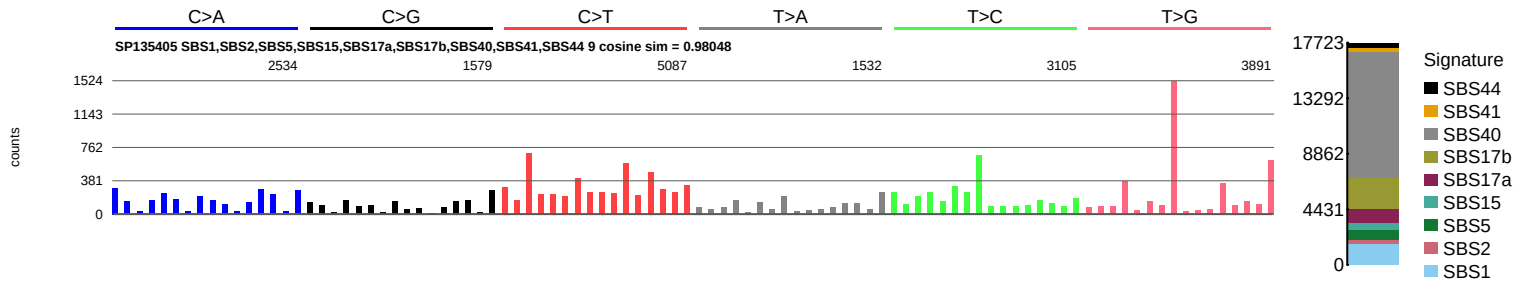
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98049
2637
1572
5046
1464
3137
3868
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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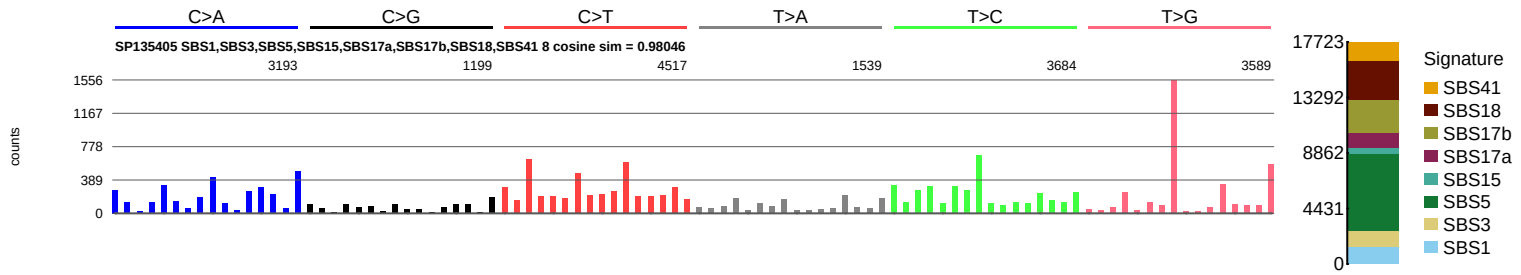
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98048
2534
1579
5087
1532
3105
3891
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98046
3193
1199
4517
1539
3684
3589
counts
1556
1167
778
389
0
17723
13292
8862
4431
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Signature
SBS41
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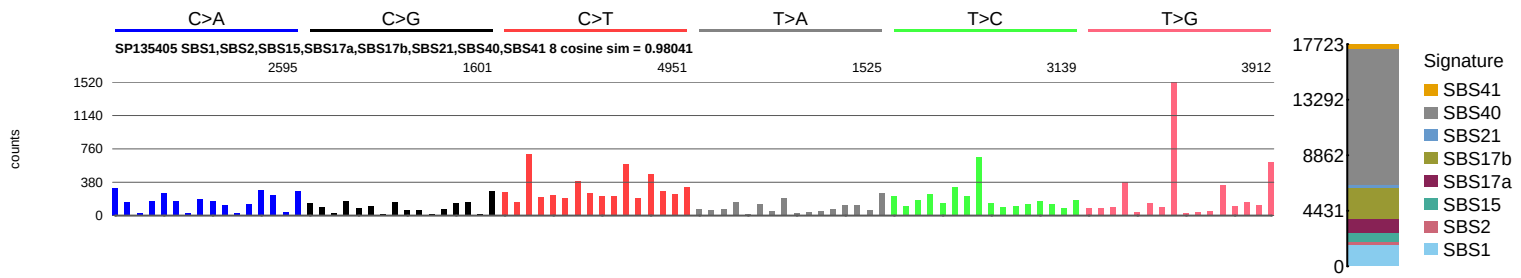
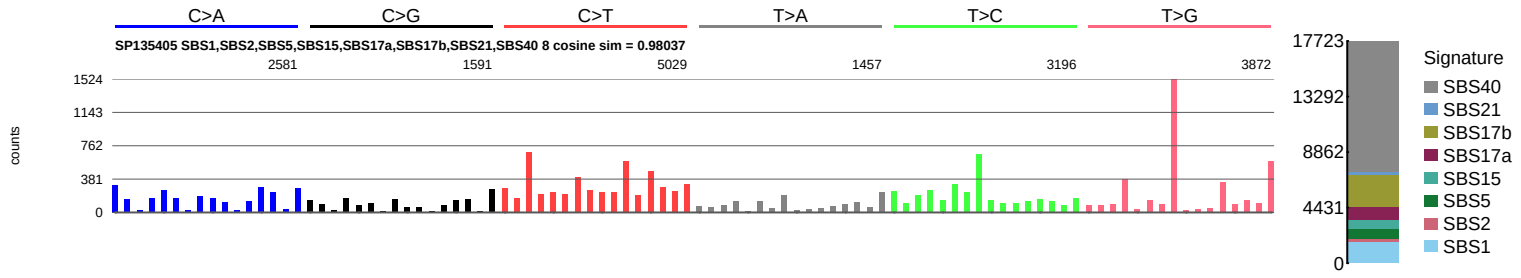
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98041
2595
1601
4951
1525
3139
3912
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98037
2581
1591
5029
1457
3196
3872
counts
1524
1143
762
381
0
17723
13292
8862
4431
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Signature
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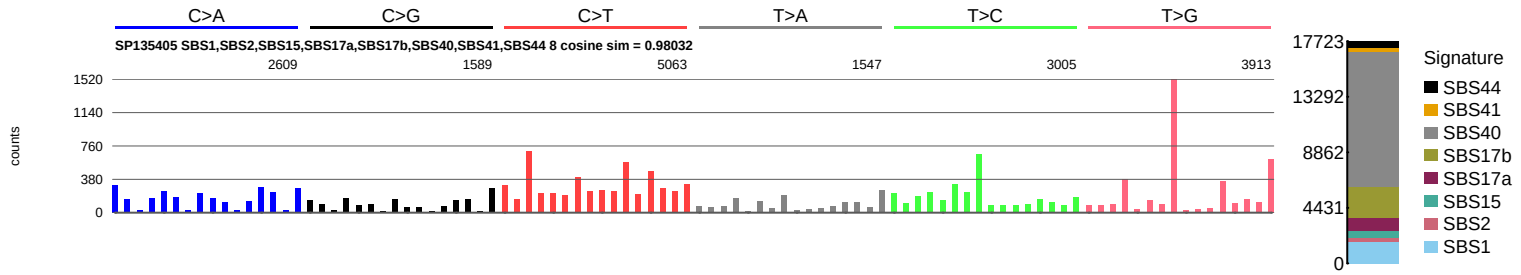
SBS2
SBS1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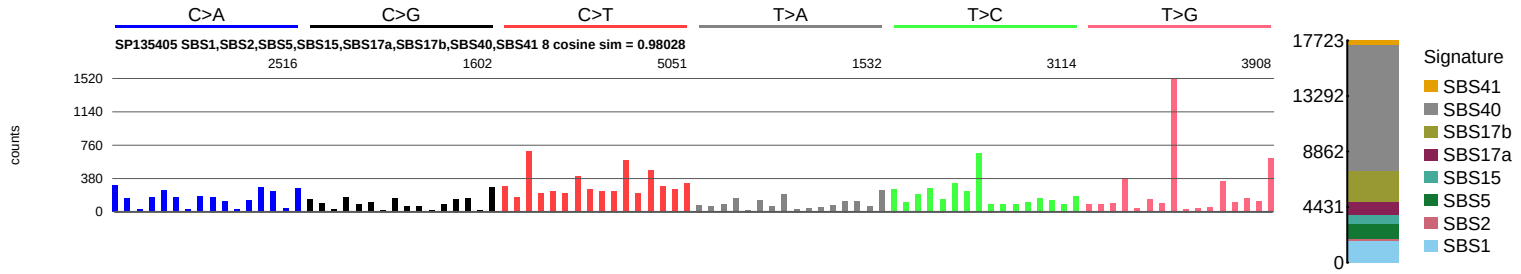
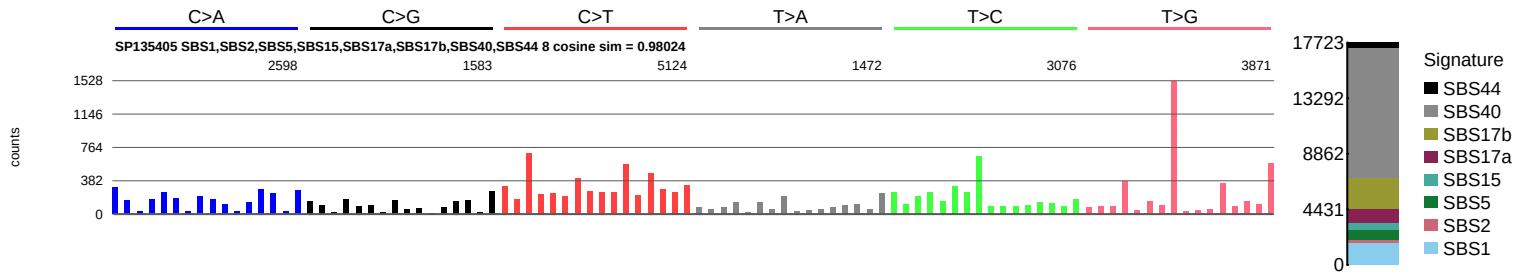
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98032
2609
1589
5063
1547
3005
3913
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98028
2516
1602
5051
1532
3114
3908
counts
1520
1140
760
380
0
17723
13292
8862
4431
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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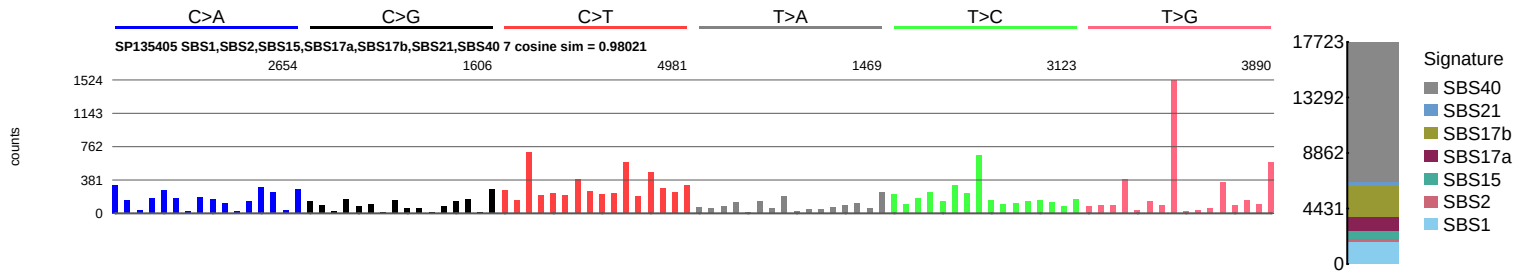
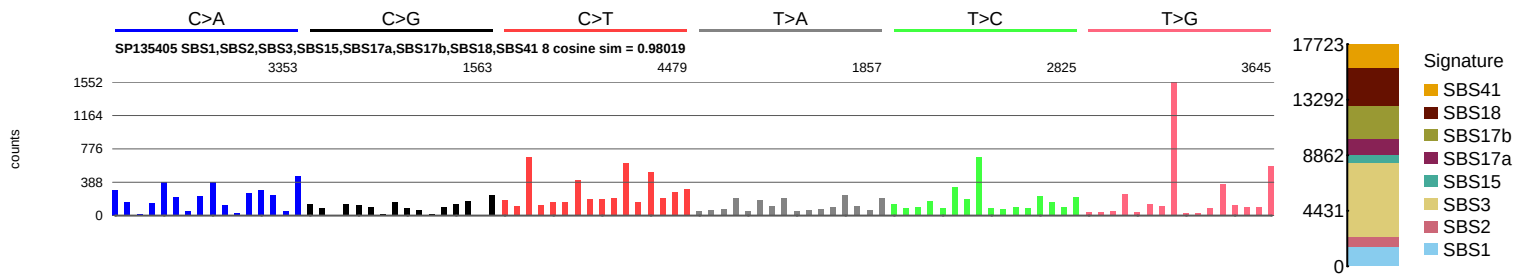
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98024
2598
1583
5124
1472
3076
3871
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98021
2654
1606
4981
1469
3123
3890
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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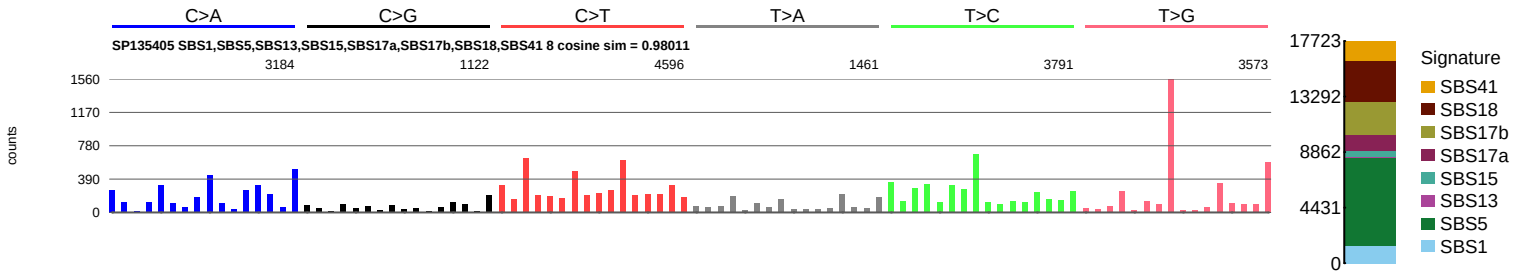
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98019
3353
1563
4479
1857
2825
3645
counts
1552
1164
776
388
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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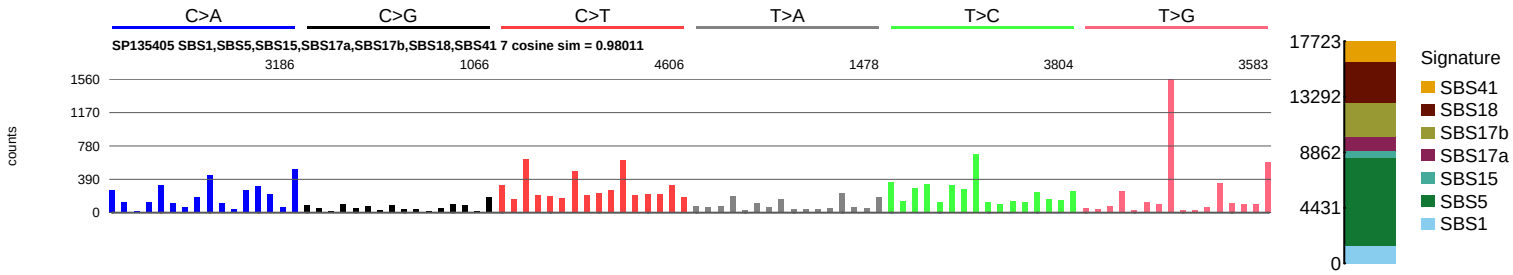
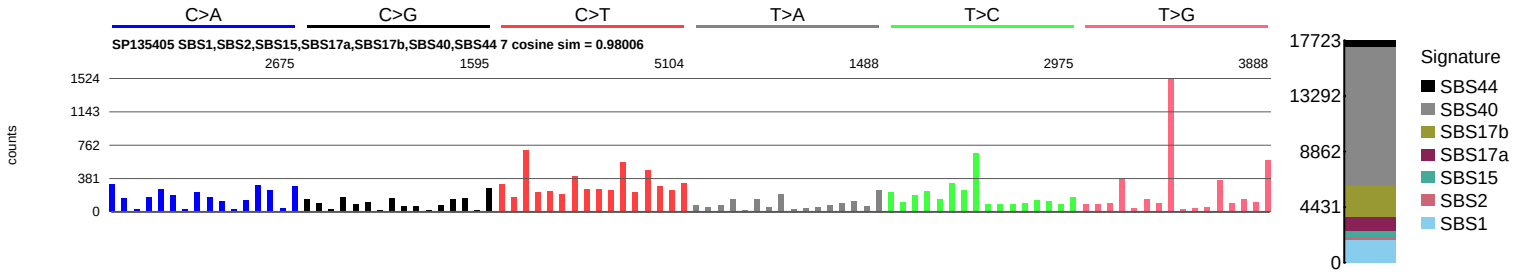
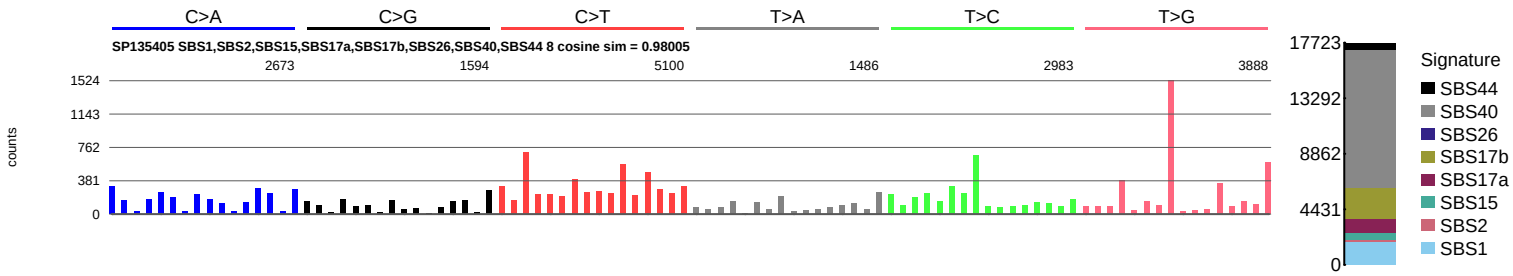
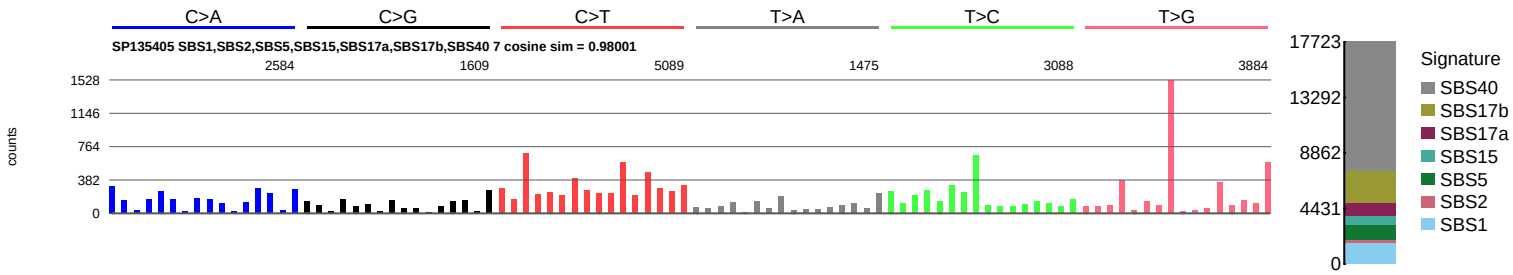
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98011
3184
1122
4596
1461
3791
3573
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98011
3186
1066
4606
1478
3804
3583
counts
1560
1170
780
390
0
17723
13292
8862
4431
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98006
2675
1595
5104
1488
2975
3888
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98005
2673
1594
5100
1486
2983
3888
counts
1524
1143
762
381
0
17723
13292
8862
4431
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135405 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98001
2584
1609
5089
1475
3088
3884
counts
1528
1146
764
382
0
17723
13292
8862
4431
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1